UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-K

(Mark One)

☒ ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

For the fiscal year ended December 31, 2023

or

☐ TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

For the transition period from to

Commission file number 1-442

THE BOEING COMPANY

(Exact name of registrant as specified in its charter)

Delaware	**91-0425694**
(State or other jurisdiction of incorporation or organization)	**(I.R.S. Employer Identification No.)**
929 Long Bridge Drive Arlington, VA	**22202**
(Address of principal executive offices)	**(Zip Code)**

Registrant's telephone number, including area code (703)-465-3500

Securities registered pursuant to Section 12(b) of the Act:

Common Stock, $5.00 Par Value	BA	New York Stock Exchange
(Title of each class)	(Trading Symbol)	(Name of each exchange on which registered)

Securities registered pursuant to Section 12(g) of the Act:None

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☐ No ☒
Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. Yes ☐ No ☒
Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐
Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§ 232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐
Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large Accelerated Filer	☒	Accelerated filer	☐
Non-accelerated filer	☐	Smaller reporting company	☐
		Emerging growth company	☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐
Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☒
If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements.☐
Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant's executive officers during the relevant recovery period pursuant to § 240.10D-1(b).☐
Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☒

As of June 30, 2023, there were 602,885,744 common shares outstanding held by non-affiliates of the registrant, and the aggregate market value of the common shares (based upon the closing price of these shares on the New York Stock Exchange) was approximately $127.3 billion.

The number of shares of the registrant's common stock outstanding as of January 24, 2024 was 610,135,205.

DOCUMENTS INCORPORATED BY REFERENCE

Part III incorporates information by reference to the registrant's definitive proxy statement, to be filed with the Securities and Exchange Commission within 120 days after the close of the fiscal year ended December 31, 2023.

THE BOEING COMPANY

Index to the Form 10-K

For the Fiscal Year Ended December 31, 2023

			Page
PART I			
	Item 1.	Business	1
	Item 1A.	Risk Factors	6
	Item 1B.	Unresolved Staff Comments	17
	Item 1C.	Cybersecurity	17
	Item 2.	Properties	19
	Item 3.	Legal Proceedings	20
	Item 4.	Mine Safety Disclosures	20
PART II			
	Item 5.	Market for Registrant's Common Equity, Related Stockholder Matters and Issuer Purchases of Equity Securities	21
	Item 6.	[Reserved]	21
	Item 7.	Management's Discussion and Analysis of Financial Condition and Results of Operations	22
	Item 7A.	Quantitative and Qualitative Disclosures About Market Risk	49
	Item 8.	Financial Statements and Supplementary Data	50
	Item 9.	Changes in and Disagreements With Accountants on Accounting and Financial Disclosure	117
	Item 9A.	Controls and Procedures	117
	Item 9B.	Other Information	117
	Item 9C.	Disclosure Regarding Foreign Jurisdictions that Prevent Inspections	117
PART III			
	Item 10.	Directors, Executive Officers and Corporate Governance	118
	Item 11.	Executive Compensation	121
	Item 12.	Security Ownership of Certain Beneficial Owners and Management and Related Stockholder Matters	122
	Item 13.	Certain Relationships and Related Transactions, and Director Independence	122
	Item 14.	Principal Accountant Fees and Services	123
PART IV			
	Item 15.	Exhibits, Financial Statement Schedules	123
	Item 16.	Form 10-K Summary	126
	Signatures		127

PART I

Item 1. Business

The Boeing Company, together with its subsidiaries (herein referred to as “Boeing,” the “Company,” “we,” “us,” “our”), is one of the world’s major aerospace firms.

We are organized based on the products and services we offer. We operate in three reportable segments:

- Commercial Airplanes (BCA);
- Defense, Space & Security (BDS);
- Global Services (BGS).

Commercial Airplanes Segment

This segment develops, produces and markets commercial jet aircraft principally to the commercial airline industry worldwide. We are a leading producer of commercial aircraft and offer a family of commercial jetliners designed to meet a broad spectrum of global passenger and cargo requirements of airlines. This family of commercial jet aircraft in production includes the 737 narrow-body model and the 767, 777 and 787 wide-body models. Development continues on the 777X program and the 737-7 and 737-10 derivatives.

Defense, Space & Security Segment

This segment engages in the research, development, production and modification of manned and unmanned military aircraft and weapons systems for strike, surveillance and mobility, including fighter and trainer aircraft; vertical lift, including rotorcraft and tilt-rotor aircraft; and commercial derivative aircraft, including anti-submarine and tanker aircraft. In addition, this segment engages in the research, development, production and modification of the following products and related services: strategic defense and intelligence systems, including strategic missile and defense systems, command, control, communications, computers, intelligence, surveillance and reconnaissance (C4ISR), cyber and information solutions, and intelligence systems, satellite systems, including government and commercial satellites and space exploration.

Global Services Segment

This segment provides services to our commercial and defense customers worldwide. Global Services sustains aerospace platforms and systems with a full spectrum of products and services, including supply chain and logistics management, engineering, maintenance and modifications, upgrades and conversions, spare parts, pilot and maintenance training systems and services, technical and maintenance documents, and data analytics and digital services.

Intellectual Property

We own numerous patents and have licenses for the use of patents owned by others, which relate to our products and their manufacture. In addition to owning a large portfolio of intellectual property, we also license intellectual property to and from third parties. For example, the U.S. government has licenses in our patents that are developed in performance of government contracts, and it may use or authorize others to use the inventions covered by such patents for government purposes. Unpatented research, development and engineering skills, as well as certain trademarks, trade secrets and other intellectual property rights, also make an important contribution to our business. While our intellectual

property rights in the aggregate are important to the operation of each of our businesses, we do not believe that our business would be materially affected by the expiration of any particular intellectual property right or termination of any particular intellectual property patent license agreement.

Human Capital

As of December 31, 2023 and 2022, Boeing's total workforce was approximately 171,000 and 156,000 with 14% and 13% located outside of the U.S.

As of December 31, 2023, our workforce included approximately 57,000 union members. Our principal collective bargaining agreements and their current status are summarized in the following table:

Union	Percent of our Employees Represented	Status of Major Agreements with Union
The International Association of Machinists and Aerospace Workers (IAM)	21%	We have two major agreements; one with IAM District 751 (Washington) expiring in September 2024 and one with IAM District 837 (Missouri) expiring in July 2025.
The Society of Professional Engineering Employees in Aerospace (SPEEA)	10%	We have two major agreements; one with SPEEA Professional and one with SPEEA Technical, both expiring in October 2026.
The United Automobile, Aerospace and Agricultural Implement Workers of America (UAW)	1%	We have one major agreement with UAW District 1069 (Pennsylvania) expiring in April 2027.

Guided by our values, we are committed to creating a company where everyone is included and respected, and where we support each other in reaching our full potential. We are committed to diverse representation across all levels of our workforce to reflect the vibrant and thriving diversity of the communities in which we live and work. In May 2023, we released our third Global Equity, Diversity and Inclusion report with our workforce composition. As of December 2022, our global workforce was comprised of approximately 24% women, and our U.S. workforce was comprised of 35% racial and ethnic minorities and 15% U.S. veterans. We also support Business Resource Groups open to all employees with more than 15,000 participants across 176 chapters globally that focus on gender, race and ethnicity, generations, gender identity, sexual orientation, disability or veteran status. These groups help foster inclusion among all teammates, build awareness, recruit and retain a diverse workforce and support the company in successfully operating in a global, multicultural business environment.

To attract and retain the best-qualified talent, we offer competitive benefits, including market-competitive compensation, healthcare, paid time off, parental leave, retirement benefits, tuition assistance, employee skills development, leadership development and rotation programs. In 2023, our voluntary resignation rate was approximately 3%. Additionally, we hired approximately 23,000 new employees in 2023 for critical skills and had an offer acceptance rate of 82%.

Employees are encouraged to provide feedback about their experience through ongoing employee engagement activities. Boeing actively listens to its employees via surveys ranging from pre-hire to exiting the company. These voluntary surveys provide aggregate trend reports for the company to address in real time and ensure Boeing maintains an employee-focused experience and culture. We also invest in rewarding performance and have established a multi-level recognition program for the purpose of acknowledging the achievements of excellent individual or team performance.

We are committed to supporting our employees' continuous development of professional, technical and leadership skills through access to digital learning resources and through partnerships with leading professional/technical societies and organizations around the world. For 2023, Boeing employees completed approximately 6.9 million hours of learning. We offer the ability for our people to pursue degree programs, professional certificates and individual courses in strategic fields of study from approximately 500 accredited colleges and universities, online and across the globe through our tuition assistance program. Approximately 13,000 Boeing employees leveraged these programs in 2023.

Safety, quality, integrity and sustainability are at the core of how Boeing operates. We aspire to achieve zero workplace injuries and provide a safe, open and accountable work environment for our employees. Employees are also required on an annual basis to sign the Boeing Code of Conduct to reaffirm their commitment to do their work in a compliant and ethical manner. We provide several channels for all employees to speak up, ask for guidance and report concerns related to ethics or safety violations. We address employee concerns and take appropriate actions that uphold our Boeing values.

Competition

The commercial jet aircraft market and the airline industry remain extremely competitive. We face aggressive international competitors who are intent on increasing their market share, such as Airbus and entrants from China. We are focused on improving our products and processes and continuing cost reduction efforts. We intend to continue to compete with other aircraft manufacturers by providing customers with airplanes and services that deliver superior design, safety, quality, efficiency and value to customers around the world.

BDS faces strong competition primarily from Lockheed Martin Corporation, Northrop Grumman Corporation, Raytheon Technologies Corporation, General Dynamics Corporation and SpaceX. Non-U.S. companies such as BAE Systems and Airbus Group continue to build a strategic presence in the U.S. market by strengthening their North American operations and partnering with U.S. defense companies. In addition, certain competitors have occasionally formed teams with other competitors to address specific customer requirements. BDS expects the trend of strong competition to continue into 2024.

The commercial and defense services markets are extremely challenging and are made up of many of the same strong U.S. and non-U.S. competitors facing BCA and BDS along with other competitors in those markets. BGS leverages our extensive services network offering products and services which span the life cycle of our defense and commercial aircraft programs: training, fleet services and logistics, maintenance and engineering, modifications and upgrades, as well as the daily cycle of gate-to-gate operations. BGS expects the market to remain highly competitive in 2024, and intends to grow market share by leveraging a high level of customer satisfaction and productivity.

Regulatory Matters

Our businesses are heavily regulated in most of our markets. We work with numerous U.S. government agencies and entities, including but not limited to, all of the branches of the U.S. military, the National Aeronautics and Space Administration (NASA), the Federal Aviation Administration (FAA) and the Department of Homeland Security. Similar government authorities exist in our non-U.S. markets.

Government Contracts. The U.S. government, and other governments, may terminate any of our government contracts at their convenience, as well as for default based on our failure to meet specified performance requirements. If any of our U.S. government contracts were to be terminated for convenience, we generally would be entitled to receive payment for work completed and allowable termination or cancellation costs. If any of our government contracts were to be terminated for default, generally the U.S. government would pay only for the work that has been accepted and could require us

to pay the difference between the original contract price and the cost to re-procure the contract items, net of the work accepted from the original contract. The U.S. government can also hold us liable for damages resulting from the default.

Commercial Aircraft. In the U.S., our commercial aircraft products are required to comply with FAA regulations governing production and quality systems, airworthiness and installation approvals, repair procedures and continuing operational safety. On January 10, 2024, the FAA notified us that it has initiated an investigation into our quality control system. This was followed by the FAA announcing actions to increase its oversight of us, including conducting (1) an audit involving the 737-9 production line and suppliers to evaluate compliance with approved quality procedures, (2) increased monitoring of 737-9 in-service events, and (3) an assessment of safety risks around delegated authority and quality oversight, and examination of options to move these functions under independent third parties. On January 24, 2024, the FAA stated that it will not approve production rate increases or additional production lines for the 737 MAX until it is satisfied that we are in full compliance with required quality control procedures. New aircraft models and new derivative aircraft are required to obtain FAA certification prior to entry into service. Outside the U.S., similar requirements exist for airworthiness, installation and operational approvals. These requirements are generally administered by the national aviation authorities of each country and, in the case of Europe, coordinated by the European Union Aviation Safety Agency.

Environmental. We are subject to various federal, state, local and non-U.S. laws and regulations relating to environmental protection, including the discharge, treatment, storage, disposal and remediation of hazardous substances and wastes. We could also be affected by laws and regulations relating to climate change, including laws limiting or otherwise related to greenhouse gas emissions. These laws and regulations could lead to increased environmental compliance expenditures, increased energy and raw materials costs and new and/or additional investment in designs and technologies. We continually assess our compliance status and management of environmental matters to ensure our operations are in compliance with all applicable environmental laws and regulations. Investigation, remediation and operation and maintenance costs associated with environmental compliance and management of sites are a normal, recurring part of our operations. These costs often are allowable costs under our contracts with the U.S. government. It is reasonably possible that costs incurred to ensure continued environmental compliance could have a material impact on our results of operations, financial condition or cash flows if additional work requirements or more stringent clean-up standards are imposed by regulators, new areas of soil, air and groundwater contamination are discovered and/or expansions of work scope are prompted by the results of investigations. For additional information relating to environmental contingencies, see Note 13 to our Consolidated Financial Statements.

Non-U.S. Sales. Our non-U.S. sales are subject to both U.S. and non-U.S. governmental regulations and procurement policies and practices, including regulations relating to import-export control, tariffs, investment, exchange controls, anti-corruption and repatriation of earnings. Non-U.S. sales are also subject to varying currency, political and economic risks.

Raw Materials, Parts and Subassemblies

We are highly dependent on the availability of essential materials, parts and subassemblies from our suppliers and subcontractors. The most important raw materials required for our aerospace products are aluminum (sheet, plate, forgings and extrusions), titanium (sheet, plate, forgings and extrusions) and composites (including carbon and boron). Although alternative sources generally exist for these raw materials, qualification of the sources could take a year or more. As a result of the Russia Ukraine war, we ceased purchasing titanium from Russia. This has not disrupted our operations as we have been able to use inventory on hand and identify alternative sources. Many major components and product equipment items are procured or subcontracted on a sole-source basis. We continue to work with a small number of sole-source suppliers to ensure continuity of supply for certain items.

Suppliers

We are dependent upon the ability of a large number of U.S. and non-U.S. suppliers and subcontractors to meet performance specifications, quality standards and delivery schedules at our anticipated costs. While we maintain an extensive qualification and performance surveillance system to control risk associated with such reliance on third parties, failure of suppliers or subcontractors to meet commitments has and could continue to adversely affect product quality, production schedules and program/contract profitability, thereby jeopardizing our ability to fulfill commitments to our customers. We are also dependent on the availability of energy sources, such as electricity, at affordable prices. The current conflict in Israel and the Gaza Strip has the potential to impact certain of our suppliers, and has impacted some operations for our airline and lessor customers. We are closely monitoring developments, supporting our employees and customers, and will take mitigating actions as appropriate.

Seasonality

No material portion of our business is considered to be seasonal.

Executive Officers of the Registrant

See “Item 10. Directors, Executive Officers and Corporate Governance” in Part III.

Other Information

Boeing was originally incorporated in the State of Washington in 1916 and reincorporated in Delaware in 1934. Our principal executive offices are located at 929 Long Bridge Drive, Arlington, Virginia 22202, and our telephone number is (703) 465-3500.

General information about us can be found at www.boeing.com. The information contained on or connected to our website is not incorporated by reference into this Annual Report on Form 10-K and should not be considered part of this or any other report filed with the Securities and Exchange Commission (SEC). Our Annual Report on Form 10-K, Quarterly Reports on Form 10-Q and Current Reports on Form 8-K, as well as any amendments to those reports, are available free of charge through our website as soon as reasonably practicable after we file them with, or furnish them to, the SEC. The SEC maintains a website at www.sec.gov that contains reports, proxy statements and other information regarding SEC registrants, including Boeing.

Forward-Looking Statements

This report, as well as our annual report to shareholders, quarterly reports and other filings we make with the SEC, press and earnings releases and other written and oral communications, contain “forward-looking statements” within the meaning of the Private Securities Litigation Reform Act of 1995. Words such as “may,” “should,” “expects,” “intends,” “projects,” “plans,” “believes,” “estimates,” “targets,” “anticipates” and similar expressions generally identify these forward-looking statements. Examples of forward-looking statements include statements relating to our future financial condition and operating results, as well as any other statement that does not directly relate to any historical or current fact.

Forward-looking statements are based on expectations and assumptions that we believe to be reasonable when made, but that may not prove to be accurate. These statements are not guarantees and are subject to risks, uncertainties and changes in circumstances that are difficult to predict. Many factors, including those set forth in the “Risk Factors” section below and other important factors disclosed in this report and from time to time in our other filings with the SEC, could cause actual results to differ materially and adversely from these forward-looking statements. Any forward-looking statement

speaks only as of the date on which it is made, and we assume no obligation to update or revise any forward-looking statement whether as a result of new information, future events or otherwise, except as required by law.

Item 1A. Risk Factors

An investment in our common stock or debt securities involves risks and uncertainties, and our actual results and future trends may differ materially from our past or projected future performance. We urge investors to consider carefully the risk factors described below in evaluating the information contained in this report.

Risks Related to Our Business and Operations

We depend heavily on commercial airlines, subjecting us to unique risks.

Market conditions have a significant impact on demand for our commercial aircraft and related services. The commercial aircraft market is predominantly driven by long-term trends in airline passenger and cargo traffic. The principal factors underlying long-term traffic growth are sustained economic growth and political stability both in developed and emerging markets. Demand for our commercial aircraft is further influenced by airline profitability, availability of aircraft financing, world trade policies, government-to-government relations, technological advances, price and other competitive factors, fuel prices, terrorism, pandemics, epidemics and environmental regulations. Historically, the airline industry has been cyclical and very competitive and has experienced significant profit swings and constant challenges to be more cost competitive. Significant deterioration in the global economic environment, the airline industry generally or the financial stability of one or more of our major customers could result in fewer new orders for aircraft or services, or could cause customers to seek to postpone or cancel contractual orders and/or payments to us, which could result in lower revenues, profitability and cash flows and a reduction in our contractual backlog. In addition, because our commercial aircraft backlog consists of aircraft scheduled for delivery over a period of several years, any of these macroeconomic, industry or customer impacts could unexpectedly affect deliveries over a long period.

We enter into firm fixed-price aircraft sales contracts with indexed price escalation clauses, which could subject us to losses if we have cost overruns or if increases in our costs exceed the applicable escalation rate. Commercial aircraft sales contracts are often entered into years before the aircraft are delivered. In order to help account for economic fluctuations between the contract date and delivery date, aircraft pricing generally consists of a fixed amount as modified by price escalation formulas derived from labor, commodity and other price indices. Our revenue estimates are based on current expectations with respect to these escalation formulas, but the actual escalation amounts are outside of our control. Escalation factors can fluctuate significantly from period to period. Changes in escalation amounts can significantly impact revenues and operating margins in our BCA business.

We derive a significant portion of our revenues from a limited number of commercial airlines. We can make no assurance that any customer will exercise purchase options, fulfill existing purchase commitments or purchase additional products or services from us. In addition, fleet decisions, airline consolidations or financial challenges involving any of our major commercial airline customers could significantly reduce our revenues and limit our opportunity to generate profits from those customers. Airlines also are experiencing increased fuel and other costs, and the global economy has experienced high inflation.

Our Commercial Airplanes business depends on our ability to maintain a healthy production system, ensure every airplane in our production system conforms to exacting specifications,

achieve planned production rate targets, successfully develop and certify new aircraft or new derivative aircraft, and meet or exceed stringent performance and reliability standards.

The commercial aircraft business is extremely complex, involving extensive coordination and integration with U.S. and non-U.S. suppliers, highly-skilled labor performed by thousands of employees of ours and other partners, and stringent and evolving regulatory requirements and performance and reliability standards. We have experienced and may continue to experience production quality issues, including in our supply chain.

On January 10, 2024, the FAA notified us that it has initiated an investigation into our quality control system. This was followed by the FAA announcing actions to increase its oversight of us, including conducting (1) an audit involving the 737-9 production line and suppliers to evaluate compliance with approved quality procedures, (2) increased monitoring of 737-9 in-service events, and (3) an assessment of safety risks around delegated authority and quality oversight, and examination of options to move these functions under independent third parties. On January 24, 2024, the FAA stated that it will not approve production rate increases or additional production lines for the 737 MAX until it is satisfied that we are in full compliance with required quality control procedures. We are currently unable to reasonably estimate what impact the January 5, 2024 Alaska Airlines accident and the related FAA actions will have on our financial position, results of operations and cash flows.

The introduction of new aircraft programs and/or derivatives, such as the 777X, 737-7 and 737-10, involves risks associated with meeting development, testing, certification and production schedules. We are following the lead of the FAA as we work through the certification process, and the FAA will ultimately determine the timing of certification and entry into service. In addition, the development schedules of the 737-7 and 737-10 could be impacted by actions resulting from the Alaska Airlines accident. If we experience delays in achieving certification and/or incorporating safety enhancements, our financial position, results of operations and cash flows would be adversely impacted.

A number of our customers have contractual remedies, including compensation for late deliveries or rights to reject individual airplane deliveries based on delivery delays. Delays on the 737, 777X and 787 programs have resulted in, and may continue to result in, customers having the right to terminate orders, be compensated for late deliveries and/or substitute orders for other Boeing aircraft.

We must minimize disruption caused by production changes, achieve operational stability and implement productivity improvements in order to meet customer demand and maintain our profitability. We have previously announced plans to adjust production rates on several of our commercial aircraft programs. In addition, we continue to seek opportunities to reduce the costs of building our aircraft, including working with our suppliers to reduce supplier costs, identifying and implementing productivity improvements and optimizing how we manage inventory. If production rate changes at any of our commercial aircraft assembly facilities are delayed or create significant disruption to our production system, or if our suppliers cannot timely deliver components that comply with design specifications to us at the cost and rates necessary to achieve our targets, we may be unable to meet delivery schedules and/or the financial performance of one or more of our programs may suffer.

Operational challenges impacting the production system for one or more of our commercial aircraft programs could result in additional production delays and/or failure to meet customer demand for new aircraft, either of which would negatively impact our revenues and operating margins. Our commercial aircraft production system is extremely complex. Operational issues, including delays or defects in supplier components, failure to meet internal performance plans, or delays or failures to achieve required regulatory approval, could result in additional out-of-sequence work and increased production costs, as well as delayed deliveries to customers, impacts to aircraft performance and/or increased warranty or fleet support costs. We and our suppliers are experiencing supply chain disruptions and constraints, labor instability and inflationary pressures. We continue to monitor the health and stability of

the supply chain. These factors have and may continue to reduce overall productivity and adversely impact our financial position, results of operations and cash flows.

If our commercial aircraft fail to satisfy performance and reliability requirements and/or potentially required sustainability standards, we could face additional costs and/or lower revenues. Developing and manufacturing commercial aircraft that meet or exceed our performance and reliability standards and/or potentially required sustainability standards, as well as those of customers and regulatory agencies, can be costly and technologically challenging. These challenges are particularly significant with newer aircraft programs. Any failure of any Boeing aircraft to satisfy performance or reliability requirements could result in disruption to our operations, higher costs and/or lower revenues.

Changes in levels of U.S. government defense spending or acquisition priorities, as well as significant delays in U.S. government appropriations, could negatively impact our business, financial position and results of operations.

We derive a substantial portion of our revenue from the U.S. government, primarily from defense related programs with the United States Department of Defense (U.S. DoD). Levels of U.S. defense spending are very difficult to predict and may be impacted by numerous factors such as the evolving nature of the national security threat environment, U.S. national security strategy, U.S. foreign policy, the domestic political environment, macroeconomic conditions and the ability of the U.S. government to enact relevant legislation such as authorization and appropriations bills. The government may also constrain discretionary spending by instituting enforceable spending caps.

The timeliness of annual appropriations for U.S. government departments and agencies remains a recurrent risk. Congress may fund government departments and agencies with one or more continuing resolutions, which could delay new programs or competitions and/or negatively impact the execution of certain program activities. A lapse in appropriations for government departments or agencies would result in a full or partial government shutdown, which could impact our operations. In the event of a prolonged shutdown, requirements to furlough employees in the U.S. DoD, the Department of Transportation, including the FAA, or other government agencies could result in payment delays, impair our ability to deliver commercial airplanes or perform work on existing contracts, delays in the certification of new aircraft or otherwise impact our operations, negatively impact future orders, and/or cause other disruptions or delays. There is uncertainty regarding which government functions would shut down or continue operations during a lapse in appropriations, and corresponding uncertainty regarding the extent or magnitude of potential impacts to our operations. For additional information on U.S. government appropriations and budgets, see “Management’s Discussion & Analysis - Additional Considerations - U.S. Government Funding” on page 28 of this Form 10-K.

In addition, there continues to be uncertainty with respect to future acquisition priorities and program-level appropriations for the U.S. DoD and other government agencies (including NASA), including changes to national security and defense priorities, and tension between modernization investments, sustainment investments, and investments in new technologies or emergent capabilities. Future investment priority changes or budget cuts, including changes associated with the authorizations and appropriations process, could result in reductions, cancellations, and/or delays of existing contracts or programs or future program opportunities. Any of these impacts could have a material effect on our financial position, results of operations and/or cash flows.

As a result of the significant ongoing uncertainty with respect to both U.S. defense spending and the evolving nature of the national security threat environment, we also expect the U.S. DoD to continue to emphasize affordability, innovation, cybersecurity and delivery of technical data and software in its procurement processes, including the implementation of cybersecurity compliance requirements on the Defense Industrial Base, for which the supply chain may not be fully prepared. If we and our suppliers

are unable to adjust to these changing acquisition priorities and policies, our revenues and market share could be impacted.

Our ability to deliver products and services that satisfy customer requirements is heavily dependent on the performance and financial stability of our subcontractors and suppliers, as well as on the availability of highly skilled labor, raw materials and other components.

We rely on other companies, including U.S. and non-U.S. subcontractors and suppliers, to provide and produce raw materials, integrated components and sub-assemblies, and production commodities and to perform some of the services that we provide to our customers. Many of our suppliers are experiencing inflationary pressures, as well as resource constraints and disruptions due to production quality issues, global supply chain constraints, and labor instability. If one or more of our suppliers or subcontractors continue to experience financial difficulties, delivery delays or other performance problems, we may be unable to meet commitments to our customers and our financial position, results of operations and cash flows may continue to be adversely impacted. In addition, if one or more of the raw materials on which we depend (such as aluminum, titanium or composites) becomes unavailable to us or our suppliers, or is available only at very high prices, we may be unable to deliver one or more of our products in a timely fashion or at budgeted costs. We continue proactively working to ensure sufficient material and parts to avoid potential near-term production disruptions, while also working to mitigate the risk of future impacts from disruptions to our supply chain. In some instances, we depend upon a single source of supply. Any service disruption from one of these suppliers, either due to circumstances beyond the supplier's control, such as geopolitical developments, or as a result of performance problems or financial difficulties, could have a material adverse effect on our ability to meet commitments to our customers or increase our operating costs.

Some of our and our suppliers' workforces are represented by labor unions, which may lead to work stoppages.

Approximately 57,000 employees, which constitute 33% of our total workforce, were union represented as of December 31, 2023 under collective bargaining agreements with varying durations and expiration dates. For additional information on our principal collective bargaining agreements, see "Business – Human Capital" on page 2 of this Form 10-K. We experienced a work stoppage in 2008 when a labor strike halted commercial aircraft and certain BDS program production. We may experience additional work stoppages in the future, which could adversely affect our business. We cannot predict how stable our union relationships, currently with 10 U.S. labor organizations and 4 non-U.S. labor organizations, will be or whether we will be able to meet the unions' requirements without impacting our financial condition. The unions may also limit our flexibility in managing our workforce and operations. Union actions at suppliers can also affect us. Work stoppages and instability in our union relationships could delay the production and/or development of our products, which could strain relationships with customers and result in lower revenues.

Competition within our markets and with respect to our products and services may reduce our future contracts and sales.

The markets in which we operate are highly competitive and one or more of our competitors may have more extensive or more specialized engineering, manufacturing and marketing capabilities than we do in some areas. In our BCA business, we face aggressive international competition intent on increasing market share. In our BDS business, we anticipate that the effects of defense industry consolidation, shifting acquisition and budget priorities, and continued cost pressure at our U.S. DoD and non-U.S. customers will intensify competition for many of our BDS products. Our BGS segment faces competition from many of the same strong U.S. and non-U.S. competitors facing BCA and BDS. Furthermore, we are facing increased international competition and cross-border consolidation of competition, and U.S. procurement and compliance requirements that could limit our ability to be cost-competitive in the

international market. There can be no assurance that we will be able to compete successfully against our current or future competitors or that the competitive pressures we face will not result in reduced revenues and market share.

We derive a significant portion of our revenues from non-U.S. sales and are subject to the risks of doing business in other countries.

In 2023, non-U.S. customers, which include foreign military sales (FMS), accounted for approximately 42% of our revenues. We expect that non-U.S. sales will continue to account for a significant portion of our revenues for the foreseeable future. We are subject to risks of doing business internationally, including:

- changes in regulatory requirements or other executive branch actions, such as Executive Orders;
- changes in the global trade environment, including disputes with authorities in non-U.S. jurisdictions, including international trade authorities, that could impact sales and/or delivery of products and services outside the U.S. and/or impose costs on our customers in the form of tariffs, duties or penalties attributable to the importation of Boeing products and services;
- changes to U.S. and non-U.S. government policies, including sourcing restrictions, requirements to expend a portion of program funds locally and governmental industrial cooperation or participation requirements;
- fluctuations in international currency exchange rates;
- volatility in international political and economic environments and changes in non-U.S. national priorities and budgets, which can lead to delays or fluctuations in orders;
- the complexity and necessity of using non-U.S. representatives and consultants;
- the uncertainty of the ability of non-U.S. customers to finance purchases, including the availability of financing from the Export-Import Bank of the United States;
- uncertainties and restrictions concerning the availability of funding credit or guarantees;
- imposition of domestic and international taxes, export controls, tariffs, embargoes, sanctions (such as those imposed on Russia) and other trade restrictions;
- the difficulty of management and operation of an enterprise spread over many countries;
- compliance with a variety of non-U.S. laws, as well as U.S. laws affecting the activities of U.S. companies abroad; and
- unforeseen developments and conditions, including terrorism, war, epidemics and international tensions and conflicts.

While the impact of these factors is difficult to predict, any one or more of these factors could adversely affect our operations in the future. For example, since 2018, the U.S. and China have imposed tariffs on each other’s imports. Certain aircraft parts and components that Boeing procures are subject to these tariffs. We are mitigating import costs through Duty Drawback Customs procedures. Overall, the U.S.-China trade relationship remains stalled as economic and national security concerns continue to be a challenge. China is a significant market for commercial aircraft and we have long-standing relationships with our Chinese customers, who represent a key component of our commercial aircraft backlog. If we are unable to deliver aircraft to customers in China consistent with our assumptions and/or obtain additional orders from China in the future, we may experience reduced deliveries and/or lower market share. Impacts from future potential deterioration in geopolitical or trade relations between the U.S. and one or more other countries could have a material adverse impact on our financial position, results of operations and/or cash flows.

We use estimates and make assumptions in accounting for contracts and programs. Changes in our estimates and/or assumptions could adversely affect our future financial results.

Contract and program accounting require judgment relative to assessing risks, estimating revenues and costs and making assumptions for schedule and technical issues. Due to the size and nature of many of our contracts and programs, the estimation of total revenues and cost at completion is complicated and subject to many variables. Assumptions have to be made regarding the length of time to complete the contract or program because costs also include expected increases in wages and employee benefits, material prices and allocated fixed costs. Incentives or penalties related to performance on contracts are considered in estimating sales and profit rates and are recorded when there is sufficient information for us to assess anticipated performance. Customer and supplier claims and assertions are also assessed and considered in estimating revenues, costs and profit rates. Estimates of future award fees are also included in revenues and profit rates.

With respect to each of our commercial aircraft programs, inventoriable production costs (including overhead), program tooling and other non-recurring costs and routine warranty costs are accumulated and charged as cost of sales by program instead of by individual units or contracts. A program consists of the estimated number of units (accounting quantity) of a product to be produced in a continuing, long-term production effort for delivery under existing and anticipated contracts limited by the ability to make reasonably dependable estimates. To establish the relationship of sales to cost of sales, program accounting requires estimates of (a) the number of units to be produced and sold in a program, (b) the period over which the units can reasonably be expected to be produced and (c) the units' expected sales prices, production costs, program tooling and other non-recurring costs, and routine warranty costs for the total program. Several factors determine accounting quantity, including firm orders, letters of intent from prospective customers and market studies. Changes to customer or model mix, production costs and rates, learning curve, changes to price escalation indices, costs of derivative aircraft, supplier performance, customer and supplier negotiations/settlements, supplier claims and/or certification issues can impact these estimates. In addition, on development programs such as the 777X, 737-7 and 737-10 we are subject to risks with respect to the timing and conditions of aircraft certification, including potential gaps between when aircraft are certified in various jurisdictions, changes in certification processes and our estimates with respect to the timing of future certifications, which could have an impact on overall program status. Any such change in estimates relating to program accounting may adversely affect future financial performance.

Because of the significance of the judgments and estimation processes described above, materially different revenues and profit amounts could be recorded if we used different assumptions, revised our estimates, or if the underlying circumstances were to change. Changes in underlying assumptions, circumstances or estimates may adversely affect future period financial performance. For additional information on our accounting policies for recognizing sales and profits, see our discussion under "Management's Discussion and Analysis – Critical Accounting Estimates – Accounting for Long-term Contracts/Program Accounting" on pages 46 - 47 and Note 1 to our Consolidated Financial Statements on pages 57 - 67 of this Form 10-K.

We may not realize the anticipated benefits of mergers, acquisitions, joint ventures/strategic alliances or divestitures.

As part of our business strategy, we may merge with or acquire businesses and/or form joint ventures and strategic alliances. Whether we realize the anticipated benefits from these acquisitions and related activities depends, in part, upon our ability to integrate the operations of the acquired business, the performance of the underlying product and service portfolio, and the performance of the management team and other personnel of the acquired operations. Accordingly, our financial results could be adversely affected by unanticipated performance issues, legacy liabilities, transaction-related charges, amortization of expenses related to intangibles, charges for impairment of long-term assets, credit

guarantees, partner performance and indemnifications. Consolidations of joint ventures could also impact our reported results of operations or financial position. While we believe that we have established appropriate and adequate procedures and processes to mitigate these risks, there is no assurance that these transactions will be successful. We also may make strategic divestitures from time to time. These transactions may result in continued financial involvement in the divested businesses, such as through guarantees or other financial arrangements, following the transaction. Nonperformance by those divested businesses could affect our future financial results through additional payment obligations, higher costs or asset write-downs.

Risks Related to Our Contracts

We conduct a significant portion of our business pursuant to U.S. government contracts, which are subject to unique risks.

In 2023, 37% of our revenues were earned pursuant to U.S. government contracts, which include Foreign Military Sales (FMS) through the U.S. government. Business conducted pursuant to such contracts is subject to extensive procurement regulations and other unique risks.

Our sales to the U.S. government are subject to extensive procurement regulations, and changes to those regulations could increase our costs. New procurement regulations or climate or cyber-related contractual disclosures, or changes to existing requirements, could increase our compliance costs or otherwise have a material impact on the operating margins of our BDS and BGS businesses. These requirements may also result in withheld payments and/or reduced future business if we fail to comply. For example, proposals to raise domestic content thresholds for our U.S. government contracts could have negative impacts on our business. Compliance costs attributable to current and potential future procurement regulations such as these could negatively impact our financial position, results of operations and/or cash flows.

The U.S. government may modify, curtail or terminate one or more of our contracts. The U.S. government contracting party may modify, curtail or terminate its contracts and subcontracts with us, without prior notice and either at its convenience or for default based on performance. In addition, funding pursuant to our U.S. government contracts may be reduced or withheld as part of the U.S. Congressional appropriations process due to changes in U.S. national security strategy and/or priorities, fiscal constraints, including enforceable spending caps, a sequester or a lack of funding available to pay incurred obligations, or for other reasons. Further uncertainty with respect to ongoing programs could also result in the event that the U.S. government finances its operations through temporary funding measures such as “Continuing Resolutions” rather than full-year appropriations. Any loss or anticipated loss or reduction of expected funding and/or modification, curtailment or termination of one or more large programs could have a material adverse effect on our financial position, results of operations and/or cash flows.

We are subject to U.S. government inquiries and investigations, including periodic audits of costs that we determine are reimbursable under U.S. government contracts. U.S. government agencies, including the Defense Contract Audit Agency and the Defense Contract Management Agency, routinely audit government contractors. These agencies review our performance under contracts, cost structure and compliance with applicable laws, regulations and standards, as well as the adequacy of and our compliance with our internal control systems and policies. Any costs found to be misclassified or inaccurately allocated to a specific contract will be deemed non-reimbursable, and to the extent already reimbursed, must be refunded. Any inadequacies in our systems and policies could result in withholds on billed receivables, penalties and reduced future business. Furthermore, if any audit, inquiry or investigation uncovers improper or illegal activities, we could be subject to civil and criminal penalties and administrative sanctions, including termination of contracts, forfeiture of profits, suspension of payments, fines and suspension or debarment from doing business with the U.S. government. We also

could suffer reputational harm if allegations of impropriety were made against us, even if such allegations are later determined to be false.

We enter into fixed-price contracts, which could subject us to losses if we have cost overruns.

Our BDS and BGS defense businesses generated approximately 58% and 65% of their 2023 revenues from fixed-price contracts. While fixed-price contracts enable us to benefit from performance improvements, cost reductions and efficiencies, they also subject us to the risk of reduced margins or incurring losses if we are unable to achieve estimated costs and revenues. If our estimated costs exceed our estimated price, we recognize reach-forward losses which can significantly affect our reported results. For example, during the year ended December 31, 2023, BDS recorded $1,585 million of additional losses on its five most significant fixed-price development programs (Commercial Crew, KC-46A Tanker, MQ-25, T-7A Red Hawk, and VC-25B Presidential Aircraft). We continue to experience production disruptions and inefficiencies due to technical challenges, supplier disruption and factory performance. These factors have contributed to significant earnings charges on a number of fixed-price development programs which are expected to adversely affect cash flows in future periods, and may result in future earnings charges and adverse cash flow effects. Production and supplier disruptions, inefficiencies, technical challenges, quality issues and labor instability also contributed to lower earnings on fixed-price production programs in 2023. New programs could also have risk for reach-forward loss upon contract award and during the period of contract performance. The long-term nature of many of our contracts makes the process of estimating costs and revenues on fixed-price contracts inherently risky. Fixed-price contracts often contain price incentives and penalties tied to performance, which can be difficult to estimate and have significant impacts on margins. In addition, some of our contracts have specific provisions relating to cost, schedule and performance.

Estimating costs to complete fixed-price development contracts is generally subject to more uncertainty than fixed-price production contracts. Many of these development programs have highly complex designs and technical challenges. In addition, technical or quality issues could lead to schedule delays and cost impacts, which could increase our estimated cost to perform the work or reduce our estimated price, either of which could result in a material charge or otherwise adversely affect our financial condition.

We enter into cost-type contracts, which also carry risks.

Our BDS and BGS defense businesses generated approximately 42% and 35% of their 2023 revenues from cost-type contracting arrangements. Some of these are development programs that have complex design and technical challenges. These cost-type programs typically have award or incentive fees that are subject to uncertainty and may be earned over extended periods. In these cases the associated financial risks are primarily reduced award or incentive fees, lower profit rates or program cancellation if cost, schedule or technical performance issues arise. Examples of programs with cost-type contracts include Ground-based Midcourse Defense, Proprietary and Space Launch System programs.

We enter into contracts that include in-orbit incentive payments that subject us to risks.

Contracts in the commercial satellite industry and certain government satellite contracts include in-orbit incentive payments. These in-orbit payments may be paid over time after final satellite acceptance or paid in full prior to final satellite acceptance. In both cases, the in-orbit incentive payment is at risk if the satellite does not perform to specifications for up to 15 years after acceptance. The net present value of in-orbit incentive fees we ultimately expect to realize is recognized as revenue in the construction period. If the satellite fails to meet contractual performance criteria, customers will not be obligated to continue making in-orbit payments and/or we may be required to provide refunds to the customer and incur significant charges.

Risks Related to Cybersecurity and Business Disruptions

Unauthorized access to our, our customers' and/or our suppliers' information and systems could negatively impact our business.

We rely extensively on information technology systems and networks to operate our company and meet our business objectives. We face various cyber security threats, including attempts to gain unauthorized access to our systems and networks, denial-of-service attacks, threats to our information technology infrastructure, ransomware and phishing attacks, and attempts to gain unauthorized access to our company-, customer- and employee-sensitive information. These threats come from a variety of actors some of which are highly organized and sophisticated such as nation-state actors and criminal enterprises. In addition, the techniques used in cyberattacks evolve rapidly, including from emerging technologies, such as advanced forms of automation and artificial intelligence. As cyber threats increase in volume and sophistication, the risk to the security of these systems and networks – and to the confidentiality, integrity, and availability of the data they house – continues to evolve, requiring constant vigilance and concerted, company-wide risk management efforts.

A cyber-related attack or security breach, whether experienced directly or through our supply chain or third party-service providers, could, among other serious consequences, result in loss of intellectual property; allow unauthorized access to or cause the publication of various categories of sensitive, proprietary or customer data; cause disruption or degradation of our business operations; compromise our products or services; and/or result in reputational harm. To address these risks, we maintain an extensive network of technical security controls, policy enforcement mechanisms, monitoring systems, contractual arrangements, tools and related services, and management and Board oversight. While these measures are designed to prevent, detect, respond to, and mitigate unauthorized activity, there is no guarantee that they will be sufficient to prevent or mitigate the risk of a cyber-related attack or incident, or allow us to detect, report or respond adequately in a timely manner.

We have experienced, and may in the future experience, whether directly or through our supply chain, third-party service providers or other channels, cybersecurity incidents. While prior cyber-related attacks and incidents (including those at our wholly-owned subsidiaries Boeing Distribution, Inc. in 2023 and Jeppesen Inc. in 2022) have not materially affected our business strategy, results of operations or financial condition, there is no guarantee that a future cyber-related attack or incident would not result in significant operational, regulatory, or financial impacts that could materially affect our business strategy, results of operations or financial condition.

In addition, we manage information and information technology systems for certain customers and suppliers. Many of these customers and suppliers face similar security threats. If we were unable to protect against the unauthorized access, release or corruption of our customers' or suppliers' confidential, classified or personally identifiable information, we could suffer a loss of business, face regulatory actions or face financial or other losses that could materially affect our business strategy, results of operations or financial condition.

Business disruptions could seriously affect our future sales and financial condition or increase our costs and expenses.

Our business may be impacted by disruptions including threats to physical security or our information technology systems, extreme weather (including effects of climate change) or other acts of nature, and pandemics or other public health crises. Any of these disruptions could affect our internal operations or our suppliers' operations and delay delivery of products and services to our customers. Any significant production delays, or any destruction, manipulation or improper use of Boeing's or our suppliers' data, information systems or networks could impact our sales, increase our expenses and/or have an adverse effect on the reputation of Boeing and of our products and services.

Risks Related to Legal and Regulatory Matters

The outcome of litigation and of government inquiries and investigations involving our business is unpredictable, and an adverse decision in any such matter could have a material effect on our financial position and results of operations.

We are involved in a number of litigation matters. These matters may divert financial and management resources that would otherwise be used to benefit our operations. No assurances can be given that the results of these matters will be favorable to us. An adverse resolution of any of these lawsuits, or future lawsuits, could have a material impact on our financial position and results of operations. In addition, we are subject to extensive regulation under the laws of the United States and its various states, as well as other jurisdictions in which we operate and/or market our products. As a result, we are sometimes subject to government inquiries and investigations due, among other things, to our business relationships with the U.S. government, the heavily regulated nature of our industry, and in the case of environmental proceedings, our current or past ownership of certain property. Any such inquiry or investigation could result in an adverse ruling against us, which could have a material impact on our financial position, results of operations and/or cash flows.

Our operations expose us to the risk of material environmental liabilities.

We are subject to various U.S. federal, state, local and non-U.S. laws and regulations related to environmental protection, including the discharge, treatment, storage, disposal and remediation of pollutants, hazardous substances and wastes. We could incur substantial costs, including cleanup costs, fines and civil or criminal sanctions, as well as third-party claims for property damage or personal injury, if we were to violate or become liable under environmental laws or regulations. In some cases, we are subject to such costs due to environmental impacts attributable to our current or past manufacturing operations or the operations of companies we have acquired. In other cases, we are subject to such costs due to an indemnification agreement between us and a third party relating to such environmental liabilities. In all cases, our current liabilities and ongoing cost assessments are based on current laws and regulations. New laws and regulations, more stringent enforcement of existing laws and regulations, the discovery of previously unknown contamination or the imposition of new remediation requirements could result in additional costs. For additional information relating to environmental contingencies, see Note 13 to our Consolidated Financial Statements.

We may be adversely affected by global climate change or by legal, regulatory or market responses to such change.

Increasing stakeholder environmental, social and governance (ESG) expectations, physical and transition risks associated with climate change, emerging ESG regulation, contractual requirements, and policy requirements may pose risk to our market outlook, brand and reputation, financial outlook, cost of capital, global supply chain and production continuity, which may impact our ability to achieve long-term business objectives. Changes in environmental and climate change laws or regulations could lead to additional operational restrictions and compliance requirements upon us or our products, require new or additional investments in production systems or product designs, result in additional carbon offset investments or otherwise negatively impact our business and/or competitive position. Increasingly stringent aircraft performance standards and requirements including but not limited to manufacturing and product air pollutant emissions, potential carbon pricing mechanisms, and sustainability disclosure requirements in the U.S. and other jurisdictions may result in increased costs or reputational risks and could limit our ability to manufacture and/or market certain of our products at acceptable costs, or at all. For example, certain jurisdictions including the State of California and the European Union have enacted legislation which would require more stringent greenhouse gas emissions and climate risk reporting. Physical impacts of climate change, increasing global chemical restrictions and bans, and

water and waste requirements may drive increased costs to us and our suppliers and impact our production continuity and data facilities.

Finally, from time to time, in alignment with our sustainability priorities, we establish and publicly announce goals and commitments to improve our environmental performance, such as our operational goals in areas of GHG emissions, energy, water and waste. If we fail to achieve or inadequately report our progress toward achieving such goals and commitments, the resulting negative publicity could adversely affect our reputation and/or our access to capital.

Risks Related to Financing and Liquidity

We may be unable to obtain debt to fund our operations and contractual commitments at competitive rates, on commercially reasonable terms or in sufficient amounts.

We depend, in part, upon the issuance of debt to fund our operations and contractual commitments. As of December 31, 2023, our debt totaled $52.3 billion of which approximately $17.7 billion of principal payments on outstanding debt will become due over the next three years. In addition, as of December 31, 2023, our airplane financing commitments totaled $17.0 billion. If we require additional funding in order to pay off existing debt, address further impacts to our business related to market developments, fund outstanding financing commitments or meet other business requirements, our market liquidity may not be sufficient. These risks will be particularly acute if we are subject to further credit rating downgrades such as those we experienced in 2020. A number of factors could cause us to incur increased borrowing costs and to have greater difficulty accessing public and private markets for debt. These factors include disruptions or declines in the global capital markets and/or a decline in our financial performance, outlook or credit ratings and/or changes in demand for our products and services. The occurrence of any or all of these events may adversely affect our ability to fund our operations and contractual or financing commitments.

Substantial pension and other postretirement benefit obligations have a material impact on our earnings, shareholders' equity and cash flows from operations, and could have significant adverse impacts in future periods.

Many of our employees have earned benefits under defined benefit pension plans. Potential pension contributions include both mandatory amounts required under the Employee Retirement Income Security Act and discretionary contributions to improve the plans' funded status. The extent of future contributions depends heavily on market factors such as the discount rate and the actual return on plan assets. We estimate future contributions to these plans using assumptions with respect to these and other items. Changes to those assumptions could have a significant effect on future contributions as well as on our annual pension costs and/or result in a significant change to shareholders' equity. For U.S. government contracts, we allocate pension costs to individual contracts based on U.S. Cost Accounting Standards, which can also affect contract profitability. We also provide other postretirement benefits to certain of our employees, consisting principally of health care coverage for eligible retirees and qualifying dependents. Our estimates of future costs associated with these benefits are also subject to assumptions, including estimates of the level of medical cost increases. For a discussion regarding how our financial statements can be affected by pension and other postretirement plan accounting policies, see “Management's Discussion and Analysis – Critical Accounting Estimates – Pension Plans” on pages 47 - 48 of this Form 10-K. Although under Generally Accepted Accounting Principles in the United States of America (GAAP) the timing of periodic pension and other postretirement benefit expense and plan contributions are not directly related, the key economic factors that affect GAAP expense would also likely affect the amount of cash or stock we would contribute to our plans.

Our insurance coverage may be inadequate to cover all significant risk exposures.

We are exposed to liabilities that are unique to the products and services we provide. We maintain insurance for certain risks and, in some circumstances, we may receive indemnification from the U.S. government. The amount of our insurance coverage may not cover all claims or liabilities, and we may be forced to bear substantial costs. For example, liabilities arising from the use of certain of our products, such as aircraft technologies, space systems, spacecraft, satellites, missile systems, weapons, cybersecurity, border security systems, anti-terrorism technologies and/or air traffic management systems may not be insurable on commercially reasonable terms. While some of these products are shielded from liability within the U.S. under the SAFETY Act provisions of the 2002 Homeland Security Act, no such protection is available outside the U.S., potentially resulting in significant liabilities. The amount of insurance coverage we maintain may be inadequate to cover these or other claims or liabilities.

A significant portion of our customer financing portfolio is concentrated among certain customers and in certain types of Boeing aircraft, which exposes us to concentration risks.

A significant portion of our customer financing portfolio, which is comprised of financing receivables and operating lease equipment, is concentrated among certain customers and in distinct geographic regions. Our portfolio is also concentrated by varying degrees across Boeing aircraft product types, most notably 717 aircraft, and among customers that we believe have less than investment-grade credit. If one or more customers holding a significant portion of our portfolio assets experiences financial difficulties or otherwise defaults on or does not renew its leases with us at their expiration, and we are unable to redeploy the aircraft on reasonable terms, or if the types of aircraft that are concentrated in our portfolio suffer greater than expected declines in value, our financial position, results of operations and/or cash flows could be materially adversely affected.

Item 1B. Unresolved Staff Comments

Not applicable

Item 1C. Cybersecurity

Risk Management and Strategy

Our cybersecurity strategy prioritizes detection, analysis and response to known, anticipated or unexpected threats; effective management of security risks; and resiliency against incidents. Our cybersecurity risk management processes include technical security controls, policy enforcement mechanisms, monitoring systems, employee training, contractual arrangements, tools and related services from third-party providers, and management oversight to assess, identify and manage material risks from cybersecurity threats. We implement risk-based controls to protect our information, the information of our customers, suppliers, and other third parties, our information systems, our business operations, and our products and related services. We have adopted security-control principles based on the National Institute of Standards and Technology (NIST) Cybersecurity Framework, other industry-recognized standards, and contractual requirements, as applicable. We also leverage government partnerships, industry and government associations, third-party benchmarking, the results from regular internal and third-party audits, threat intelligence feeds, and other similar resources to inform our cybersecurity processes and allocate resources.

We maintain security programs that include physical, administrative and technical safeguards, and we maintain plans and procedures whose objective is to help us prevent and timely and effectively respond to cybersecurity threats or incidents. Through our cybersecurity risk management process, we continuously monitor cybersecurity vulnerabilities and potential attack vectors to company systems as well as our aerospace products and services, and we evaluate the potential operational and financial

effects of any threat and of cybersecurity countermeasures made to defend against such threats. We continue to integrate our cyber practice into our Enterprise Risk Management program and our Compliance Risk Management program, both of which are overseen by our Board of Directors and provide central, standardized frameworks for identifying and tracking cyber-related business and compliance risks across the Company. Risks from cybersecurity threats to our products and services are also overseen by our Board of Directors. In addition, we periodically engage third-party consultants to assist us in assessing, enhancing, implementing, and monitoring our cybersecurity risk management programs and responding to any incidents.

As part of our cybersecurity risk management process, we conduct “tabletop” exercises during which we simulate cybersecurity incidents to ensure that we are prepared to respond to such an incident and to highlight any areas for potential improvement in our cyber incident preparedness. These exercises are conducted at both the technical level and senior management level, which has included participation by a member of our Board of Directors. In addition, all employees are required to pass a mandatory cybersecurity training course on an annual basis and receive monthly phishing simulations to provide “experiential learning” on how to recognize phishing attempts.

We have established a cybersecurity supply chain risk management program, which is a cross-functional program that forms part of our Enterprise Risk Management program and is supported by our security, compliance, and supply chain organizations. Through this evolving program, we assess the risks from cybersecurity threats that impact select suppliers and third-party service providers with whom we share personal identifying and confidential information. We continue to evolve our oversight processes to mature how we identify and manage cybersecurity risks associated with the products or services we procure from such suppliers. We generally require our suppliers to adopt security-control principles based on industry-recognized standards.

We have experienced, and may in the future experience, whether directly or through our supply chain or other channels, cybersecurity incidents. While prior incidents have not materially affected our business strategy, results of operations or financial condition, and although our processes are designed to help prevent, detect, respond to, and mitigate the impact of such incidents, there is no guarantee that a future cyber incident would not materially affect our business strategy, results of operations or financial condition. See “Risks Related to Cybersecurity and Business Disruptions” in “Risk Factors” on page 14 of this Form 10-K.

Governance

Our Board of Directors has overall responsibility for risk oversight, with its committees assisting the Board in performing this function based on their respective areas of expertise. Our Board of Directors has delegated oversight of risks related to cybersecurity to two Board committees, the Audit Committee and the Aerospace Safety Committee, and each committee reports on its activities and findings to the full Board after each meeting. The Audit Committee is charged with reviewing our cybersecurity processes for assessing key strategic, operational, and compliance risks. Our Chief Information Officer and Senior Vice President, Information Technology & Data Analytics (CIO) and our Chief Security Officer (CSO) provide presentations to the Audit Committee on cybersecurity risks at each of its bimonthly meetings. These briefings include assessments of cyber risks, the threat landscape, updates on incidents, and reports on our investments in cybersecurity risk mitigation and governance. In addition, the Audit Committee has designated one of its members with expertise in cyber risk management to meet regularly with management and review our cybersecurity strategy and key initiatives and progress toward our objectives. In the event of a potentially material cybersecurity event, the Chair of the Audit Committee is notified and briefed, and meetings of the Audit Committee and/or full Board of Directors would be held, as appropriate. The Aerospace Safety Committee provides oversight of the risks from cybersecurity threats related to our aerospace products and services. The Aerospace Safety Committee receives regular updates and reports from senior management, including the Chief

Engineer, the Chief Aerospace Safety Officer, and the Chief Product Security Engineer, who provide briefings on significant cybersecurity threats or incidents that may pose a risk to the safe operation of our aerospace products. Both committees brief the full Board on cybersecurity matters discussed during committee meetings, and the CIO provides annual briefings to the Board on information technology and data analytics related matters, including cybersecurity.

At the management level, we have established a Global Security Governance Council (the Council) to further strengthen our cybersecurity risk management activities across the Company, including the prevention, detection, mitigation, and remediation of cybersecurity incidents. The Council is responsible for developing and coordinating enterprise cybersecurity policy and strategy, and for providing guidance to key management and oversight bodies.

Richard Puckett, as our CSO, serves as the chair of the Council. He is responsible for overseeing a unified security program that provides cybersecurity, fire and protection operations, physical security, insider threat, and classified security. Mr. Puckett has nearly 30 years of experience in the cybersecurity industry, including, prior to joining Boeing in 2022, as Chief Information Security Officer of SAP SE and Thomson Reuters Corporation, Vice President, Product and Commercial Security of General Electric, Inc., and Senior Security Architect at Cisco Systems, Inc. He reports directly to the CIO and meets regularly with other members of senior management and the Audit Committee.

The Council also includes, among other senior executives, our Chief Engineer, Chief Information Officer, Chief Aerospace Safety Officer and Chief Product Security Engineer, who each have several decades of business and senior leadership experience managing risks in their respective fields, collectively covering all aspects of cybersecurity, data and analytics, product security engineering, enterprise engineering, safety and the technical integrity of our products and services.

The Council meets monthly and updates key members of the Company's Executive Council on progress towards specific cybersecurity objectives. A strong partnership exists between Information Technology, Enterprise Security, Corporate Audit, and Legal so that identified issues are addressed in a timely manner and incidents are reported to the appropriate regulatory bodies as required.

Item 2. Properties

We had approximately 89 million square feet of floor space on December 31, 2023 for manufacturing, warehousing, engineering, administration and other productive uses, of which approximately 86% was located in the United States. The following table provides a summary of the floor space by business as of December 31, 2023:

(Square feet in thousands)	Owned	Leased	Government Owned	Total
Commercial Airplanes	39,919	7,795		47,714
Defense, Space & Security	22,849	4,404		27,253
Global Services	1,265	7,004		8,269
Other[1]	2,205	2,809	315	5,329
Total	66,238	22,012	315	88,565

[1] Other includes sites used for corporate offices, enterprise research and development and common internal services.

At December 31, 2023, the combined square footage at the following major locations totaled more than 82 million square feet:

- Commercial Airplanes – Greater Seattle, WA; China; Greater Charleston, SC; Greater Los Angeles, CA; Greater Portland, OR; Greater Salt Lake City, UT; Australia: Canada and Germany
- Defense, Space & Security – Greater St. Louis, MO; Greater Seattle, WA; Greater Los Angeles, CA; Philadelphia, PA; Mesa, AZ; Huntsville, AL; Oklahoma City, OK; Heath, OH; Greater Washington, DC; Australia; Houston, TX; Kennedy Space Center and Greater Portland, OR
- Global Services – San Antonio, TX; Greater Dallas, TX; Great Britain; Greater Miami, FL; China; Jacksonville, FL; and Germany
- Other – India; Chicago, IL; Greater Los Angeles, CA; Greater St. Louis, MO; and Greater Washington, DC.

Most runways and taxiways that we use are located on airport properties owned by others and are used jointly with others. Our rights to use such facilities are provided for under long-term leases with municipal, county or other government authorities. In addition, the U.S. government furnishes us certain office space, installations and equipment at U.S. government bases for use in connection with various contract activities.

Item 3. Legal Proceedings

Currently, we are involved in a number of legal proceedings. For a discussion of contingencies related to legal proceedings, see Note 21 to our Consolidated Financial Statements, which is hereby incorporated by reference.

Item 4. Mine Safety Disclosures

Not applicable

PART II

Item 5. Market for Registrant's Common Equity, Related Stockholder Matters and Issuer Purchases of Equity Securities

The principal market for our common stock is the New York Stock Exchange where it trades under the symbol BA. As of January 24, 2024, there were 84,633 shareholders of record.

Issuer Purchases of Equity Securities

The following table provides information about purchases we made during the quarter ended December 31, 2023 of equity securities that are registered by us pursuant to Section 12 of the Exchange Act:

(Dollars in millions, except per share data)

	(a)	(b)	(c)	(d)
	Total Number of Shares Purchased[1]	Average Price Paid per Share	Total Number of Shares Purchased as Part of Publicly Announced Plans or Programs	Approximate Dollar Value of Shares That May Yet be Purchased Under the Plans or Programs
10/1/2023 thru 10/31/2023	7,546	$190.17		
11/1/2023 thru 11/30/2023	12,373	192.12		
12/1/2023 thru 12/31/2023	1,428,309	246.44		
Total	1,448,228	$245.68		

(1) A total of 1,448,228 shares were transferred to us from employees in satisfaction of minimum tax withholding obligations associated with the vesting of restricted stock units during the period. We did not purchase any shares of our common stock in the open market pursuant to a repurchase program.

Item 6. [Reserved]

Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations

Consolidated Results of Operations and Financial Condition

Overview

We are a global market leader in the design, development, manufacture, sale, service and support of commercial jetliners, military aircraft, satellites, missile defense, human space flight and launch systems and services. We are one of the two major manufacturers of 100+ seat airplanes for the worldwide commercial airline industry and one of the largest defense contractors in the U.S. While our principal operations are in the U.S., we conduct operations in an expanding number of countries and rely on an extensive network of non-U.S. partners, key suppliers and subcontractors.

Our strategy is centered on successful execution in healthy core businesses – Commercial Airplanes (BCA), Defense, Space & Security (BDS) and Global Services (BGS). BCA is committed to being the leader in commercial aviation by offering airplanes and services that deliver superior design, safety, quality, efficiency and value to customers around the world. BDS integrates its resources in defense, intelligence, communications, security, space and services to deliver capability-driven solutions to customers at reduced costs. Our BDS strategy is to leverage our core businesses to capture key next-generation programs while expanding our presence in adjacent and international markets. BGS provides support for commercial and defense through innovative, comprehensive and cost-competitive product and service solutions.

Business Environment and Trends

In 2023, global air traffic largely recovered to 2019 levels with domestic travel continuing to be the most robust and the single-aisle market following closely. International travel has mostly recovered and the wide-body market continues to be paced by the international travel recovery. The transition in the international commercial market from recovery to normal market conditions is progressing slowly as China international travel remains below 2019 levels. We are experiencing strong demand from our airline customers globally.

We and our suppliers are experiencing supply chain disruptions as a result of production quality issues, global supply chain constraints, and labor instability. We and our suppliers are also experiencing inflationary pressures. We continue to monitor the health and stability of the supply chain. These factors have reduced overall productivity and adversely impacted our financial position, results of operations and cash flows.

Airline financial performance, which influences demand for new capacity, has benefited from the resilient demand for travel. The International Air Transport Association (IATA) is estimating 2023 industry-wide profit of $23.3 billion, up from its forecast of $4.6 billion a year ago, primarily driven by North America, Europe and the Middle East. For 2024, IATA is forecasting $25.7 billion in profits for the industry globally. The overall outlook continues to stabilize as we face uncertainties in the environment in the near- to medium-term as airlines are facing persistently high and volatile cost of fuel and tight labor conditions. The global economy is expecting an easing of inflation and interest rates, with regional economic and geopolitical difficulties adding uncertainty to the outlook and the financial viability of some airlines and regions.

The long-term outlook for the industry remains positive due to the fundamental drivers of air travel demand: economic growth, increasing propensity to travel due to increased trade, globalization and improved airline services driven by liberalization of air traffic rights between countries. Our Commercial Market Outlook forecast projects a 3.5% growth rate in the global fleet over a 20-year period. Based on long-term global economic growth projections of 2.6% in average annual gross domestic product, we

project demand for approximately 42,595 new airplanes over the next 20 years. The industry remains vulnerable to exogenous developments including fuel price spikes, credit market shocks, acts of terrorism, natural disasters, conflicts, epidemics, pandemics and increased global environmental regulations.

At BDS, we continue to see stable demand reflecting the important role our products and services have in ensuring our national security. Outside of the U.S., we are seeing similar solid demand as governments prioritize security, defense technology and global cooperation given evolving threats. We continue to experience production disruptions and inefficiencies due to technical challenges, supplier disruption and factory performance. These factors have contributed to significant earnings charges on fixed-price development programs as well as on a number of mature programs which are continuing to adversely affect margins and cash flows.

At BGS, we expect commercial revenues to remain strong in future quarters as the commercial airline industry has largely recovered and transitions to growth. The demand outlook for our government services business remains stable.

Consolidated Results of Operations

The following table summarizes key indicators of consolidated results of operations:

(Dollars in millions, except per share data)

Years ended December 31,	**2023**	2022	2021
Revenues	**$77,794**	$66,608	$62,286
GAAP			
Loss from operations	**($773)**	($3,519)	($2,870)
Operating margins	**(1.0) %**	(5.3) %	(4.6) %
Effective income tax rate	**(11.8) %**	(0.6) %	14.8 %
Net loss attributable to Boeing Shareholders	**($2,222)**	($4,935)	($4,202)
Diluted loss per share	**($3.67)**	($8.30)	($7.15)
Non-GAAP [(1)]			
Core operating loss	**($1,829)**	($4,662)	($4,043)
Core operating margins	**(2.4 %)**	(7.0 %)	(6.5 %)
Core loss per share	**($5.81)**	($11.06)	($9.44)

(1) These measures exclude certain components of pension and other postretirement benefit expense. See pages 43 - 45 for important information about these non-GAAP measures and reconciliations to the most directly comparable GAAP measures.

Revenues

The following table summarizes Revenues:

(Dollars in millions)

Years ended December 31,	**2023**	2022	2021
Commercial Airplanes	**$33,901**	$26,026	$19,714
Defense, Space & Security	**24,933**	23,162	26,540
Global Services	**19,127**	17,611	16,328
Unallocated items, eliminations and other	**(167)**	(191)	(296)
Total	**$77,794**	$66,608	$62,286

Revenues increased by $11,186 million in 2023 compared with 2022 driven by higher revenues at all three operating segments. BCA revenues increased by $7,875 million primarily driven by higher 787 deliveries. BDS revenues increased by $1,771 million primarily due to higher revenues on fixed-price development programs. BGS revenues increased by $1,516 million primarily due to higher commercial services revenue driven by market recovery across the commercial portfolio.

Revenues increased by $4,322 million in 2022 compared with 2021 driven by higher revenues at BCA and BGS, partially offset by lower revenues at BDS. BCA revenues increased by $6,312 million primarily driven by higher 737 and 787 deliveries. BGS revenues increased by $1,283 million primarily due to higher commercial services volume, partially offset by lower government services volume and performance. BDS revenues decreased by $3,378 million primarily due to charges on fixed-price development programs, unfavorable performance across other defense programs, and lower P-8 and weapons volume.

Revenues will continue to be significantly impacted until the global supply chain stabilizes, labor instability diminishes, and deliveries ramp up.

Loss From Operations

The following table summarizes Loss from operations:

(Dollars in millions)

Years ended December 31,	**2023**	2022	2021
Commercial Airplanes	**($1,635)**	($2,341)	($6,377)
Defense, Space & Security	**(1,764)**	(3,544)	1,544
Global Services	**3,329**	2,727	2,017
Segment operating loss	**(70)**	(3,158)	(2,816)
Unallocated items, eliminations and other	**(1,759)**	(1,504)	(1,227)
Pension FAS/CAS service cost adjustment	**799**	849	882
Postretirement FAS/CAS service cost adjustment	**257**	294	291
Loss from operations (GAAP)	**($773)**	($3,519)	($2,870)
FAS/CAS service cost adjustment [(1)]	**(1,056)**	(1,143)	(1,173)
Core operating loss (Non-GAAP)[(2)]	**($1,829)**	($4,662)	($4,043)

(1) The FAS/CAS service cost adjustment represents the difference between the FAS pension and postretirement service costs calculated under GAAP and costs allocated to the business segments.

(2) Core operating loss is a non-GAAP measure that excludes the FAS/CAS service cost adjustment. See pages 43 - 45.

Loss from operations decreased by $2,746 million in 2023 compared with 2022. BDS loss from operations decreased by $1,780 million compared to the same period in 2022, primarily due to a reduction in net unfavorable cumulative contract catch-up adjustments, which were $2,328 million better than the net unfavorable impact in the prior year. BCA loss from operations decreased by $706 million reflecting higher deliveries and lower period expenses including lower abnormal production costs, partially offset by higher spending on research and development. BGS earnings from operations increased by $602 million in 2023 compared with 2022 primarily due to higher commercial services revenue. Loss from operations on Unallocated items, eliminations and other increased by $255 million in 2023 primarily due to higher deferred compensation expense.

Loss from operations increased by $649 million in 2022 compared with 2021. BDS had a loss from operations of $3,544 million compared with earnings of $1,544 million during 2021, primarily due to charges on development programs. BCA loss from operations decreased by $4,036 million primarily due to the absence in 2022 of the $3,460 million reach-forward loss taken on the 787 program in 2021, higher 737 deliveries and lower abnormal production costs, partially offset by higher research and development spending, charges related to the war in Ukraine and other period expenses. BGS earnings from operations increased by $710 million in 2022 compared with 2021 primarily due to higher commercial services volume and favorable mix, partially offset by lower government services performance.

Core operating loss decreased by $2,833 million in 2023 compared with 2022 and increased by $619 million in 2022 compared with 2021 primarily due to changes in Segment operating loss as described above.

Unallocated Items, Eliminations and Other The most significant items included in Unallocated items, eliminations and other (expense)/income are shown in the following table:

(Dollars in millions)

Years ended December 31,	**2023**	2022	2021
Share-based plans	**$62**	($114)	($174)
Deferred compensation	**(188)**	117	(126)
Amortization of previously capitalized interest	**(95)**	(95)	(107)
Research and development expense, net	**(315)**	(278)	(184)
Eliminations and other unallocated items	**(1,223)**	(1,134)	(636)
Unallocated items, eliminations and other	**($1,759)**	($1,504)	($1,227)

Share-based plans expense decreased by $176 million in 2023 and $60 million in 2022, primarily due to fewer share-based grants and the timing of corporate allocations in 2023. The lower expense in 2022 compared to 2021 was due to decreased grants of restricted stock units (RSUs) and other share-based compensation.

Deferred compensation expense increased by $305 million in 2023, and decreased by $243 million in 2022, primarily driven by changes in broad stock market conditions.

Research and development expense increased by $37 million in 2023 and increased by $94 million in 2022 primarily due to spending on enterprise product development.

Eliminations and other unallocated items was largely unchanged in 2023. Eliminations and other unallocated expense increased by $498 million in 2022 primarily due to a $200 million settlement with the Securities and Exchange Commission related to the 737 MAX accidents, lower income from operating investments and an increase in environmental remediation expense.

Net periodic pension benefit costs included in Loss from operations were as follows:

(Dollars in millions)	Pension		
Years ended December 31,	**2023**	2022	2021
Allocated to business segments	**($801)**	($852)	($885)
Pension FAS/CAS service cost adjustment	**799**	849	882
Net periodic pension benefit cost included in Loss from operations	**($2)**	($3)	($3)

The pension FAS/CAS service cost adjustment recognized in Loss from operations in 2023 decreased by $50 million compared with 2022 and decreased by $33 million in 2022 compared with 2021 due to changes in allocated pension cost year over year. Net periodic benefit cost included in Loss from operations in 2023 was largely consistent with 2022 and 2021.
For additional discussion related to Postretirement Plans, see Note 16 to our Consolidated Financial Statements.

Other Earnings Items

(Dollars in millions)

Years ended December 31,	**2023**	2022	2021
Loss from operations	**($773)**	($3,519)	($2,870)
Other income, net	**1,227**	1,058	551
Interest and debt expense	**(2,459)**	(2,561)	(2,714)
Loss before income taxes	**(2,005)**	(5,022)	(5,033)
Income tax (expense)/benefit	**(237)**	(31)	743
Net loss from continuing operations	**(2,242)**	(5,053)	(4,290)
Less: net loss attributable to noncontrolling interest	**(20)**	(118)	(88)
Net loss attributable to Boeing Shareholders	**($2,222)**	($4,935)	($4,202)

Non-operating pension income included in Other income, net was $529 million in 2023, $881 million in 2022 and $528 million in 2021. The decreased income in 2023 compared to 2022 was primarily due to higher interest cost and lower expected return on plan assets, partially offset by lower amortization of net actuarial losses. The increased income in 2022 compared to 2021 was primarily due to lower amortization of net actuarial losses in 2022 and a settlement loss recorded in 2021.

Non-operating postretirement income included in Other income, net was $58 million in 2023 and 2022, and $1 million in 2021. The increased income in 2022 was due to lower amortization of net actuarial losses.

Interest and debt expense decreased by $102 million in 2023 and $153 million 2022 primarily due to lower average debt balances.

For additional discussion related to Income Taxes, see Note 4 to our Consolidated Financial Statements.

Total Costs and Expenses (“Cost of Sales”)

Cost of sales, for both products and services, consists primarily of raw materials, parts, sub-assemblies, labor, overhead and subcontracting costs. Our BCA segment predominantly uses program accounting to account for cost of sales. Under program accounting, cost of sales for each commercial aircraft program equals the product of (i) revenue recognized in connection with customer deliveries and (ii) the

estimated cost of sales percentage applicable to the total remaining program. For long-term contracts, the amount reported as cost of sales is recognized as incurred. Substantially all contracts at our BDS segment and certain contracts at our BGS segment are long-term contracts with the U.S. government and other customers that generally extend over several years. Cost of sales for commercial spare parts is recorded at average cost.

The following table summarizes cost of sales:

(Dollars in millions)

Years ended December 31	**2023**	2022	Change	2022	2021	Change
Cost of sales	**$70,070**	$63,078	$6,992	$63,078	$59,237	$3,841
Cost of sales as a % of Revenues	**90.1 %**	94.7 %	(4.6)%	94.7 %	95.1 %	(0.4)%

Cost of sales increased by $6,992 million in 2023 compared with 2022, primarily due to higher revenues at BCA and BGS, partially offset by lower development charges at BDS. Cost of sales as a percentage of Revenues decreased in 2023 compared to 2022 primarily due to lower charges on BDS development programs.

Cost of sales increased by $3,841 million in 2022 compared with 2021, primarily due to charges recorded at BDS and higher revenues at BCA. Cost of sales as a percentage of Revenues remained largely consistent in 2022 compared to 2021.

Research and Development

The following table summarizes our Research and development expense:

(Dollars in millions)

Years ended December 31,	**2023**	2022	2021
Commercial Airplanes	**$2,036**	$1,510	$1,140
Defense, Space & Security	**919**	945	818
Global Services	**107**	119	107
Other	**315**	278	184
Total	**$3,377**	$2,852	$2,249

Research and development expense increased by $525 million in 2023 compared with 2022 primarily due to higher research and development expenditures on the 777X program as well as other BCA and enterprise investments in product development.

Research and development expense increased by $603 million in 2022 compared with 2021 primarily due to higher research and development expenditures on 777X, 737 MAX, as well as BCA and enterprise investments in product development.

Backlog

Our backlog at December 31 was as follows:

(Dollars in millions)

Years ended December 31,	**2023**	2022
Commercial Airplanes	**$440,507**	$329,824
Defense, Space & Security	**59,012**	54,373
Global Services	**19,869**	19,338
Unallocated items, eliminations and other	**807**	846
Total Backlog	**$520,195**	$404,381
Contractual backlog	**$497,094**	$381,977
Unobligated backlog	**23,101**	22,404
Total Backlog	**$520,195**	$404,381

Contractual backlog of unfilled orders excludes purchase options, announced orders for which definitive contracts have not been executed, orders where customers have the unilateral right to terminate, and unobligated U.S. and non-U.S. government contract funding. The increase in contractual backlog during 2023 was primarily due to increases in BCA and BDS backlog. If we are unable to deliver aircraft to customers in China consistent with our assumptions, and/or entry into service of the 777X, 737-7 and/or 737-10 is further delayed, we may experience reductions to backlog and/or significant order cancellations.

Unobligated backlog includes U.S. and non-U.S. government definitive contracts for which funding has not been authorized. Unobligated backlog was largely unchanged in 2023.

Additional Considerations

U.S. Government Funding The Continuing Resolution enacted on January 19, 2024, continues federal funding at fiscal year 2023 appropriated levels through March 1, 2024, for selected departments and agencies, including the Department of Transportation, and through March 8, 2024, for the remaining departments and agencies, including the United States Department of Defense (U.S. DoD) and the National Aeronautics and Space Administration (NASA). Congress and the President must enact either full-year fiscal year 2024 (FY24) appropriations bills or an additional Continuing Resolution to fund government departments and agencies after these dates, or a partial or full government shutdown could result. U.S. government discretionary spending in FY24 and 2025 (FY25), including defense spending, was capped by the Fiscal Responsibility Act of 2023 (FRA). Additionally, a Continuing Resolution for FY24 or FY25 in place on April 30 of the relevant fiscal year, would trigger a sequester under the FRA.

Global Trade We continually monitor the global trade environment in response to geopolitical economic developments, as well as changes in tariffs, trade agreements or sanctions that may impact the Company.

The current state of U.S.-China relations remains an ongoing watch item. Since 2018, the U.S. and China have imposed tariffs on each other's imports. Certain aircraft parts and components that Boeing procures are subject to these tariffs. We are mitigating import costs through Duty Drawback Customs procedures. China is a significant market for commercial aircraft and we have long-standing relationships with our Chinese customers, who represent a key component of our commercial aircraft backlog. Overall, the U.S.-China trade relationship remains stalled as economic and national security concerns continue to be a challenge.

Beginning in June 2018, the U.S. Government imposed tariffs on steel and aluminum imports. In response to these tariffs, several major U.S. trading partners have imposed, or announced their intention to impose, tariffs on U.S. goods. The U.S. has subsequently reached agreements with Mexico, Canada, Japan, the United Kingdom, and the European Union, to ease or remove tariffs on steel and/or aluminum. We continue to monitor the potential for any extra costs that may result from the remaining global tariffs.

We are complying with all U.S. and other government export control restrictions and sanctions imposed on certain businesses and individuals in Russia. We continue to monitor and evaluate additional sanctions and export restrictions that may be imposed by the U.S. Government or other governments, as well as any responses from Russia that could affect our supply chain, business partners or customers, for any additional impacts to our business.

Supply Chain We and our suppliers are experiencing supply chain disruptions as a result of global supply chain constraints and labor instability. We and our suppliers are also experiencing inflationary pressures. We continue to monitor the health and stability of the supply chain. These factors have reduced overall productivity and adversely impacted our financial position, results of operations and cash flows.

The current conflict in Israel and the Gaza Strip has the potential to impact certain of our suppliers, and has impacted some operations for our airline and lessor customers. We are closely monitoring developments, supporting our employees and customers, and will take mitigating actions as appropriate.

Segment Results of Operations and Financial Condition

Commercial Airplanes

Business Environment and Trends

Airline Industry Environment See Overview to Management's Discussion and Analysis of Financial Condition and Results of Operations for a discussion of the airline industry environment.

Industry Competitiveness The commercial aircraft market and the airline industry both remain extremely competitive. Continued access to global markets remains vital to our ability to fully realize our sales potential and long-term investment returns. Approximately 78% of BCA's total backlog, in dollar terms, is with non-U.S. airlines. We face aggressive international competitors who are intent on increasing their market share. They offer competitive products and have access to most of the same customers and suppliers. With government support, Airbus has historically invested heavily to create a family of products to compete with ours. After the acquisition of a majority share of Bombardier's C Series (now A220) in 2018, Airbus continues to expand in the 100-150 seat transcontinental market. Other competitors are also in different phases of developing commercial jet aircraft, including Commercial Aircraft Corporation of China, Ltd. (COMAC), which delivered its first C919 aircraft in 2022. Some of these competitors have historically enjoyed access to government-provided financial support, including "launch aid," which greatly reduces the cost and commercial risks associated with airplane development activities. This has enabled the development of airplanes without broad commercial viability; others to be brought to market more quickly than otherwise possible; and many offered for sale below market-based prices. Competitors continue to make improvements in efficiency, which may result in funding product development, gaining market share and improving earnings. This market environment has resulted in intense pressures on pricing and other competitive factors, and we expect these pressures to continue or intensify in the coming years.

Results of Operations

(Dollars in millions)

Years ended December 31,	**2023**	2022	2021
Revenues	**$33,901**	$26,026	$19,714
% of total company revenues	**44 %**	39 %	32 %
Loss from operations	**($1,635)**	($2,341)	($6,377)
Operating margins	**(4.8)%**	(9.0)%	(32.3)%
Research and development	**$2,036**	$1,510	$1,140

Revenues

BCA revenues increased by $7,875 million in 2023 compared with 2022 primarily due to higher 787 deliveries in 2023.

BCA revenues increased by $6,312 million in 2022 compared with 2021 primarily due to higher 737 and 787 deliveries in 2022.

BCA deliveries, including intercompany deliveries, as of December 31 were as follows:

	737 *	**747**	**767** *	**777**	**787**	**Total**
2023						
Cumulative deliveries	**8,528**	**1,573**	**1,303**	**1,727**	**1,110**	
Deliveries	**396** (9)	**1**	**32** (14)	**26**	**73**	**528**
2022						
Cumulative deliveries	8,132	1,572	1,271	1,701	1,037	
Deliveries	387 (13)	5	33 (15)	24	31	480
2021						
Cumulative deliveries	7,745	1,567	1,238	1,677	1,006	
Deliveries	263 (16)	7	32 (13)	24	14	340

* Intercompany deliveries identified by parentheses

Loss From Operations

BCA loss from operations was $1,635 million in 2023 compared with $2,341 million in 2022 reflecting higher deliveries and lower period expenses including lower abnormal production costs, partially offset by higher spending on research and development. Abnormal production costs in 2023 were $1,527 million, including $1,014 million related to the 787 program and $513 million related to the 777X program. Abnormal production costs in 2022 were $1,753 million, including $1,240 million related to the 787 program, $325 million related to the 777X program, and $188 million related to the 737 program.

BCA loss from operations was $2,341 million in 2022 compared with $6,377 million in 2021. The 2021 loss includes a reach-forward loss on the 787 program of $3,460 million. The improved performance in 2022 also reflects higher 737 deliveries and lower abnormal production costs, partially offset by higher research and development spending, charges related to the war in Ukraine and other period expenses. Abnormal production costs in 2021 were $2,355 million, including $1,887 million related to the 737 program and $468 million related to the 787 program.

Backlog

Our total backlog represents the estimated transaction prices on unsatisfied and partially satisfied performance obligations to our customers where we believe it is probable that we will collect the consideration due and where no contingencies remain before we and the customer are required to perform. Backlog does not include prospective orders where customer-controlled contingencies remain, such as the customer receiving approval from its board of directors, shareholders or government or completing financing arrangements. All such contingencies must be satisfied or have expired prior to recording a new firm order even if satisfying such conditions is highly probable. Backlog excludes options and Boeing customer financing orders as well as orders where customers have the unilateral right to terminate. A number of our customers may have contractual remedies, including rights to reject individual airplane deliveries if the actual delivery date is significantly later than the contractual delivery date. We address customer claims and requests for other contractual relief as they arise. The value of orders in backlog is adjusted as changes to price and schedule are agreed to with customers and is reported in accordance with the requirements of Accounting Standards Codification (ASC) 606.

BCA total backlog of $440,507 million at December 31, 2023 increased from $329,824 million at December 31, 2022, reflecting new orders in excess of deliveries and a decrease in the value of existing orders that in our assessment do not meet the accounting requirements of ASC 606 for inclusion in backlog, partially offset by order cancellations. Aircraft order cancellations during the year ended December 31, 2023 totaled $12,925 million and primarily relate to 737 and 777X aircraft. The net ASC 606 adjustments for the year ended December 31, 2023 resulted in an increase to backlog of $20,605 million primarily due to a net decrease of 777X and 737 aircraft. ASC 606 adjustments include consideration of aircraft orders where a customer-controlled contingency may exist, as well as an assessment of whether the customer is committed to perform, impacts of geopolitical events or related sanctions, or whether it is probable that the customer will pay the full amount of consideration when it is due. If we are unable to deliver aircraft to customers in China consistent with our assumptions, and/or entry into service of the 777X, 737-7 and/or 737-10 is further delayed, we may experience reductions to backlog and/or significant order cancellations.

Accounting Quantity The accounting quantity is our estimate of the quantity of airplanes that will be produced for delivery under existing and anticipated contracts. The determination of the accounting quantity is limited by the ability to make reasonably dependable estimates of the revenue and cost of existing and anticipated contracts. It is a key determinant of the gross margins we recognize on sales of individual airplanes throughout a program's life. Estimation of each program's accounting quantity takes into account several factors that are indicative of the demand for that program, including firm orders, letters of intent from prospective customers and market studies. We review our program accounting quantities quarterly.

The accounting quantity for each program may include units that have been delivered, undelivered units under contract and units anticipated to be under contract in the reasonable future (anticipated orders). In developing total program estimates, all of these items within the accounting quantity must be considered.

The following table provides details of the accounting quantities and firm orders by program as of December 31. Cumulative firm orders represent the cumulative number of commercial jet aircraft deliveries plus undelivered firm orders. Firm orders include military derivative aircraft that are not included in program accounting quantities. All revenues and costs associated with military derivative aircraft production are reported in the BDS segment.

	Program						
	737	**747**	**767**	**777**	**777X**	**787**	**†**
2023							
Program accounting quantities	**11,600**	**1,574**	**1,279**	**1,790**	**500**	**1,700**	
Undelivered units under firm orders	**4,332**		**104**	**48**	**416**	**726**	**(8)**
Cumulative firm orders	**12,860**	**1,573**	**1,407**	**1,775**	**416**	**1,836**	
2022							
Program accounting quantities	10,800	1,574	1,267	1,790	400	1,600	
Undelivered units under firm orders	3,653	1	106	69	244	505	(8)
Cumulative firm orders	11,785	1,573	1,377	1,770	244	1,542	
2021							
Program accounting quantities	10,400	1,574	1,243	1,750	350	1,500	
Undelivered units under firm orders	3,414	6	108	58	253	411	(14)
Cumulative firm orders	11,159	1,573	1,346	1,735	253	1,417	

† Boeing customer financing aircraft orders are identified in parentheses.

Program Highlights

737 Program The accounting quantity for the 737 program increased by 800 units during 2023 due to the program's normal progress of obtaining additional orders and delivering airplanes. We are currently producing at a rate of 38 per month.

On January 5, 2024, an Alaska Airlines 737-9 flight made an emergency landing after a mid-exit door plug detached in flight. Following the accident, the Federal Aviation Administration (FAA) grounded and required inspections of all 737-9 aircraft with a mid-exit door plug, which constitute the large majority of the approximately 220 737-9 aircraft in the in-service fleet. On January 24, 2024, the FAA approved an enhanced maintenance and inspection process that must be performed on each of the grounded 737-9 aircraft. Our 737-9 operators have begun returning their fleets to service, and many 737-9s have completed inspections and resumed revenue flights. All 737-9 aircraft in production will undergo this same enhanced inspection process prior to delivery.

On January 10, 2024, the FAA notified Boeing that the FAA has initiated an investigation into Boeing's quality control system. This was followed by the FAA announcing actions to increase its oversight of Boeing, including conducting:

1. An audit involving the 737-9 production line and its suppliers to evaluate Boeing's compliance with approved quality procedures,
2. Increased monitoring of Boeing's 737-9 in-service events, and
3. An assessment of safety risks around delegated authority and quality oversight, and examination of options to move these functions under independent third parties.

On January 24, 2024, the FAA stated that it will not approve production rate increases or additional production lines for the 737 MAX until it is satisfied that Boeing is in full compliance with required quality control procedures.

We are following the lead of the FAA as we work through the certification process of the 737-7 and 737-10 models. We continue to work with the FAA on open actions to support 737-7 certification. During the fourth quarter of 2023, the 737-10 program received approval from the FAA to begin the first phase of FAA certification flight testing. At December 31, 2023, we had approximately 35 737-7 and 737-10 aircraft in inventory. We are now planning to incorporate engineering solutions to the de-icing systems on the 737-7 and 737-10, which will delay certification and first deliveries.

We are currently unable to reasonably estimate what impact the accident, the related FAA actions and certification delays will have on our financial position, results of operations and cash flows.

During the third quarter of 2023, we discovered non-conforming holes in the aft pressure dome of certain 737-7, 737-8 and 737 military derivative aircraft. Rework on non-conforming fuselages with our supplier is complete and newly built aircraft meet our specifications. We do not expect inspection of completed aircraft in inventory to result in significant rework or production disruption.

As of December 31, 2023, we had approximately 140 737-8 aircraft in inventory that were produced prior to 2023, including 85 aircraft for customers in China. Return-to-service of the China 737 MAX fleet is complete. While there continues to be uncertainty, we are continuing to work with airlines and government officials on delivery timing and expect to deliver most of the aircraft in inventory by the end of 2024. In the event that we are unable to deliver aircraft consistent with our assumptions, our financial position, results of operations and cash flows could be adversely affected.

See further discussion of the 737 MAX in Note 7, Note 13 and Note 23 to our Consolidated Financial Statements .

747 Program We completed production of the 747 in the fourth quarter of 2022 and delivery of the last aircraft occurred in February 2023. Ending production of the 747 did not have a material impact on our financial position, results of operations or cash flows.

767 Program The accounting quantity for the 767 program increased by 12 units during 2023 due to the program's normal progress of obtaining additional orders and delivering airplanes. The 767 assembly line includes the commercial program and a derivative to support the KC-46A Tanker program. The commercial program has near break-even gross margins. We are currently producing at a combined rate of 3 aircraft per month.

777 and 777X Programs The accounting quantity for the 777X program increased by 100 units during the year ended December 31, 2023 due to obtaining additional orders for the 777-9 and 777-8. We are currently producing at a combined production rate of 4 per month for the 777/777X programs.

We are following the lead of the FAA as we work through the certification process including obtaining approval from the FAA to begin certification flight testing. We expect the first delivery of the 777-9 to occur in 2025 and the 777-8 freighter to occur in 2027. First delivery of the 777-8 passenger aircraft is not expected to occur before 2030.

In April 2022, we decided to pause production of the 777-9 during 2022 and 2023, which resulted in cumulative abnormal production costs of $0.8 billion. In the fourth quarter of 2023, the 777X program resumed production.

The level of profitability on the 777X program will be subject to a number of factors. These factors include aircraft certification requirements and timing, change incorporation on completed aircraft, production disruption due to labor instability and supply chain disruption, customer negotiations, further

production rate adjustments for the 777X or other commercial aircraft programs, and contraction of the accounting quantity. One or more of these factors could result in reach-forward losses in future periods.

787 Program The accounting quantity for the 787 program increased by 100 units during 2023 due to the program's normal progress of obtaining additional orders and delivering airplanes.

During 2023, we delivered 73 aircraft and increased the production rate to 5 per month beginning in October 2023. At December 31, 2023 and 2022, we had approximately 50 and 90 aircraft in inventory that require rework which we expect to complete by the end of 2024.

Beginning in 2021, the 787 program lowered production rates and paused deliveries in order to improve production quality and implement changes in the production process designed to ensure that newly-built aircraft meet our specifications. In the third quarter of 2021, we determined that production rates below 5 per month represented abnormally low production rates. This resulted in abnormal production costs, which we expensed as incurred through the third quarter of 2023. We also determined that the inspections and rework costs on inventoried aircraft are excessive and should also be accounted for as abnormal production costs. Cumulative abnormal costs recorded through December 31, 2023 totaled $2.7 billion. The costs associated with the remaining rework are not expected to be significant.

Fleet Support We provide the operators of our commercial aircraft with assistance and services to facilitate efficient and safe airplane operation. Collectively known as fleet support services, these activities and services begin prior to airplane delivery and continue throughout the operational life of the airplane. They include flight and maintenance training, field service support, engineering services, information services and systems and technical data and documents. The costs for fleet support are expensed as incurred and have historically been approximately 1% of total consolidated costs of products and services.

Additional Considerations

The development and ongoing production of commercial aircraft is extremely complex, involving extensive coordination and integration with suppliers and highly-skilled labor from employees and other partners. Meeting or exceeding our performance and reliability standards, as well as those of customers and regulators, can be costly and technologically challenging. In addition, the introduction of new aircraft and derivatives, such as the 777X, 737-7 and 737-10, involves increased risks associated with meeting development, production and certification schedules. These challenges include significant global regulatory scrutiny of all development aircraft. As a result, our ability to deliver aircraft on time, satisfy performance and reliability standards and achieve or maintain, as applicable, program profitability is subject to significant risks. Factors that could result in lower margins (or a material charge if an airplane program has or is determined to have reach-forward losses) include: changes to the program accounting quantity, customer and model mix, production costs and rates, changes to price escalation factors due to changes in the inflation rate or other economic indicators, performance or reliability issues involving completed aircraft, capital expenditures and other costs associated with increasing or adding new production capacity, learning curve, additional change incorporation, rework or safety enhancements, operational and supply chain challenges, achieving anticipated cost reductions, additional regulatory requirements in connection with certification in one or more jurisdictions, flight test and certification schedules, costs, schedule and demand for new airplanes and derivatives and status of customer claims, supplier claims or assertions and other contractual negotiations. While we believe the cost and revenue estimates incorporated in the consolidated financial statements are appropriate, the technical complexity of our airplane programs creates financial risk as additional completion costs may become necessary or scheduled delivery dates could be extended, which could trigger termination provisions, order cancellations or other financially significant exposure.

Defense, Space & Security

Business Environment and Trends

United States Government Defense Environment Overview

In March 2023, the U.S. government released the President's budget request for FY24, which requested $842 billion in funding for the U.S. DoD and $27 billion for NASA. The President's budget request does not request funding for F/A-18, V-22, or P-8 production aircraft. The P-8 program continues to pursue U.S. and non-U.S. sales opportunities. In addition, there is ongoing uncertainty with respect to program-level appropriations for the U.S. DoD, NASA and other government agencies for FY24 and beyond.

Future budget cuts or investment priority changes, including changes associated with the authorizations and appropriations process, could result in reductions, cancellations and/or delays of existing contracts or programs. Any of these impacts could have a material effect on our results of operations, financial position, and/or cash flows.

Non-U.S. Defense Environment Overview The non-U.S. market continues to be driven by complex and evolving security challenges and the need to modernize aging equipment and inventories. BDS expects that it will continue to have a wide range of opportunities across Asia, Europe and the Middle East given the diverse regional threats. At the end of 2023, 29% of BDS backlog was attributable to non-U.S. customers.

Results of Operations

(Dollars in millions)

Years ended December 31,	**2023**	2022	2021
Revenues	**$24,933**	$23,162	$26,540
% of total company revenues	**32 %**	35 %	43 %
(Loss)/earnings from operations	**($1,764)**	($3,544)	$1,544
Operating margins	**(7.1)%**	(15.3)%	5.8 %

Since our operating cycle is long-term and involves many different types of development and production contracts with varying delivery and milestone schedules, the operating results of a particular period may not be indicative of future operating results. In addition, depending on the customer and their funding sources, our orders might be structured as annual follow-on contracts, or as one large multi-year order or long-term award. As a result, period-to-period comparisons of backlog are not necessarily indicative of future workloads. The following discussions of comparative results among periods should be viewed in this context.

Deliveries of new-build production units, including remanufactures and modifications, were as follows:

Years ended December 31,	**2023**	2022	2021
F/A-18 Models	**22**	14	21
F-15 Models	**9**	12	16
T-7A Red Hawk	**3**		
CH-47 Chinook (New)	**11**	19	15
CH-47 Chinook (Remanufactured)	**9**	9	5
AH-64 Apache (New)	**20**	25	27
AH-64 Apache (Remanufactured)	**57**	50	56
MH-139 Grey Wolf	**2**	4	
KC-46 Tanker	**13**	15	13
P-8 Models	**11**	12	16
Commercial Satellites	**5**	4	
Military Satellites		1	
Total	**162**	165	169

Revenues

BDS revenues in 2023 increased by $1,771 million compared with 2022. This increase is not indicative of future projected revenue trends. Revenues related to BDS' five major fixed-price development programs increased by $1,767 million in 2023 compared with 2022. This increase reflects lower unfavorable net cumulative contract catch-up adjustments in 2023 as well as higher costs incurred in 2023 to complete these contracts. Net unfavorable cumulative contract catch-up adjustments in 2023 were $648 million better than in 2022 largely due to lower charges on development programs in 2023, partially offset by unfavorable performance on other programs.

BDS revenues in 2022 decreased by $3,378 million compared with 2021 primarily due to charges on development programs. Unfavorable performance across other defense programs and lower P-8 and weapons volume also contributed to the decrease in revenue. Cumulative contract catch-up adjustments in 2022 were $1,858 million more unfavorable than the prior year largely due to charges on development programs.

(Loss)/earnings From Operations

BDS loss from operations in 2023 of $1,764 million decreased by $1,780 million compared with $3,544 million in 2022. The decrease is primarily due to $2,863 million of lower charges in 2023 on fixed-price development programs that were partially offset by lower earnings across other programs including satellites and F-15, as well as higher period expenses. During 2023, losses incurred on the five fixed-price development programs totaled $1,585 million compared with $4,448 million in 2022. Charges on fixed-price development programs in 2023 included VC-25B ($482 million), KC-46A Tanker ($309 million), Commercial Crew ($288 million), T-7A Red Hawk ($275 million) and MQ-25 ($231 million). Loss from operations in 2023 includes a $315 million impact from an agreement with one of our satellite customers which includes customer considerations as well as increased costs to enhance the constellation and meet lifecycle commitments. Net unfavorable cumulative contract catch-up adjustments were $2,328 million better than in 2022.

BDS loss from operations in 2022 of $3,544 million decreased by $5,088 million compared with earnings from operations of $1,544 million in 2021 primarily due to unfavorable impacts of cumulative contract catch-up adjustments ($4,284 million more unfavorable in 2022 than 2021). Volume and mix and higher research and development also contributed to the year over year earnings decline. Charges

on fixed-price development programs in 2022 included VC-25B ($1,452 million), KC-46A Tanker ($1,374 million), MQ-25 ($579 million), T-7A Red Hawk Production Options ($552 million), T-7A Red Hawk Engineering and Manufacturing Development (EMD) ($203 million), and Commercial Crew ($288 million). These were partially offset by charges on the KC-46A Tanker ($402 million), VC-25B ($318 million), and Commercial Crew ($214 million) recognized in 2021. The net unfavorable cumulative contract catch-up adjustments represent losses incurred on these development and other programs. See further discussion of fixed-price contracts in Note 13 to our Consolidated Financial Statements.

BDS (loss)/earnings from operations includes our share of income from equity method investments of $44 million, $13 million and $53 million primarily from our United Launch Alliance and other joint ventures in 2023, 2022 and 2021, respectively.

Backlog

Total backlog of $59,012 million at December 31, 2023 was $4,639 million higher than December 31, 2022 due to the timing of awards and revenue recognized on contracts awarded in prior years.

Additional Considerations

Our BDS business includes a variety of development programs which have complex design and technical challenges. Some of these programs have cost-type contracting arrangements. In these cases, the associated financial risks are primarily reduced award or incentive fees, lower profit rates, or program cancellation if cost, schedule or technical performance issues arise. Examples of these programs include Ground-based Midcourse Defense, Proprietary and Space Launch System programs.

Some of our development programs are contracted on a fixed-price basis. Examples of fixed-price development programs include Commercial Crew, KC-46A Tanker, MQ-25, T-7A Red Hawk, VC-25B, and commercial and military satellites. A number of our ongoing fixed-price development programs have reach-forward losses. New programs could also have risk for reach-forward loss upon contract award and during the period of contract performance. Many development programs have highly complex designs. As technical or quality issues arise during development, we may experience schedule delays and cost impacts, which could increase our estimated cost to perform the work or reduce our estimated price, either of which could result in a material charge or otherwise adversely affect our financial condition. These programs are ongoing, and while we believe the cost and fee estimates incorporated in the financial statements are appropriate, the technical complexity of these programs creates financial risk as additional completion costs may become necessary or scheduled delivery dates could be extended, which could trigger termination provisions or other financially significant exposure. Risk remains that we may be required to record additional reach-forward losses in future periods.

Global Services

Business Environment and Trends

The aerospace markets we serve include parts distribution, logistics and other inventory services; maintenance, engineering and upgrades; training and professional services; and data analytics and digital services. In 2023, commercial services volume at BGS exceeded pre-pandemic levels. We expect BGS commercial revenues to remain strong in future quarters as the commercial airline industry transitions from recovery to growth.

Over the long-term, as the size of the worldwide commercial airline fleet continues to grow, so does demand for aftermarket services designed to increase efficiency and extend the economic lives of aircraft. Airlines are using data analytics to plan flight operations and predictive maintenance to improve

their productivity and efficiency. Airlines continue to look for opportunities to reduce the size and cost of their spare parts inventory, frequently outsourcing spares management to third parties.

The demand outlook for our government services business has remained stable in 2023. Government services market segments are growing on pace with related fleets, but vary based on the utilization and age of the aircraft. The U.S. government services market is the single largest individual market, comprising over 50 percent of the government services markets served. Over the next decade, we expect U.S. growth to remain flat and non-U.S. fleets, led by Middle East and Asia Pacific customers, to add rotorcraft and commercial derivative aircraft at faster rates. We expect approximately 30 percent of the worldwide fleet of military aircraft to be retired and replaced over the next ten years, driving increased demand for services to maintain aging aircraft and enhance aircraft capability.

BGS' major customer, the U.S. government, remains subject to the spending limits and uncertainty, which could restrict the execution of certain program activities and delay new programs or competitions.

Industry Competitiveness Aviation services is a competitive market with many domestic and international competitors. This market environment has resulted in intense pressures on pricing, and we expect these pressures to continue or intensify in the coming years. Continued access to global markets remains vital to our ability to fully realize our sales growth potential and long-term investment returns.

Results of Operations

(Dollars in millions)

Years ended December 31,	**2023**	2022	2021
Revenues	**$19,127**	$17,611	$16,328
% of total company revenues	**25 %**	26 %	26 %
Earnings from operations	**$3,329**	$2,727	$2,017
Operating margins	**17.4 %**	15.5 %	12.4 %

Revenues

BGS revenues in 2023 increased by $1,516 million compared with 2022 primarily due to higher commercial services revenue driven by market recovery across the commercial portfolio. The net unfavorable impact of cumulative contract catch-up adjustments in 2023 was $16 million worse than the net favorable impact in the prior year comparable period.

BGS revenues in 2022 increased by $1,283 million compared with 2021 primarily due to higher commercial services volume, partially offset by lower government services volume and performance. The decrease in government services volume is partly driven by the discontinuation of an engine distribution agreement in the second quarter of 2022. The net favorable impact of cumulative contract catch-up adjustments in 2022 was $137 million lower than the prior year.

Earnings From Operations

BGS earnings from operations in 2023 increased by $602 million compared with 2022, primarily due to higher commercial services revenue. The net unfavorable impact of cumulative contract catch-up adjustments in 2023 was $9 million higher than the prior year.

BGS earnings from operations in 2022 increased by $710 million compared with 2021, primarily due to higher commercial services volume and favorable mix, partially offset by lower government services performance. The net unfavorable impact of cumulative contract catch-up adjustments in 2022 was $148 million worse than the net favorable impact in the prior year.

Backlog

BGS total backlog of $19,869 million at December 31, 2023 increased by 3% from $19,338 million at December 31, 2022, primarily due to the timing of awards, partially offset by revenue recognized on contracts awarded in prior years.

Liquidity and Capital Resources

Cash Flow Summary

(Dollars in millions)

Years ended December 31,	**2023**	2022	2021
Net loss	**($2,242)**	($5,053)	($4,290)
Non-cash items	**4,113**	4,426	7,851
Changes in assets and liabilities	**4,089**	4,139	(6,977)
Net cash provided/(used) by operating activities	**5,960**	3,512	(3,416)
Net cash (used)/provided by investing activities	**(2,437)**	4,370	9,324
Net cash used by financing activities	**(5,487)**	(1,266)	(5,600)
Effect of exchange rate changes on cash and cash equivalents	**30**	(73)	(39)
Net (decrease)/increase in cash & cash equivalents, including restricted	**(1,934)**	6,543	269
Cash & cash equivalents, including restricted, at beginning of year	**14,647**	8,104	7,835
Cash & cash equivalents, including restricted, at end of year	**$12,713**	$14,647	$8,104

Operating Activities Net cash provided by operating activities was $6.0 billion during 2023 compared with $3.5 billion during 2022. Net cash provided by operating activities in 2022 included a $1.5 billion income tax refund. The year-over-year improvement in cash provided by operating activities reflects increases in revenues at BCA and BGS, higher Advances and progress billings ($3.3 billion) and lower payments to 737 MAX customers ($0.6 billion), partially offset by increased Inventories ($2.1 billion). Increases in Accrued liabilities in both years primarily reflects accrued reach-forward losses on BDS programs. Changes in assets and liabilities for 2023 decreased by $0.1 billion compared to 2022 primarily driven by unfavorable changes in Inventories ($2.1 billion) and Accrued liabilities ($2.2 billion), partially offset by increases in Advances and progress billings ($3.3 billion). The change in Inventories was primarily driven by increased production on the 737 and 777X programs, partially offset by increased deliveries for the 787 program in 2023. The change in Accrued liabilities was primarily driven by higher accrued losses on BDS fixed-price development programs recorded in the prior year. Cash provided by Advances and progress billings was $3.4 billion in 2023 as compared with $0.1 billion in 2022. Concessions paid to 737 MAX customers totaled $0.4 billion in 2023 and $1.0 billion in 2022.

The $6.9 billion improvement in cash provided by operating activities in 2022 was primarily driven by improved changes in assets and liabilities of $11.1 billion, partially offset by lower non-cash items of $3.4 billion and higher net loss of $0.8 billion. Changes in assets and liabilities for 2022 improved by $11.1 billion compared with 2021 primarily driven by favorable changes in Accrued liabilities ($6.6 billion), Accounts payable ($4.6 billion) and Inventories ($1.5 billion), partially offset by a decrease in Advances and progress billings ($2.4 billion) in 2022. The increase in Accrued liabilities was primarily driven by the accrued losses on BDS fixed-price development programs, lower payments to 737 MAX customers in 2022, and a $0.7 billion payment in 2021 consistent with the terms of the Deferred Prosecution Agreement between Boeing and the U.S. Department of Justice. Concessions paid to 737 MAX customers totaled $1.0 billion and $2.5 billion during 2022 and 2021. Growth in Accounts payable in 2022 was a source of cash while reductions in Accounts payable in 2021 was a use of cash,

generally reflecting increases in production rates. Inventory improvements were driven by higher 737 MAX deliveries and resumption of 787 deliveries in 2022. Additionally, in 2022 and 2021 we received income tax refunds of $1.5 billion and $1.7 billion. Cash provided by Advances and progress billings was $0.1 billion in 2022 as compared with $2.5 billion in 2021. The $3.4 billion reduction in non-cash items in 2022 was primarily driven by the $3.5 billion reach-forward loss on the 787 program that was recorded in 2021. Net loss for 2022 was $5.1 billion compared with net loss of $4.3 billion in 2021. The $0.8 billion year-over-year increase in the net loss was primarily driven by the absence of an income tax benefit in 2022.

Payables to suppliers who elected to participate in supply chain financing programs increased by $0.4 billion in 2023, increased by $0.2 billion in 2022, and decreased by $1.5 billion in 2021. Supply chain financing is not material to our overall liquidity. The decrease in 2021 was primarily due to reductions in commercial purchases from suppliers.

Investing Activities Cash used by investing activities during 2023 was $2.4 billion, compared with cash provided of $4.4 billion during 2022 and $9.3 billion during 2021. The increase in use of cash in 2023 compared to 2022 was primarily due to net contributions to investments of $0.7 billion in 2023 compared to net proceeds from investments of $5.6 billion in 2022. The decrease in cash inflows in 2022 compared to 2021 was primarily due to $4.2 billion of higher net proceeds from investments in 2021. Capital expenditures totaled $1.5 billion in 2023, compared with $1.2 billion in 2022 and $1.0 billion in 2021. We expect capital expenditures to grow in 2024 compared with 2023.

Financing Activities Cash used by financing activities was $5.5 billion during 2023, compared with $1.3 billion during 2022, and $5.6 billion in 2021. The increase of $4.2 billion compared with 2022 primarily reflects higher net debt repayments in 2023. During 2023, debt repayments net of new borrowings were $5.1 billion compared with $1.3 billion in 2022 and $5.6 billion in 2021.

At December 31, 2023 and 2022 debt balances totaled $52.3 billion and $57.0 billion, of which $5.2 billion were classified as short-term for both periods.

We had 1.7 million, 0.2 million and 0.3 million shares transferred to us from employee tax withholdings in 2023, 2022 and 2021, respectively. The increase in 2023 was primarily due to the vesting of a one-time RSU grant awarded to most employees in December 2020. In 2020, we announced the suspension of our dividend until further notice. As a result, we did not pay any dividends in 2023, 2022 and 2021.

Capital Resources

The following table summarizes certain cash requirements for known contractual and other obligations as of December 31, 2023, and the estimated timing thereof. See Note 12 for future operating lease payments.

(Dollars in millions)	Current	Long-term	Total
Long-term debt (including current portion)	$5,212	$47,587	$52,799
Interest on debt	2,146	29,254	31,400
Pension and other postretirement	495	7,582	8,077
Purchase obligations	61,963	79,159	141,122

We expect to be able to fund our cash requirements through cash and short-term investments and cash provided by operations, as well as continued access to capital markets. At December 31, 2023, we had $12.7 billion of cash, $3.3 billion of short-term investments, and $10.0 billion of unused borrowing capacity on revolving credit line agreements. In the third quarter of 2023, we entered into a $3.0 billion five-year revolving credit agreement expiring in August 2028 and a $0.8 billion 364-day revolving credit agreement expiring in August 2024. The 364-day credit facility has a one-year term out option which allows us to extend the maturity of any borrowings until August 2025. The legacy three-year revolving credit agreement expiring in August 2025, which consists of $3.0 billion of total commitments, and the legacy five-year revolving credit agreement expiring in October 2024, as amended, which consists of $3.2 billion of total commitments, each remain in effect. We anticipate that these credit lines will remain undrawn and primarily serve as back-up liquidity to support our general corporate borrowing needs.

Our credit ratings were downgraded in 2020 and remained unchanged as of December 31, 2023. During the fourth quarter of 2023, Fitch upgraded our credit rating outlook from stable to positive driven by declining inventory levels, improving production and delivery profile, growing backlog, and forecasted positive free cash flow. During the third quarter of 2023, S&P upgraded the outlook on our credit rating from negative to stable primarily driven by improving deliveries and expected increases in production. During the first quarter of 2023, Moody's upgraded the outlook on our credit rating from negative to stable primarily driven by an improvement in operating cash flow and a reduction of 737 and 787 aircraft in inventory.

We expect to be able to access capital markets when we require additional funding in order to pay off existing debt, address further impacts to our business related to market developments, fund outstanding financing commitments or meet other business requirements. A number of factors could cause us to incur increased borrowing costs and to have greater difficulty accessing public and private markets for debt. These factors include disruptions or declines in the global capital markets and/or a decline in our financial performance, outlook or credit ratings, and/or associated changes in demand for our products and services. These risks will be particularly acute if we are subject to further credit rating downgrades such as those we experienced in 2020. The occurrence of any or all of these events may adversely affect our ability to fund our operations and financing or contractual commitments.

Any future borrowings may affect our credit ratings and are subject to various debt covenants. At December 31, 2023, we were in compliance with the covenants for our debt and credit facilities. The most restrictive covenants include a limitation on mortgage debt and sale and leaseback transactions as a percentage of consolidated net tangible assets (as defined in the credit agreements) and a limitation on consolidated debt as a percentage of total capital (as defined in the credit agreements). When considering debt covenants, we continue to have substantial borrowing capacity.

Pension and Other Postretirement Benefits Pension cash requirements are based on an estimate of our minimum funding requirements, pursuant to Employee Retirement Income Security Act (ERISA) regulations, although we may make additional discretionary contributions. Estimates of other postretirement benefits are based on both our estimated future benefit payments and the estimated contributions to plans that are funded through trusts.

At December 31, 2023 and 2022, our pension plans were $5.4 billion and $5.3 billion underfunded as measured under U.S. generally accepted accounting principles (GAAP). On an ERISA basis, our plans are more than 100% funded at December 31, 2023. We do not expect to make significant contributions to our pension plans in 2024. We may be required to make higher contributions to our pension plans in future years.

For the foreseeable future, we are using common stock in lieu of cash to fund Company contributions to our 401(k) plans.

Purchase Obligations Purchase obligations represent contractual agreements to purchase goods or services that are legally binding; specify a fixed, minimum or range of quantities; specify a fixed, minimum, variable or indexed price provision; and specify approximate timing of the transaction. Purchase obligations include amounts recorded as well as amounts that are not recorded on the Consolidated Statements of Financial Position.

Purchase obligations not recorded on the Consolidated Statements of Financial Position include agreements for inventory procurement, tooling costs, electricity and natural gas contracts, property, plant and equipment, information technology software and hardware, and other miscellaneous production related obligations. The most significant obligation relates to inventory procurement contracts. We have entered into certain significant inventory procurement contracts that specify determinable prices and quantities, and long-term delivery timeframes. In addition, we purchase raw materials on behalf of our suppliers. These agreements require suppliers and vendors to be prepared to build and deliver items in sufficient time to meet our production schedules. The need for such arrangements with suppliers and vendors arises from the extended production planning horizon for many of our products. A significant portion of these inventory commitments is supported by firm contracts with customers and/or has historically resulted in settlement through reimbursement from customers for penalty payments to the supplier should the customer not take delivery. These amounts are also included in our forecasts of costs for program and contract accounting. Some inventory procurement contracts may include escalation adjustments. In these limited cases, we have included our best estimate of the effect of the escalation adjustment in the amounts disclosed in the table above.

Purchase obligations recorded on the Consolidated Statements of Financial Position primarily include accounts payable and certain other current and long-term liabilities, including accrued compensation.

We have entered into various industrial participation agreements with certain customers outside of the U.S. to facilitate economic flow back and/or technology or skills transfer to their businesses or government agencies as the result of their procurement of goods and/or services from us. These commitments may be satisfied by our local operations there, placement of direct work or vendor orders for supplies, opportunities to bid on supply contracts, transfer of technology or other forms of assistance. However, in some instances, our commitments may be satisfied through other parties (such as our vendors) who purchase supplies from our non-U.S. customers. In certain cases, penalties could be imposed if we do not meet our industrial participation commitments. During 2023, we incurred no such penalties. As of December 31, 2023, we had outstanding industrial participation agreements totaling $24.5 billion that extend through 2034. Purchase order commitments associated with industrial participation agreements are included in purchase obligations. To be eligible for such a purchase order commitment from us, a non-U.S. supplier must have sufficient capability to meet our requirements and must be competitive in cost, quality and schedule.

Off-Balance Sheet Arrangements We are a party to certain off-balance sheet arrangements including certain guarantees. For discussion of these arrangements, see Note 14 to our Consolidated Financial Statements.

Commercial Commitments

The following table summarizes our commercial commitments outstanding as of December 31, 2023.

(Dollars in millions)	Total Amounts Committed/Maximum Amount of Loss	Less than 1 year	1-3 years	4-5 years	After 5 years
Standby letters of credit and surety bonds	$4,548	$3,150	$1,210	$25	$163
Commercial aircraft financing commitments	17,003	1,946	7,927	4,062	3,068
Total commercial commitments	$21,551	$5,096	$9,137	$4,087	$3,231

Commercial aircraft financing commitments include commitments to provide financing related to aircraft on order, under option for deliveries or proposed as part of sales campaigns or refinancing with respect to delivered aircraft, based on estimated earliest potential funding dates. Customer financing commitments totaled $17.0 billion and $16.1 billion at December 31, 2023 and 2022. The increase relates to new financing commitments. We anticipate that we will not be required to fund a significant portion of our financing commitments as we continue to work with third party financiers to provide alternative financing to customers. Historically, we have not been required to fund significant amounts of outstanding commitments. However, there can be no assurances that we will not be required to fund greater amounts than historically required. See Note 13 to our Consolidated Financial Statements.

Contingent Obligations

We have significant contingent obligations that arise in the ordinary course of business, which include the following:

Legal Various legal proceedings, claims and investigations are pending against us. Legal contingencies are discussed in Note 21 to our Consolidated Financial Statements.

Environmental Remediation We are involved with various environmental remediation activities and have recorded a liability of $844 million at December 31, 2023. For additional information, see Note 13 to our Consolidated Financial Statements.

Non-GAAP Measures

Core Operating Earnings/(Loss), Core Operating Margin and Core Earnings/(Loss) Per Share

Our Consolidated Financial Statements are prepared in accordance with GAAP which we supplement with certain non-GAAP financial information. These non-GAAP measures should not be considered in isolation or as a substitute for the related GAAP measures, and other companies may define such measures differently. We encourage investors to review our financial statements and publicly-filed reports in their entirety and not to rely on any single financial measure. Core operating earnings/(loss), core operating margin and core earnings/(loss) per share exclude the FAS/CAS service cost adjustment. The FAS/CAS service cost adjustment represents the difference between the Financial Accounting Standards (FAS) pension and postretirement service costs calculated under GAAP and costs allocated to the business segments. Core earnings/(loss) per share excludes both the FAS/CAS service cost adjustment and non-operating pension and postretirement expenses. Non-operating pension and postretirement expenses represent the components of net periodic benefit costs other than service cost. Pension costs, comprising service and prior service costs computed in accordance with GAAP are allocated to BCA and certain BGS businesses supporting commercial customers. Pension costs allocated to BDS and BGS businesses supporting government customers are computed in accordance with U.S. Government Cost Accounting Standards (CAS), which employ different actuarial

assumptions and accounting conventions than GAAP. CAS costs are allocable to government contracts. Other postretirement benefit costs are allocated to all business segments based on CAS, which is generally based on benefits paid.

The Pension FAS/CAS service cost adjustments recognized in Loss from operations were benefits of $799 million in 2023, $849 million in 2022 and $882 million in 2021. The lower benefits in 2023 and 2022 were primarily due to reductions in allocated pension cost year over year. The non-operating pension expense included in Other income, net was a benefit of $529 million in 2023, $881 million in 2022 and $528 million in 2021. The lower benefits in 2023 were primarily due to higher interest cost and lower expected return on plan assets, offset by lower amortization of net actuarial losses. The higher benefits in 2022 were primarily due to lower amortization of net actuarial losses and a settlement loss that was recorded in 2021. For further discussion of pension and other postretirement costs, see the Management's Discussion and Analysis on page 26 of this Form 10-K and see Note 22 to our Consolidated Financial Statements.

Management uses core operating earnings/(loss), core operating margin and core earnings/(loss) per share for purposes of evaluating and forecasting underlying business performance. Management believes these core earnings measures provide investors additional insights into operational performance as unallocated pension and other postretirement benefit cost primarily represent costs driven by market factors and costs not allocable to U.S. government contracts.

Reconciliation of Non-GAAP Measures to GAAP Measures

The table below reconciles the non-GAAP financial measures of core operating earnings/(loss), core operating margins and core earnings/(loss) per share with the most directly comparable GAAP financial measures of Earnings/(loss) from operations, operating margins and Diluted earnings/(loss) per share.

(Dollars in millions, except per share data)

Years ended December 31,	**2023**	2022	2021
Revenues	**$77,794**	$66,608	$62,286
Loss from operations, as reported	**($773)**	($3,519)	($2,870)
Operating margins	**(1.0)%**	(5.3)%	(4.6)%
Pension FAS/CAS service cost adjustment [1]	**($799)**	($849)	($882)
Postretirement FAS/CAS service cost adjustment [1]	**(257)**	(294)	(291)
FAS/CAS service cost adjustment [1]	**($1,056)**	($1,143)	($1,173)
Core operating loss (non-GAAP)	**($1,829)**	($4,662)	($4,043)
Core operating margins (non-GAAP)	**(2.4)%**	(7.0)%	(6.5)%
Diluted loss per share, as reported	**($3.67)**	($8.30)	($7.15)
Pension FAS/CAS service cost adjustment [1]	**(1.32)**	(1.43)	(1.50)
Postretirement FAS/CAS service cost adjustment [1]	**(0.42)**	(0.49)	(0.49)
Non-operating pension expense [2]	**(0.87)**	(1.47)	(0.91)
Non-operating postretirement expense [2]	**(0.10)**	(0.10)	
Provision for deferred income taxes on adjustments [3]	**0.57**	0.73	0.61
Core loss per share (non-GAAP)	**($5.81)**	($11.06)	($9.44)
Weighted average diluted shares (in millions)	**606.1**	595.2	588.0

[1] FAS/CAS service cost adjustment represents the difference between the FAS pension and postretirement service costs calculated under GAAP and costs allocated to the business segments. This adjustment is excluded from Core operating loss (non-GAAP)).

[2] Non-operating pension and postretirement expenses represent the components of net periodic benefit costs other than service cost. These expenses are included in Other income, net and are excluded from Core loss per share (non-GAAP).

[3] The income tax impact is calculated using the U.S. corporate statutory tax rate.

Critical Accounting Estimates

Accounting for Long-term Contracts

Substantially all contracts at BDS and certain contracts at BGS are long-term contracts. Our long-term contracts typically represent a single distinct performance obligation due to the highly interdependent and interrelated nature of the underlying goods and/or services and the significant service of integration that we provide.

Accounting for long-term contracts involves a judgmental process of estimating the total revenue, costs, and profit for each performance obligation. Cost of sales is recognized as incurred, and revenue is determined by adding a proportionate amount of the estimated profit to the amount reported as cost of sales.

Due to the size, duration and nature of many of our long-term contracts, the estimation of total revenues and costs through completion is complicated and subject to many variables. Total revenue estimates are based on negotiated contract prices and quantities, modified by our assumptions regarding contract options, change orders, incentive and award fee provisions associated with technical performance, and price adjustment clauses (such as inflation or index-based clauses). The majority of these long-term contracts are with the U.S. government where the price is generally based on the estimated cost to produce the product or service plus profit. Federal Acquisition Regulations provide guidance on the types of cost that will be reimbursed in establishing contract price. Total cost estimates are largely based on negotiated or estimated purchase contract terms, historical performance trends, business base and other economic projections. Factors that influence these estimates include inflationary trends, technical and schedule risk, internal and supplier performance trends, production quality, labor instability, global supply chain constraints, business volume assumptions, asset utilization, and anticipated labor agreements.

Revenue and cost estimates for all significant long-term contract performance obligations are reviewed and reassessed quarterly. Changes in these estimates could result in recognition of cumulative catch-up adjustments to the contract's inception to date revenues, cost of sales and profit in the period in which such changes are made. Changes in revenue and cost estimates could also result in a reach-forward loss or an adjustment to a reach-forward loss which would be recorded immediately in earnings. Net cumulative catch-up adjustments for changes in estimated revenues and costs at completion across all long-term contracts, including the impact of estimated losses on unexercised options, increased Loss from operations by $2,943 million, $5,253 million and $880 million in 2023, 2022 and 2021, respectively, and were primarily due to losses recognized on the VC-25B, KC-46A Tanker, Commercial Crew, T-7A Red Hawk and MQ-25 programs. These are all fixed-price development programs, and there is ongoing risk that similar losses may have to be recognized in future periods on these and/or other programs.

Due to the significance of judgment in the estimation process described above, it is likely that materially different earnings could be recorded if we used different assumptions or if the underlying circumstances were to change. Changes in underlying assumptions/estimates, internal and supplier performance, inflationary trends, or other circumstances may adversely or positively affect financial performance in future periods. If the combined gross margins for our profitable long-term contracts had been estimated to be higher or lower by 1% during 2023, it would have increased or decreased pre-tax income for the year by approximately $300 million.

Program Accounting

Program accounting requires the demonstrated ability to reliably estimate revenues, costs and gross profit margin for the defined program accounting quantity. A program consists of the estimated number of units (accounting quantity) of a product to be produced in a continuing, long-term production effort for

delivery under existing and anticipated contracts. The determination of the accounting quantity is limited by the ability to make reasonably dependable estimates.

Factors that must be estimated include program accounting quantity, sales price, production rates, labor and employee benefit costs, material costs, procured part costs, major component costs, overhead costs, program tooling and other non-recurring costs, and warranty costs. Estimation of the accounting quantity for each program takes into account several factors that are indicative of the demand for the particular program, such as firm orders, letters of intent from prospective customers and market studies. Total estimated program sales are determined by estimating the model mix and sales price for all unsold units within the accounting quantity, added together with the sales prices for all undelivered units under contract. The sales prices for all undelivered units within the accounting quantity include an escalation adjustment for inflation that is updated quarterly. Cost estimates are based largely on negotiated and anticipated contracts with suppliers, historical performance trends, and business base and other economic projections. Factors that influence these estimates include production rates, internal and supplier performance trends, production quality, labor instability, global supply chain constraints, learning curve, change incorporation, rework or safety enhancements, regulatory requirements, flight test and certification requirements and schedules, performance or reliability issues involving completed aircraft, customer and/or supplier claims or assertions, asset utilization, anticipated labor agreements, and inflationary or deflationary trends.

The introduction of new aircraft and derivatives, such as the 777X, 737-7 and 737-10, involves increased risks associated with meeting development, certification, and production schedules. These challenges include significant global regulatory scrutiny of all development aircraft. As a result, our ability to deliver aircraft on time, satisfy performance and reliability standards and achieve or maintain, as applicable, program profitability is subject to significant risks.

To ensure reliability in our estimates, we employ a rigorous estimating process that is reviewed and updated on a quarterly basis. This includes reassessing the accounting quantity. Changes in estimates of program gross profit margins are normally recognized on a prospective basis; however, when estimated costs to complete a program plus costs already included in inventory exceed estimated revenues from the program, a loss is recorded in the current period. Reductions to the estimated loss are included in the gross profit margin for undelivered units in the accounting quantity whereas increases to the estimated loss are recorded as an earnings charge in the period in which the loss is determined.

The 767, 777X, and 787 programs had near break-even or single digit margins at December 31, 2023. Adverse changes to the revenue and/or cost estimates for these programs could result in earnings charges in future periods.

Due to the significance of judgment in the estimation process described above, it is reasonably possible that changes in underlying circumstances or assumptions could have a material effect on program gross margins. If the combined gross margin percentages for our commercial airplane programs had been estimated to be 1% higher or lower it would have an approximately $330 million impact on operating earnings for the year ended December 31, 2023.

Pension Plans

Many of our employees have earned benefits under defined benefit pension plans. The majority of employees that had participated in defined benefit pension plans have transitioned to a company-funded defined contribution retirement savings plan. Accounting rules require an annual measurement of our projected obligation and plan assets. These measurements are based upon several assumptions. Future changes in assumptions or differences between actual and expected outcomes can significantly affect our future annual expense, projected benefit obligation and Shareholders' equity.

The projected benefit obligation is sensitive to discount rates. The projected benefit obligation would decrease by $1,280 million or increase by $1,425 million if the discount rate increased or decreased by 25 basis points. A 25 basis point change in the discount rate would not have a significant impact on pension cost. However, net periodic pension cost is sensitive to changes in the expected long-term rate of asset return. A decrease or increase of 25 basis points in the expected long-term rate of asset return would have increased or decreased 2023 net periodic pension cost by $150 million. See Note 16 of the Notes to our Consolidated Financial Statements, which includes the discount rate and expected long-term rate of asset return assumptions for the last three years.

Deferred Income Taxes – Valuation Allowance

The Company had deferred income tax assets of $14,743 million at December 31, 2023 that can be used in future years to offset taxable income and reduce income taxes payable. The Company had deferred income tax liabilities of $10,363 million at December 31, 2023 that will partially offset deferred income tax assets and result in higher taxable income in future years and increase income taxes payable. Tax law determines whether future reversals of temporary differences will result in taxable and deductible amounts that offset each other in future years. The particular years in which temporary differences result in taxable or deductible amounts generally are determined by the timing of the recovery of the related asset or settlement of the related liability.

On a quarterly basis, we assess the likelihood that we will be able to recover our deferred tax assets against future sources of taxable income and reduce the carrying amounts of deferred tax assets by recording a valuation allowance if, based on the available evidence, it is more likely than not (defined as a likelihood of more than 50%) that all or a portion of such assets will not be realized.

This assessment takes into account both positive and negative evidence. A recent history of financial reporting losses is heavily weighted as a source of objectively verifiable negative evidence. Due to our recent history of losses, we determined we could not include future projected earnings in our analysis. Rather, we use systematic and logical methods to estimate when deferred tax liabilities will reverse and generate taxable income and when deferred tax assets will reverse and generate tax deductions. The selection of methodologies and assessment of when temporary differences will result in taxable or deductible amounts involves significant management judgment and is inherently complex and subjective. We believe that the methodologies we use are reasonable and can be replicated on a consistent basis in future periods.

Deferred tax liabilities represent the assumed source of future taxable income and the majority are assumed to generate taxable amounts during the next five years. Deferred tax assets include amounts related to pension and other postretirement benefits that are assumed to generate significant deductible amounts beyond five years. The Company's valuation allowance of $4,550 million at December 31, 2023 primarily relates to pension and other postretirement benefit obligation deferred tax assets, tax net operating losses, tax credits and interest carryforwards that are assumed to reverse beyond the period in which reversals of deferred tax liabilities are assumed to occur. During 2023, the Company increased the valuation allowance by $1,388 million primarily due to tax credits and other carryforwards generated in 2023 that cannot be realized in 2023. Until the Company generates sustained levels of profitability, additional valuation allowances may have to be recorded with corresponding adverse impacts on earnings and/or other comprehensive income.

For additional information regarding income taxes, see Note 4 of the Notes to the Consolidated Financial Statements.

Item 7A. Quantitative and Qualitative Disclosures About Market Risk

Interest Rate Risk

We have financial instruments that are subject to interest rate risk, principally fixed-rate debt obligations. The investors in our fixed-rate debt obligations do not generally have the right to demand we pay off these obligations prior to maturity. Therefore, exposure to interest rate risk is not believed to be material for our fixed-rate debt.

Foreign Currency Exchange Rate Risk

We are subject to foreign currency exchange rate risk relating to receipts from customers and payments to suppliers in foreign currencies. We use foreign currency forward contracts to hedge the price risk associated with firmly committed and forecasted foreign denominated payments and receipts related to our ongoing business. Foreign currency forward contracts are sensitive to changes in foreign currency exchange rates. At December 31, 2023, a 10% increase or decrease in the exchange rate in our portfolio of foreign currency contracts would have increased or decreased our unrealized losses by $361 million. Consistent with the use of these contracts to neutralize the effect of exchange rate fluctuations, such unrealized losses or gains would be offset by corresponding gains or losses, respectively, in the remeasurement of the underlying transactions being hedged. When taken together, these forward currency contracts and the offsetting underlying commitments do not create material market risk.

Commodity Price Risk

We are subject to commodity price risk relating to commodity purchase contracts for items used in production that are subject to changes in the market price. We use commodity swaps and commodity purchase contracts to hedge against these potentially unfavorable price changes. Our commodity purchase contracts and derivatives are both sensitive to changes in the market price. At December 31, 2023, a 10% increase or decrease in the market price in our commodity derivatives would have increased or decreased our unrealized losses by $37 million. Consistent with the use of these contracts to neutralize the effect of market price fluctuations, such unrealized losses or gains would be offset by corresponding gains or losses, respectively, in the remeasurement of the underlying transactions being hedged. When taken together, these commodity purchase contracts and the offsetting swaps do not create material market risk.

Market Risk

Participants in deferred compensation plans can diversify the deferred amounts among investment funds which are subject to potential changes in fair value. As of December 31, 2023, the deferred compensation liability, which is being marked to market, was $1.6 billion. A 10% change in the fair value of these investment funds would increase or decrease the liability by $164 million. Changes in the liability are recorded in operating earnings.

Item 8. Financial Statements and Supplementary Data

Index to the Consolidated Financial Statements

	Page
Consolidated Statements of Operations	51
Consolidated Statements of Comprehensive Income	52
Consolidated Statements of Financial Position	53
Consolidated Statements of Cash Flows	54
Consolidated Statements of Equity	55
Summary of Business Segment Data	56
Note 1 - Summary of Significant Accounting Policies	57
Note 2 - Goodwill and Acquired Intangibles	68
Note 3 - Earnings Per Share	68
Note 4 - Income Taxes	70
Note 5 - Accounts Receivable	73
Note 6 - Allowance for Losses on Financial Assets	73
Note 7 - Inventories	74
Note 8 - Contracts with Customers	74
Note 9 - Financing Receivables and Operating Lease Equipment	75
Note 10 - Property, Plant and Equipment	77
Note 11 - Investments	78
Note 12 - Leases	78
Note 13 - Liabilities, Commitments and Contingencies	80
Note 14 - Arrangements with Off-Balance Sheet Risk	85
Note 15 - Debt	86
Note 16 - Postretirement Plans	87
Note 17 - Share-Based Compensation and Other Compensation Arrangements	95
Note 18 - Shareholders' Equity	99
Note 19 - Derivative Financial Instruments	100
Note 20 - Fair Value Measurements	102
Note 21 - Legal Proceedings	104
Note 22 - Segment and Revenue Information	104
Note 23 - Subsequent Events	109
Reports of Independent Registered Public Accounting Firm	111

The Boeing Company and Subsidiaries
Consolidated Statements of Operations

(Dollars in millions, except per share data)

Years ended December 31,	**2023**	2022	2021
Sales of products	**$65,581**	$55,893	$51,386
Sales of services	**12,213**	10,715	10,900
Total revenues	**77,794**	66,608	62,286
Cost of products	**(59,864)**	(53,969)	(49,954)
Cost of services	**(10,206)**	(9,109)	(9,283)
Total costs and expenses	**(70,070)**	(63,078)	(59,237)
	7,724	3,530	3,049
Income/(loss) from operating investments, net	**46**	(16)	210
General and administrative expense	**(5,168)**	(4,187)	(4,157)
Research and development expense, net	**(3,377)**	(2,852)	(2,249)
Gain on dispositions, net	**2**	6	277
Loss from operations	**(773)**	(3,519)	(2,870)
Other income, net	**1,227**	1,058	551
Interest and debt expense	**(2,459)**	(2,561)	(2,714)
Loss before income taxes	**(2,005)**	(5,022)	(5,033)
Income tax (expense)/benefit	**(237)**	(31)	743
Net loss	**(2,242)**	(5,053)	(4,290)
Less: net loss attributable to noncontrolling interest	**(20)**	(118)	(88)
Net loss attributable to Boeing Shareholders	**($2,222)**	($4,935)	($4,202)
Basic loss per share	**($3.67)**	($8.30)	($7.15)
Diluted loss per share	**($3.67)**	($8.30)	($7.15)

See Notes to the Consolidated Financial Statements on pages 57 - 110.

The Boeing Company and Subsidiaries
Consolidated Statements of Comprehensive Income

(Dollars in millions)

Years ended December 31,	**2023**	2022	2021
Net loss	**($2,242)**	($5,053)	($4,290)
Other comprehensive (loss)/income, net of tax:			
Currency translation adjustments	**33**	(62)	(75)
Unrealized gain/(loss) on certain investments, net of tax of $0, $0 and $0	**2**	(1)	
Derivative instruments:			
Unrealized gain/(loss) arising during period, net of tax of ($11), $12 and ($16)	**41**	(40)	55
Reclassification adjustment for (gain)/loss included in net loss, net of tax of $1, ($3) and $2	**(5)**	10	(6)
Total unrealized gain/(loss) on derivative instruments, net of tax	**36**	(30)	49
Defined benefit pension plans & other postretirement benefits:			
Net actuarial (loss)/gain arising during the period, net of tax of $13, ($22) and ($32)	**(722)**	1,533	4,262
Amortization of actuarial (gain)/loss included in net periodic pension cost, net of tax of $0, ($11) and ($8)	**(2)**	791	1,155
Settlement (gain)/loss included in net periodic cost, net of tax of $0, $0 and ($2)		(4)	191
Amortization of prior service credits included in net periodic pension cost, net of tax of $1, $2 and $1	**(102)**	(114)	(114)
Prior service credit arising during the period, net of tax of $0, $0 and $0		(1)	
Pension and postretirement (cost)/benefit related to our equity method investments, net of tax of $0, $0 and ($2)		(3)	6
Total defined benefit pension plans & other postretirement benefits, net of tax	**(826)**	2,202	5,500
Other comprehensive (loss)/income, net of tax	**(755)**	2,109	5,474
Comprehensive (loss)/income, net of tax	**(2,997)**	(2,944)	1,184
Less: Comprehensive loss related to noncontrolling interest	**(20)**	(118)	(88)
Comprehensive (loss)/income attributable to Boeing Shareholders, net of tax	**($2,977)**	($2,826)	$1,272

See Notes to the Consolidated Financial Statements on pages 57 - 110.

The Boeing Company and Subsidiaries
Consolidated Statements of Financial Position

(Dollars in millions, except per share data)

December 31,	**2023**	2022
Assets		
Cash and cash equivalents	**$12,691**	$14,614
Short-term and other investments	**3,274**	2,606
Accounts receivable, net	**2,649**	2,517
Unbilled receivables, net	**8,317**	8,634
Current portion of financing receivables, net	**99**	154
Inventories	**79,741**	78,151
Other current assets, net	**2,504**	2,847
Total current assets	**109,275**	109,523
Financing receivables and operating lease equipment, net	**860**	1,450
Property, plant and equipment, net	**10,661**	10,550
Goodwill	**8,093**	8,057
Acquired intangible assets, net	**2,094**	2,311
Deferred income taxes	**59**	63
Investments	**1,035**	983
Other assets, net of accumulated amortization of $ 1,046 and $949	**4,935**	4,163
Total assets	**$137,012**	$137,100
Liabilities and equity		
Accounts payable	**$11,964**	$10,200
Accrued liabilities	**22,331**	21,581
Advances and progress billings	**56,328**	53,081
Short-term debt and current portion of long-term debt	**5,204**	5,190
Total current liabilities	**95,827**	90,052
Deferred income taxes	**229**	230
Accrued retiree health care	**2,233**	2,503
Accrued pension plan liability, net	**6,516**	6,141
Other long-term liabilities	**2,332**	2,211
Long-term debt	**47,103**	51,811
Total liabilities	**154,240**	152,948
Shareholders' equity:		
Common stock, par value $ 5.00 – 1,200,000,000 shares authorized; 1,012,261,159 shares issued	**5,061**	5,061
Additional paid-in capital	**10,309**	9,947
Treasury stock, at cost	**(49,549)**	(50,814)
Retained earnings	**27,251**	29,473
Accumulated other comprehensive loss	**(10,305)**	(9,550)
Total shareholders' deficit	**(17,233)**	(15,883)
Noncontrolling interests	**5**	35
Total equity	**(17,228)**	(15,848)
Total liabilities and equity	**$137,012**	$137,100

See Notes to the Consolidated Financial Statements on pages 57 - 110.

The Boeing Company and Subsidiaries
Consolidated Statements of Cash Flows

(Dollars in millions)

Years ended December 31,	**2023**	2022	2021
Cash flows – operating activities:			
Net loss	**($2,242)**	($5,053)	($4,290)
Adjustments to reconcile net loss to net cash provided/(used) by operating activities:			
Non-cash items –			
Share-based plans expense	**690**	725	833
Treasury shares issued for 401(k) contribution	**1,515**	1,215	1,233
Depreciation and amortization	**1,861**	1,979	2,144
Investment/asset impairment charges, net	**46**	112	98
Gain on dispositions, net	**(2)**	(6)	(277)
787 reach-forward loss			3,460
Other charges and credits, net	**3**	401	360
Changes in assets and liabilities –			
Accounts receivable	**(128)**	142	(713)
Unbilled receivables	**321**	6	(586)
Advances and progress billings	**3,365**	108	2,505
Inventories	**(1,681)**	420	(1,127)
Other current assets	**389**	(591)	345
Accounts payable	**1,672**	838	(3,783)
Accrued liabilities	**779**	2,956	(3,687)
Income taxes receivable, payable and deferred	**44**	1,347	733
Other long-term liabilities	**(313)**	(158)	(206)
Pension and other postretirement plans	**(1,049)**	(1,378)	(972)
Financing receivables and operating lease equipment, net	**571**	142	210
Other	**119**	307	304
Net cash provided/(used) by operating activities	**5,960**	3,512	(3,416)
Cash flows – investing activities:			
Payments to acquire property, plant and equipment	**(1,527)**	(1,222)	(980)
Proceeds from disposals of property, plant and equipment	**27**	35	529
Acquisitions, net of cash acquired	**(70)**		(6)
Contributions to investments	**(16,448)**	(5,051)	(35,713)
Proceeds from investments	**15,739**	10,619	45,489
Other	**(158)**	(11)	5
Net cash (used)/provided by investing activities	**(2,437)**	4,370	9,324
Cash flows – financing activities:			
New borrowings	**75**	34	9,795
Debt repayments	**(5,216)**	(1,310)	(15,371)
Stock options exercised	**45**	50	42
Employee taxes on certain share-based payment arrangements	**(408)**	(40)	(66)
Other	**17**		
Net cash used by financing activities	**(5,487)**	(1,266)	(5,600)
Effect of exchange rate changes on cash and cash equivalents	**30**	(73)	(39)
Net (decrease)/increase in cash & cash equivalents, including restricted	**(1,934)**	6,543	269
Cash & cash equivalents, including restricted, at beginning of year	**14,647**	8,104	7,835
Cash & cash equivalents, including restricted, at end of year	**12,713**	14,647	8,104
Less restricted cash & cash equivalents, included in Investments	**22**	33	52
Cash and cash equivalents at end of year	**$12,691**	$14,614	$8,052

See Notes to the Consolidated Financial Statements on pages 57 - 110.

The Boeing Company and Subsidiaries
Consolidated Statements of Equity

	Boeing shareholders						
(Dollars in millions, except per share data)	Common Stock	Additional Paid-In Capital	Treasury Stock	Retained Earnings	Accumulated Other Comprehensive Loss	Non-controlling Interests	Total
Balance at January 1, 2021	$5,061	$7,787	($52,641)	$38,610	($17,133)	$241	($18,075)
Net loss				(4,202)		(88)	(4,290)
Other comprehensive income, net of tax of ($57)					5,474		5,474
Share-based compensation		833					833
Treasury shares issued for stock options exercised, net		(28)	70				42
Treasury shares issued for other share-based plans, net		(98)	35				(63)
Treasury shares issued for 401(k) contribution		558	675				1,233
Balance at December 31, 2021	$5,061	$9,052	($51,861)	$34,408	($11,659)	$153	($14,846)
Net loss				(4,935)		(118)	(5,053)
Other comprehensive income, net of tax of ($22)					2,109		2,109
Share-based compensation		725					725
Treasury shares issued for stock options exercised, net		(31)	81				50
Treasury shares issued for other share-based plans, net		(94)	46				(48)
Treasury shares issued for 401(k) contribution		295	920				1,215
Balance at December 31, 2022	**$5,061**	**$9,947**	**($50,814)**	**$29,473**	**($9,550)**	**$35**	**($15,848)**
Net loss				**(2,222)**		**(20)**	**(2,242)**
Other comprehensive loss, net of tax of $4					**(755)**		**(755)**
Share-based compensation		**690**					**690**
Treasury shares issued for stock options exercised, net		**(28)**	**73**				**45**
Treasury shares issued for other share-based plans, net		**(660)**	**304**				**(356)**
Treasury shares issued for 401(k) contribution		**627**	**888**				**1,515**
Subsidiary shares purchased from noncontrolling interests		**(267)**					**(267)**
Other changes in noncontrolling interests						**(10)**	**(10)**
Balance at December 31, 2023	**$5,061**	**$10,309**	**($49,549)**	**$27,251**	**($10,305)**	**$5**	**($17,228)**

See Notes to the Consolidated Financial Statements on pages 57 - 110.

The Boeing Company and Subsidiaries
Notes to the Consolidated Financial Statements
Summary of Business Segment Data

(Dollars in millions)

Years ended December 31,	**2023**	2022	2021
Revenues:			
Commercial Airplanes	**$33,901**	$26,026	$19,714
Defense, Space & Security	**24,933**	23,162	26,540
Global Services	**19,127**	17,611	16,328
Unallocated items, eliminations and other	**(167)**	(191)	(296)
Total revenues	**$77,794**	$66,608	$62,286
Loss from operations:			
Commercial Airplanes	**($1,635)**	($2,341)	($6,377)
Defense, Space & Security	**(1,764)**	(3,544)	1,544
Global Services	**3,329**	2,727	2,017
Segment operating loss	**(70)**	(3,158)	(2,816)
Unallocated items, eliminations and other	**(1,759)**	(1,504)	(1,227)
FAS/CAS service cost adjustment	**1,056**	1,143	1,173
Loss from operations	**(773)**	(3,519)	(2,870)
Other income, net	**1,227**	1,058	551
Interest and debt expense	**(2,459)**	(2,561)	(2,714)
Loss before income taxes	**(2,005)**	(5,022)	(5,033)
Income tax (expense)/benefit	**(237)**	(31)	743
Net loss	**(2,242)**	(5,053)	(4,290)
Less: net loss attributable to noncontrolling interest	**(20)**	(118)	(88)
Net loss attributable to Boeing Shareholders	**($2,222)**	($4,935)	($4,202)

This information is an integral part of the Notes to the Consolidated Financial Statements. See Note 22 for further segment results.

The Boeing Company and Subsidiaries
Notes to the Consolidated Financial Statements
Years ended December 31, 2023, 2022 and 2021
(Dollars in millions, except otherwise stated)

Note 1 – Summary of Significant Accounting Policies

Principles of Consolidation and Basis of Presentation

The Consolidated Financial Statements included in this report have been prepared by management of The Boeing Company (herein referred to as "Boeing," the "Company," "we," "us" or "our"). These statements include the accounts of all majority-owned subsidiaries and variable interest entities that are required to be consolidated. All significant intercompany accounts and transactions have been eliminated. As described in Note 22, we now operate in three reportable segments: Commercial Airplanes (BCA), Defense, Space & Security (BDS), and Global Services (BGS). As a result, prior period amounts have been reclassified to conform to current period presentation.

Use of Estimates

The preparation of financial statements in conformity with U.S. generally accepted accounting principles (GAAP) requires management to make estimates and assumptions that affect the reported amounts of assets and liabilities and disclosure of contingent assets and liabilities at the date of the financial statements and the reported amounts of revenues and expenses during the reporting period. Actual results could differ from those estimates.

Operating Cycle

For classification of certain current assets and liabilities, we use the duration of the related contract or program as our operating cycle, which is generally longer than one year.

Revenue and Related Cost Recognition

Commercial aircraft contracts The majority of our BCA segment revenue is derived from commercial aircraft contracts. For each contract, we determine the transaction price based on the consideration expected to be received. We allocate the transaction price to each commercial aircraft performance obligation based on relative standalone selling prices adjusted by an escalation formula as specified in the customer agreement. Revenue is recognized for each commercial aircraft performance obligation at the point in time when the aircraft is completed and accepted by the customer. We use program accounting to determine the amount reported as cost of sales.

Payments for commercial aircraft sales are received in accordance with the customer agreement, which generally includes a deposit upon order and additional payments in accordance with a payment schedule, with the balance being due immediately prior to or at aircraft delivery. Advances and progress billings (contract liabilities) are normal and customary for commercial aircraft contracts and not considered a significant financing component as they are intended to protect us from the other party failing to adequately complete some or all of its obligations under the contract.

Long-term contracts Substantially all contracts at BDS and certain contracts at BGS are long-term contracts with the U.S. government and other customers that generally extend over several years. Products sales under long-term contracts primarily include fighter jets, rotorcraft, cybersecurity products, surveillance suites, advanced weapons, missile defense, military derivative aircraft, satellite systems and modification of commercial passenger aircraft to cargo freighters. Sales of services under long-term contracts primarily include support and maintenance agreements associated with our commercial and defense products and space travel on Commercial Crew.

For each long-term contract, we determine the transaction price based on the consideration expected to be received. We allocate the transaction price to each distinct performance obligation to deliver a good or service, or a collection of goods and/or services, based on the relative standalone selling prices. A long-term contract will typically represent a single distinct performance obligation due to the highly interdependent and interrelated nature of the underlying goods and/or services and the significant service of integration that we provide. While the scope and price on certain long-term contracts may be modified over their life, the transaction price is based on current rights and obligations under the contract and does not include potential modifications until they are agreed upon with the customer. When applicable, a cumulative adjustment or separate recognition for the additional scope and price may result. Long-term contracts can be negotiated with a fixed price or a price in which we are reimbursed for costs incurred plus an agreed upon profit. The Federal Acquisition Regulations provide guidance on the types of cost that will be reimbursed in establishing the price for contracts with the U.S. government. Certain long-term contracts include in the transaction price variable consideration, such as incentive and award fees, if specified targets are achieved. The amount included in the transaction price represents the expected value, based on a weighted probability, or the most likely amount.

Long-term contract revenue is recognized over the contract term (over time) as the work progresses, either as products are produced or as services are rendered. We generally recognize revenue over time as we perform on long-term contracts because of continuous transfer of control to the customer. For U.S. government contracts, this continuous transfer of control to the customer is supported by clauses in the contract that allow the customer to unilaterally terminate the contract for convenience, pay us for costs incurred plus a reasonable profit and take control of any work in process. Similarly, for non-U.S. government contracts, the customer typically controls the work in process as evidenced either by contractual termination clauses or by our rights to payment of the transaction price associated with work performed to date on products or services that do not have an alternative use to the Company.

The accounting for long-term contracts involves a judgmental process of estimating total revenues, costs and profit for each performance obligation. Cost of sales is recognized as incurred. The amount reported as revenues is determined by adding a proportionate amount of the estimated profit to the amount reported as cost of sales. Recognizing revenue as costs are incurred provides an objective measure of progress on the long-term contract and thereby best depicts the extent of transfer of control to the customer.

For long-term contracts for which revenue is recognized over time, changes in estimated revenues, cost of sales and the related effect on operating income are recognized using a cumulative catch-up adjustment which recognizes in the current period the cumulative effect of the changes on current and prior periods based on a long-term contract's percentage-of-completion. When the current estimates of total revenues and costs at completion for a long-term contract indicate a loss, a provision for the entire reach-forward loss on the long-term contract is recognized.

The table below reflects the impact of net cumulative catch-up adjustments for changes in estimated revenues and costs at completion across all long-term contracts including the impact to Loss from operations from estimated losses on unexercised options for the years ended December 31:

	2023	2022	2021
Decrease to Revenue	**($1,706)**	($2,335)	($379)
Increase to Loss from operations	**($2,943)**	($5,253)	($880)
Increase to Diluted loss per share	**($5.43)**	($8.88)	($1.28)

Significant adjustments during the three years ended December 31, 2023 included losses on VC-25B, KC-46A Tanker, MQ-25, Commercial Crew and T-7A Red Hawk programs in addition to lower earnings on F-15 and satellites.

Due to the significance of judgment in the estimation process, changes in underlying assumptions/estimates, internal and supplier performance, inflationary trends, or other circumstances may adversely or positively affect financial performance in future periods.

Payments under long-term contracts may be received before or after revenue is recognized. The U.S. government customer typically withholds payment of a small portion of the contract price until contract completion. Therefore, long-term contracts typically generate Unbilled receivables (contract assets) but may generate Advances and progress billings (contract liabilities). Long-term contract Unbilled receivables and Advances and progress billings are not considered a significant financing component because they are intended to protect either the customer or the Company in the event that some or all of the obligations under the contract are not completed.

Commercial spare parts contracts Certain contracts at our BGS segment include sales of commercial spare parts. For each contract, we determine the transaction price based on the consideration expected to be received. The spare parts have discrete unit prices that represent fair value. We generally consider each spare part to be a separate performance obligation. Revenue is recognized for each commercial spare part performance obligation at the point in time of delivery to the customer. We may provide our customers with a right to return a commercial spare part where a customer may receive a full or partial refund, a credit applied to amounts owed, a different product in exchange, or any combination of these items. We consider the potential for customer returns in the estimated transaction price. The amount reported as cost of sales is recorded at average cost. Payments for commercial spare parts sales are typically received shortly after delivery.

Other service revenue contracts Certain contracts at our BGS segment are for sales of services to commercial customers including maintenance, training, data analytics and information-based services. We recognize revenue for these service performance obligations over time as the services are rendered. The method of measuring progress (such as straight-line or billable amount) varies depending upon which method best depicts the transfer of control to the customer based on the type of service performed. Cost of sales is recorded as incurred.

Concession sharing arrangements We account for sales concessions to our customers in consideration of their purchase of products and services as a reduction of the transaction price and the revenue that is recognized for the related performance obligations. The sales concessions incurred may be partially reimbursed by certain suppliers in accordance with concession sharing arrangements. We record these reimbursements, which are presumed to represent reductions in the price of the vendor’s products or services, as a reduction in Cost of products.

Unbilled receivables and advances and progress billings Unbilled receivables (contract assets) arise when the Company recognizes revenue for amounts which cannot yet be billed under terms of the contract with the customer. Advances and progress billings (contract liabilities) arise when the Company receives payments from customers in advance of recognizing revenue. The amount of Unbilled receivables or Advances and progress billings is determined for each contract.

Financial services revenue We record financial services revenue associated with sales-type leases, operating leases and loans in Sales of services on the Consolidated Statements of Operations. For sales-type leases, we recognize selling profit or loss at lease inception if collection of the lease payments is probable. For sales-type leases, we record financing receivables at lease inception. A financing receivable is recorded at the aggregate of future lease payments, estimated residual value of the leased equipment, and any deferred incremental direct costs less unearned income. Income is recognized over the life of the lease to approximate a level rate of return on the net investment. For notes receivable, we record financing receivables net of any unamortized discounts and deferred incremental direct costs. Interest income and amortization of any discounts are recorded ratably over the related term of the note.

Income recognition is generally suspended for financing receivables that are uncollectible. We determine that a financing receivable is uncollectible when, based on current information and events, it is probable that we will be unable to collect amounts due according to the original contractual terms. We determine a financing receivable is past due when cash has not been received upon the due date specified in the contract.

We evaluate the collectability of financing receivables at commencement and on a recurring basis. If a financing receivable is determined to be uncollectible, the customer is categorized as non-accrual status. When a customer is in non-accrual status at commencement, sales-type lease revenue is deferred until substantially all cash has been received or the customer is removed from non-accrual status. If we have a note receivable with a customer that is in non-accrual status, or a sales-type lease with a customer that changes to non-accrual status after commencement, we recognize contractual interest income as payments are received to the extent there is sufficient collateral and payments exceed past due principal payments.

Residual values, which are reviewed periodically, represent the estimated amount we expect to receive at lease termination from the disposition of the leased equipment. Actual residual values realized could differ from these estimates. Declines in estimated residual value that are deemed other-than-temporary are recognized in the period in which the declines occur.

For operating leases, revenue on leased aircraft and equipment is recorded on a straight-line basis over the term of the lease. Operating lease assets, included in Financing receivables and operating lease equipment, net, are recorded at cost and depreciated to an estimated residual value using the straight-line method over the period that we project we will hold the asset. We periodically review our estimates of residual value and recognize forecasted changes by prospectively adjusting depreciation expense. We record assets held for sale at the lower of carrying value or fair value less costs to sell. We evaluate for impairment assets under operating leases when events or changes in circumstances indicate that the expected undiscounted cash flow from the asset may be less than the carrying value. When we determine that impairment is indicated for an asset, the amount of impairment expense recorded is the excess of the carrying value over the fair value of the asset.

Reinsurance revenue Our wholly-owned insurance subsidiary, Astro Ltd., participates in a reinsurance pool for workers' compensation. The member agreements and practices of the reinsurance pool minimize any participating members' individual risk. Reinsurance revenues were $163, $129 and $126 during 2023, 2022 and 2021, respectively. Reinsurance costs related to premiums and claims paid to the reinsurance pool were $181, $134 and $129 during 2023, 2022 and 2021, respectively. Revenues and costs are presented net in Cost of sales in the Consolidated Statements of Operations.

Research and Development

Research and development includes costs incurred for experimentation, design and testing, as well as bid and proposal efforts related to government products and services, which are expensed as incurred unless the costs are related to certain contractual arrangements with customers. Costs that are incurred pursuant to such contractual arrangements are recorded over the period that revenue is recognized, consistent with our long-term contract accounting policy. We have certain research and development arrangements that meet the requirement for best efforts research and development accounting. Accordingly, the amounts funded by the customer are recognized as an offset to our research and development expense rather than as contract revenues. Research and development expense, net included bid and proposal costs of $188, $217 and $213 in 2023, 2022 and 2021, respectively.

Share-Based Compensation

We provide various forms of share-based compensation to our employees. For awards settled in shares, we measure compensation expense based on the grant-date fair value net of estimated

forfeitures. For awards settled in cash, or that may be settled in cash, we measure compensation expense based on the fair value at each reporting date net of estimated forfeitures. The expense is recognized over the requisite service period, which is generally the vesting period of the award.

Income Taxes

Provisions for U.S. federal, state and local, and non-U.S. income taxes are calculated on reported Loss before income taxes based on current tax law and also include, in the current period, the cumulative effect of any changes in tax rates from those used previously in determining deferred tax assets and liabilities. Such provisions differ from the amounts currently receivable or payable because certain items of income and expense are recognized in different time periods for financial reporting purposes than for income tax purposes. Significant judgment is required in determining income tax provisions and evaluating tax positions.

The accounting for uncertainty in income taxes requires a more-likely-than-not threshold for financial statement recognition and measurement of tax positions taken or expected to be taken in a tax return. We record a liability for the difference between the benefit recognized and measured for financial statement purposes and the tax position taken or expected to be taken on our tax return. To the extent that our assessment of such tax positions changes, the change in estimate is recorded in the period in which the determination is made. Tax-related interest and penalties are classified as a component of Income tax (expense)/benefit.

We also assess the likelihood that we will be able to recover our deferred tax assets against future sources of taxable income and reduce the carrying amounts of deferred tax assets by recording a valuation allowance if, based on the available evidence, it is more likely than not that all or a portion of such assets will not be realized. Changes in our estimates and judgments regarding realization of deferred tax assets may result in an increase or decrease to our tax expense and/or other comprehensive income, which would be recorded in the period in which the change occurs.

Postretirement Plans

Many of our employees have earned benefits under defined benefit pension plans. The majority of employees that had participated in defined benefit pension plans have transitioned to a company-funded defined contribution retirement savings plan. We also provide postretirement benefit plans other than pensions, consisting principally of health care coverage to eligible retirees and qualifying dependents. Benefits under the pension and other postretirement benefit plans are generally based on age at retirement and years of service and, for some pension plans, benefits are also based on the employee's annual earnings. The net periodic cost of our pension and other postretirement plans is determined using the projected unit credit method and several actuarial assumptions, the most significant of which are the discount rate, the long-term rate of asset return and medical trend (rate of growth for medical costs). Actuarial gains and losses, which occur when actual experience differs from actuarial assumptions, are reflected in Shareholders' equity (net of taxes). If actuarial gains and losses exceed ten percent of the greater of plan assets or plan liabilities, we amortize them over the average expected future lifetime of participants. The funded status of our pension and postretirement plans is reflected on the Consolidated Statements of Financial Position.

Postemployment Plans

We record a liability for postemployment benefits, such as severance or job training, when payment is probable, the amount is reasonably estimable, and the obligation relates to rights that have vested or accumulated.

Environmental Remediation

We are subject to federal and state requirements for protection of the environment, including those for discharge of hazardous materials and remediation of contaminated sites. We routinely assess, based on in-depth studies, expert analyses and legal reviews, our contingencies, obligations and commitments for remediation of contaminated sites, including assessments of ranges and probabilities of recoveries from other responsible parties and/or insurance carriers. Our policy is to accrue and charge to current expense identified exposures related to environmental remediation sites when it is probable that a liability has been incurred and the amount can be reasonably estimated. The amount of the liability is based on our best estimate or the low end of a range of reasonably possible exposure for investigation, cleanup and monitoring costs to be incurred. Estimated remediation costs are not discounted to present value as the timing of payments cannot be reasonably estimated. We may be able to recover a portion of the remediation costs from insurers or other third parties. Such recoveries are recorded when realization of the claim for recovery is deemed probable.

Cash and Cash Equivalents

Cash and cash equivalents consist of highly liquid instruments, such as commercial paper, time deposits, and other money market instruments, which have original maturities of three months or less. We aggregate our cash balances by bank where conditions for right of set-off are met, and reclassify any negative balances, consisting mainly of uncleared checks, to Accounts payable. Negative balances reclassified to Accounts payable were $117 and $102 at December 31, 2023 and 2022.

Inventories

Inventoried costs on commercial aircraft programs and long-term contracts include direct engineering, production and tooling and other non-recurring costs, and applicable overhead, which includes fringe benefits, production related indirect and plant management salaries and plant services, not in excess of estimated net realizable value. To the extent a material amount of such costs are related to an abnormal event or are fixed costs not appropriately attributable to our programs or contracts, they are expensed in the current period rather than inventoried. Inventoried costs include amounts relating to programs and contracts with long-term production cycles, a portion of which is not expected to be realized within one year. Included in inventory for federal government contracts is an allocation of allowable costs related to manufacturing process reengineering.

Commercial aircraft programs inventory includes deferred production costs and supplier advances. Deferred production costs represent actual costs incurred for production of early units that exceed the estimated average cost of all units in the program accounting quantity. Higher production costs are experienced at the beginning of a new or derivative aircraft program. Units produced early in a program require substantially more effort (labor and other resources) than units produced later in a program because of volume efficiencies and the effects of learning. We expect that these deferred costs will be fully recovered when all units included in the accounting quantity are delivered as the expected unit cost for later deliveries is below the estimated average cost of all units in the program. Supplier advances represent payments for parts we have contracted to receive from suppliers in the future. As parts are received, supplier advances are amortized to work in process.

The determination of net realizable value of long-term contract costs is based upon quarterly reviews that estimate costs to be incurred to complete all contract requirements. When actual contract costs and the estimate to complete exceed total estimated contract revenues, a loss provision is recorded. The determination of net realizable value of commercial aircraft program costs is based upon quarterly program reviews that estimate revenue and cost to be incurred to complete the program accounting quantity. When estimated costs to complete exceed estimated program revenues to go, a program loss provision is recorded in the current period for the estimated loss on all undelivered units in the accounting quantity.

Used aircraft purchased by our BCA segment and general stock materials are stated at cost not in excess of net realizable value. Spare parts inventory is stated at lower of average unit cost or net realizable value. We review our commercial spare parts and general stock materials quarterly to identify impaired inventory, including excess or obsolete inventory, based on historical sales trends, expected production usage, and the size and age of the aircraft fleet using the part. Impaired inventories are charged to Cost of products in the period the impairment occurs.

Included in inventory for commercial aircraft programs are amounts paid or credited in cash, or other consideration to certain airline customers, that are referred to as early issue sales consideration. Early issue sales consideration is recognized as a reduction to revenue when the delivery of the aircraft under contract occurs. If an airline customer does not perform and take delivery of the contracted aircraft, we believe that we would have the ability to recover amounts paid. However, to the extent early issue sales consideration exceeds advances and is not considered to be otherwise recoverable, it would be written off in the current period.

Precontract Costs

We may, from time to time, incur costs in excess of the amounts required for existing contracts. If we determine the costs are probable of recovery from future orders, then we capitalize the precontract costs we incur, excluding start-up costs which are expensed as incurred. Capitalized precontract costs are included in Inventories in the accompanying Consolidated Statements of Financial Position. Should future orders not materialize or we determine the costs are no longer probable of recovery, the capitalized costs would be written off.

Property, Plant and Equipment

Property, plant and equipment are recorded at cost, including applicable construction-period interest, less accumulated depreciation and are depreciated principally over the following estimated useful lives: new buildings and land improvements, from 10 to 40 years; and new machinery and equipment, from 4 to 20 years. The principal methods of depreciation are as follows: buildings and land improvements, 150% declining balance; and machinery and equipment, sum-of-the-years' digits. Capitalized internal use software is included in Other assets, net and amortized using the straight line method over 5 years. Capitalized costs of software purchased as a service are included in Other assets, net and amortized using the straight line method over the term of the hosting arrangement, which is typically no greater than 10 years. We periodically evaluate the appropriateness of remaining depreciable lives assigned to long-lived assets, including assets that may be subject to a management plan for disposition.

Long-lived assets held for sale are stated at the lower of cost or fair value less cost to sell. Long-lived assets held for use are subject to an impairment assessment whenever events or changes in circumstances indicate that the carrying amount may not be recoverable. If the carrying value is no longer recoverable based upon the undiscounted future cash flows of the asset, the amount of the impairment is the difference between the carrying amount and the fair value of the asset.

Leases We determine if an arrangement is, or contains, a lease under which we are the lessee at the inception date. Operating lease assets are included in Other assets, net, with the related liabilities included in Accrued liabilities and Other long-term liabilities. Assets under finance leases, which primarily represent computer equipment, are included in Property, plant and equipment, net, with the related liabilities included in Short-term debt and current portion of long-term debt and Long-term debt on the Consolidated Statements of Financial Position.

Operating lease assets represent our right to use an underlying asset for the lease term and lease liabilities represent our obligation to make lease payments arising from the lease. Operating lease assets and liabilities are recognized at the lease commencement date based on the estimated present value of lease payments over the lease term. We use our estimated incremental borrowing rate in

determining the present value of lease payments. Variable components of the lease payments such as fair market value adjustments, utilities and maintenance costs are expensed as incurred and not included in determining the present value. Our lease terms include options to extend or terminate the lease when it is reasonably certain that we will exercise that option. Lease expense for operating leases is recognized on a straight-line basis over the lease term.

We have real property lease agreements with lease and non-lease components which are accounted for as a single lease component.

Asset Retirement Obligations

We record all known asset retirement obligations for which the liability's fair value can be reasonably estimated, including certain asbestos removal, asset decommissioning and contractual lease restoration obligations. Recorded amounts are not material.

We also have known conditional asset retirement obligations, such as certain asbestos remediation and asset decommissioning activities to be performed in the future, that are not reasonably estimable due to insufficient information about the timing and method of settlement of the obligation. Accordingly, these obligations have not been recorded in the Consolidated Financial Statements. A liability for these obligations will be recorded in the period when sufficient information regarding timing and method of settlement becomes available to make a reasonable estimate of the liability's fair value. In addition, there may be conditional asset retirement obligations that we have not yet discovered (e.g. asbestos may exist in certain buildings but we have not become aware of it through the normal course of business), and therefore, these obligations also have not been included in the Consolidated Financial Statements.

Goodwill and Other Acquired Intangibles

Goodwill and other acquired intangible assets with indefinite lives are not amortized, but are tested for impairment annually and when an event occurs or circumstances change such that it is more likely than not that an impairment may exist. Our annual testing date is April 1.

We test goodwill for impairment by performing a qualitative assessment or using a quantitative test. If we choose to perform a qualitative assessment and determine it is more likely than not that the carrying value of the net assets is more than the fair value of the related operations, the quantitative test is then performed; otherwise, no further testing is required. For operations where the quantitative test is used, we compare the carrying value of net assets to the estimated fair value of the related operations. If the fair value is determined to be less than carrying value, the shortfall up to the carrying value of the goodwill represents the amount of goodwill impairment.

Indefinite-lived intangibles consist of a brand and trade name and in-process research and development (IPR&D) acquired in business combinations. IPR&D is initially capitalized at fair value as an intangible asset with an indefinite life and assessed for impairment thereafter. IPR&D is reclassified to finite-lived acquired intangible assets when a project is completed and then amortized on a straight-line basis over the asset's estimated useful life. We test these intangibles for impairment by comparing the carrying values to current projections of related discounted cash flows. Any excess carrying value over the amount of discounted cash flows represents the amount of the impairment.

Our finite-lived acquired intangible assets are amortized on a straight-line basis over their estimated useful lives as follows: developed technology, from 4 to 14 years; product know-how, from 6 to 30 years; customer base, from 3 to 17 years; distribution rights, from 3 to 27 years; and other, from 1 to 32 years. We evaluate the potential impairment of finite-lived acquired intangible assets whenever events or changes in circumstances indicate that the carrying amount may not be recoverable. If the carrying

value is no longer recoverable based upon the undiscounted future cash flows of the asset, the amount of the impairment is the difference between the carrying amount and the fair value of the asset.

Investments

Time deposits are held-to-maturity investments that are carried at cost.

Available-for-sale debt investments include commercial paper, U.S. government agency securities and corporate debt securities. Available-for-sale debt investments are recorded at fair value, and unrealized gains and losses are recorded, net of tax, as a component of accumulated other comprehensive income. Realized gains and losses on available-for-sale debt investments are recognized based on the specific identification method. Available-for-sale debt investments are assessed for impairment quarterly.

The equity method of accounting is used to account for investments for which we have the ability to exercise significant influence, but not control, over an investee. Significant influence is generally deemed to exist if we have an ownership interest in the voting stock of an investee of between 20% and 50%. The cumulative earnings approach is used for cash flow classification of distributions received from equity method investments.

Other Equity investments are recorded at fair value, with gains and losses recorded through net earnings. Equity investments without readily determinable fair value are measured at cost, less impairments, plus or minus observable price changes. Equity investments without readily determinable fair value are assessed for impairment quarterly.

We classify investment income and loss on our Consolidated Statements of Operations based on whether the investment is operating or non-operating in nature. Operating investments align strategically and are integrated with our operations. Earnings from operating investments, including our share of income or loss from equity method investments, dividend income from other equity investments, and any impairments or gain/loss on the disposition of these investments, are recorded in Income/(loss) from operating investments, net. Non-operating investments are those we hold for non-strategic purposes. Earnings from non-operating investments, including interest and dividends on marketable securities, and any impairments or gain/loss on the disposition of these investments are recorded in Other income, net.

Derivatives

All derivative instruments are recognized in the financial statements and measured at fair value regardless of the purpose or intent of holding them. We use derivative instruments to principally manage a variety of market risks. For our cash flow hedges, the derivative’s gain or loss is initially reported in comprehensive income and is subsequently reclassified into earnings in the same period(s) during which the hedged forecasted transaction affects earnings.

We have agreements to purchase and sell aluminum to address long-term strategic sourcing objectives and non-U.S. business requirements. We also hold certain other derivative instruments for economic purposes. These aluminum purchase and sale agreements and other derivative instruments are derivatives for accounting purposes but are not designated as hedges for accounting purposes. For these aluminum agreements and other derivative instruments not designated for hedge accounting treatment, the changes in their fair value are recorded in earnings immediately.

Allowances for Losses on Certain Financial Assets

We establish allowances for credit losses on accounts receivable, unbilled receivables, financing receivables and certain other financial assets. The adequacy of these allowances is assessed quarterly through consideration of factors such as customer credit ratings, bankruptcy filings, published or

estimated credit default rates, age of the receivable, expected loss rates and collateral exposures. Collateral exposure is the excess of the carrying value of a financial asset over the fair value of the related collateral. We determine the creditworthiness of our customers by assigning internal credit ratings based upon publicly available information and information obtained directly from the customers. Our rating categories are comparable to those used by major credit rating agencies.

Financing receivables are collateralized by security in the related asset. We use a median calculated from published collateral values from multiple third-party aircraft value publications based on the type and age of the aircraft to determine the fair value of aircraft collateral. Under certain circumstances, we apply judgment based on the attributes of the specific aircraft or equipment, usually when the features or use of the aircraft vary significantly from the more generic aircraft attributes covered by outside publications.

We have entered into agreements with certain customers that would entitle us to look beyond the specific collateral underlying the receivable for purposes of determining the collateral exposure. Should the proceeds from the sale of the underlying collateral asset resulting from a default condition be insufficient to cover the carrying value of our receivable (creating a shortfall condition), these agreements would, for example, permit us to take the actions necessary to sell or retain certain other assets in which the customer has an equity interest and use the proceeds to cover the shortfall.

Commercial Aircraft Trade-in Commitments

In conjunction with signing a definitive agreement for the sale of new commercial aircraft (Sale Aircraft), we have entered into trade-in commitments with certain customers that give them the right to trade in used aircraft at a specified price.

Exposure related to trade-in commitments may take the form of:

(1) adjustments to revenue for the difference between the contractual trade-in price in the definitive agreement and our best estimate of the fair value of the trade-in aircraft as of the date of such agreement, which would be recognized upon delivery of the Sale Aircraft, and/or

(2) charges to cost of products for adverse changes in the fair value of trade-in aircraft that occur subsequent to signing of a definitive agreement for Sale Aircraft but prior to the purchase of the used trade-in aircraft. Estimates based on current aircraft values would be included in Accrued liabilities.

The fair value of trade-in aircraft is determined using aircraft-specific data such as model, age and condition, market conditions for specific aircraft and similar models, and multiple valuation sources. This process uses our assessment of the market for each trade-in aircraft, which in most instances begins years before the return of the aircraft. There are several possible markets in which we continually pursue opportunities to place used aircraft. These markets include, but are not limited to, the resale market, which could potentially include the cost of long-term storage; the leasing market, with the potential for refurbishment costs to meet the leasing customer’s requirements; or the scrap market. Trade-in aircraft valuation varies significantly depending on which market we determine is most likely for each aircraft. On a quarterly basis, we update our valuation analysis based on the actual activities associated with placing each aircraft into a market or using current published third-party aircraft valuations based on the type and age of the aircraft, adjusted for individual attributes and known conditions.

Warranties

In conjunction with certain product sales, we provide warranties that cover factors such as non-conformance to specifications and defects in material and design. The majority of our warranties are issued by our BCA segment. Generally, aircraft sales are accompanied by a 3 to 4-year standard

warranty for systems, accessories, equipment, parts, and software manufactured by us or manufactured to certain standards under our authorization. These warranties are included in the programs' estimate at completion. On occasion we have made commitments beyond the standard warranty obligation to correct fleet-wide major issues of a particular model, resulting in additional accrued warranty expense. Warranties issued by our BDS segment principally relate to sales of military aircraft and weapons systems. These sales are generally accompanied by a six month to two-year warranty period and cover systems, accessories, equipment, parts and software manufactured by us to certain contractual specifications. Estimated costs related to standard warranties are recorded in the period in which the related product delivery occurs. The warranty liability recorded at each balance sheet date reflects the estimated number of months of warranty coverage outstanding for products delivered times the average of historical monthly warranty payments, as well as additional amounts for certain major warranty issues that exceed a normal claims level. Estimated costs of these additional warranty issues are considered changes to the initial liability estimate.

We provide guarantees to certain commercial aircraft customers which include compensation provisions for failure to meet specified aircraft performance targets. We account for these performance guarantees as warranties. The estimated liability for these warranties is based on known and anticipated operational characteristics and forecasted customer operation of the aircraft relative to contractually specified performance targets, and anticipated settlements when contractual remedies are not specified. Estimated payments are recorded as a reduction of revenue at delivery of the related aircraft. We have agreements that require certain suppliers to compensate us for amounts paid to customers for failure of supplied equipment to meet specified performance targets. Claims against suppliers under these agreements are included in Inventories and recorded as a reduction in Cost of products at delivery of the related aircraft. These performance warranties and claims against suppliers are included in estimates to complete the related programs.

Supplier Penalties

We may incur penalties to suppliers under certain circumstances such as a contract termination. We record an accrual for supplier penalties when an event occurs that makes it probable we will incur a supplier penalty and the amount is reasonably estimable.

Guarantees

At the inception of a guarantee, we record a liability in Accrued liabilities for the fair value of the guarantee. For credit guarantees, the liability is equal to the present value of the expected loss. We determine the expected loss by multiplying the creditor's default rate by the guarantee amount reduced by the expected recovery, if applicable. We also recognize a liability for the expected contingent loss at inception and adjust it each quarter.

Note 2 – Goodwill and Acquired Intangibles

Changes in the carrying amount of goodwill for the years ended December 31, 2023 and 2022 were as follows:

	Commercial Airplanes	Defense, Space & Security	Global Services	Other	Total
Balance at December 31, 2021	$1,316	$3,224	$3,443	$85	$8,068
Goodwill adjustments			(11)		(11)
Balance at December 31, 2022	**$1,316**	**$3,224**	**$3,432**	**$85**	**$8,057**
Acquisitions	**3**	**11**	**16**		**30**
Goodwill adjustments			**6**		**6**
Balance at December 31, 2023	**$1,319**	**$3,235**	**$3,454**	**$85**	**$8,093**

As of December 31, 2023 and 2022, we had indefinite-lived intangible assets with carrying amounts of $ 197 relating to trade names. As of December 31, 2023 and 2022, we had an indefinite-lived intangible asset with a carrying amount of $202 related to in process research and development for a next-generation air vehicle.

The gross carrying amounts and accumulated amortization of our acquired finite-lived intangible assets were as follows at December 31:

	2023		2022	
	Gross Carrying Amount	**Accumulated Amortization**	Gross Carrying Amount	Accumulated Amortization
Distribution rights	**$2,545**	**$1,566**	$2,546	$1,443
Product know-how	**552**	**465**	552	441
Customer base	**1,358**	**837**	1,356	777
Developed technology	**638**	**569**	621	545
Other	**280**	**241**	276	233
Total	**$5,373**	**$3,678**	$5,351	$3,439

Amortization expense for acquired finite-lived intangible assets for the years ended December 31, 2023 and 2022 was $ 235 and $241. Estimated amortization expense for the five succeeding years is as follows:

	2024	2025	2026	2027	2028
Estimated amortization expense	$224	$199	$194	$174	$147

Note 3 – Earnings Per Share

Basic and diluted earnings per share are computed using the two-class method, which is an earnings allocation method that determines earnings per share for common shares and participating securities. The undistributed earnings are allocated between common shares and participating securities as if all earnings had been distributed during the period. Participating securities and common shares have equal rights to undistributed earnings.

Basic earnings per share is calculated by taking net earnings attributable to Boeing Shareholders, less earnings available to participating securities, divided by the basic weighted average common shares outstanding.

Diluted earnings per share is calculated by taking net earnings attributable to Boeing Shareholders, less earnings available to participating securities, divided by the diluted weighted average common shares outstanding. Diluted weighted average common shares outstanding is calculated using the treasury stock method.

The elements used in the computation of basic and diluted earnings per share were as follows:

(In millions - except per share amounts)

Years ended December 31,	**2023**	2022	2021
Net loss attributable to Boeing Shareholders	**($2,222)**	($4,935)	($4,202)
Less: earnings available to participating securities			
Net loss available to common shareholders	**($2,222)**	($4,935)	($4,202)
Basic			
Basic weighted average shares outstanding	**606.1**	595.2	588.0
Less: participating securities[(1)]	**0.3**	0.3	0.4
Basic weighted average common shares outstanding	**605.8**	594.9	587.6
Diluted			
Diluted weighted average shares outstanding	**606.1**	595.2	588.0
Less: participating securities[(1)]	**0.3**	0.3	0.4
Diluted weighted average common shares outstanding	**605.8**	594.9	587.6
Net loss per share:			
Basic	**($3.67)**	($8.30)	($7.15)
Diluted	**(3.67)**	(8.30)	(7.15)

(1) Participating securities include certain instruments in our deferred compensation plan.

The following table represents potential common shares that were not included in the computation of Diluted loss per share because the effect was antidilutive based on their strike price or the performance condition was not met.

(Shares in millions)

Years ended December 31,	**2023**	2022	2021
Performance awards			2.9
Performance-based restricted stock units		0.4	0.8
Restricted stock units		1.0	0.4
Stock options	**0.8**	0.8	0.3

In addition, potential common shares of 5.7 million, 3.5 million, and 2.6 million for the years ended December 31, 2023, 2022 and 2021 were excluded from the computation of Diluted loss per share, because the effect would have been antidilutive as a result of incurring a net loss in those periods.

Note 4 – Income Taxes

The components of Loss before income taxes were:

Years ended December 31,	**2023**	2022	2021
U.S.	**($2,512)**	($5,457)	($5,475)
Non-U.S.	**507**	435	442
Total	**($2,005)**	($5,022)	($5,033)

Income tax (expense)/benefit consisted of the following:

Years ended December 31,	**2023**	2022	2021
Current tax (benefit)/expense			
U.S. federal	**$9**	($58)	($89)
Non-U.S.	**179**	142	147
U.S. state	**19**	(42)	42
Total current	**207**	42	100
Deferred tax (benefit)/expense			
U.S. federal	**6**	(62)	(855)
Non-U.S.	**5**	(3)	(12)
U.S. state	**19**	54	24
Total deferred	**30**	(11)	(843)
Total income tax expense/(benefit)	**$237**	$31	($743)

Net income tax payments/(refunds) were $204, ($1,317) and ($1,480) in 2023, 2022 and 2021, respectively.

The following is a reconciliation of the U.S. federal statutory tax to actual income tax (benefit)/expense:

Years ended December 31,	**2023**		2022		2021	
	Amount	**Rate**	Amount	Rate	Amount	Rate
U.S. federal statutory tax	**($421)**	**21.0 %**	($1,054)	21.0 %	($1,057)	21.0 %
Valuation allowance	**1,150**	**(57.3)**	1,199	(23.9)	512	(10.2)
Research and development credits	**(472)**	**23.6**	(204)	4.1	(189)	3.8
State income tax provision, net of effects on U.S. federal tax	**(75)**	**3.7**	(90)	1.8	(94)	1.9
Tax on non-U.S. activities	**35**	**(1.8)**	64	(1.3)	47	(0.9)
Impact of subsidiary shares purchased from noncontrolling interests	**(29)**	**1.5**				
Other provision adjustments	**49**	**(2.5)**	116	(2.3)	38	(0.9)
Income tax expense/(benefit)	**$237**	**(11.8)%**	$31	(0.6)%	($743)	14.7 %

Significant components of our deferred tax assets/(liabilities) at December 31 were as follows:

	2023	2022
Inventory and long-term contract methods of income recognition	**($5,115)**	($4,369)
Research expenditures	**2,873**	1,464
Federal net operating loss, credit, interest and other carryovers [1]	**2,551**	2,082
Fixed assets, intangibles and goodwill	**(1,566)**	(1,641)
Pension benefits	**1,178**	1,146
Other employee benefits	**1,162**	1,095
State net operating loss, credit, interest and other carryovers [2]	**1,137**	1,021
Accrued expenses and reserves	**956**	933
Other postretirement benefit obligations	**590**	660
737 MAX customer concessions and other considerations	**310**	425
Other	**304**	179
Gross deferred tax assets/(liabilities) before valuation allowance	**$4,380**	$2,995
Valuation allowance	**(4,550)**	(3,162)
Net deferred tax assets/(liabilities) after valuation allowance	**($170)**	($167)

[1] Of the deferred tax asset for federal net operating loss, credit, interest and other carryovers, $ 1,224 expires on or before December 31, 2043 and $1,327 may be carried over indefinitely.

[2] Of the deferred tax asset for state net operating loss, credit, interest and other carryovers, $ 575 expires on or before December 31, 2043 and $ 562 may be carried over indefinitely.

Net deferred tax assets/(liabilities) at December 31 were as follows:

	2023	2022
Deferred tax assets	**$14,743**	$12,301
Deferred tax liabilities	**(10,363)**	(9,306)
Valuation allowance	**(4,550)**	(3,162)
Net deferred tax assets/(liabilities)	**($170)**	($167)

The Company's deferred income tax assets of $ 14,743 can be used in future years to offset taxable income and reduce income taxes payable. The Company's deferred income tax liabilities of $10,363 will partially offset deferred income tax assets and result in higher taxable income in future years and increase income taxes payable. Tax law determines whether future reversals of temporary differences will result in taxable and deductible amounts that offset each other in future years. The particular years in which temporary differences result in taxable or deductible amounts generally are determined by the timing of the recovery of the related asset or settlement of the related liability. The deferred income tax assets and liabilities relate primarily to U.S. federal and state tax jurisdictions. From a U.S. federal tax perspective, the Company generated tax net operating losses in 2021 and interest carryovers in 2021, 2022, and 2023 that can be carried forward indefinitely and federal research and development credits that can be carried forward 20 years.

Throughout 2021, 2022, and 2023, the Company was in a three-year cumulative pre-tax loss position. For purposes of assessing the recoverability of deferred tax assets, the Company determined that it could not include future projected earnings in the analysis due to recent history of losses.

As of December 31, 2023 and 2022, the Company has recorded valuation allowances of $ 4,550 and $3,162 primarily for certain domestic deferred tax assets, and certain domestic net operating losses, tax

credit and interest carryforwards. To measure the valuation allowance, the Company estimated in what year each of its deferred tax assets and liabilities would reverse using systematic and logical methods to estimate the reversal patterns. Based on these methods, deferred tax liabilities are assumed to reverse and generate taxable income over the next 5 to 10 years while deferred tax assets related to pension and other postretirement benefit obligations are assumed to reverse and generate tax deductions over the next 15 to 20 years. The valuation allowance results from not having sufficient income from deferred tax liability reversals in the appropriate future periods to support the realization of deferred tax assets.

During 2023, the Company increased the valuation allowance by $ 1,388, primarily due to tax credits and other carryforwards generated in 2023 that cannot be realized in 2023. This reflects a tax expense of $1,150 recorded in continuing operations, an increase of $ 31 related to the associated federal benefit of state impacts, a tax expense of $173 included in Other comprehensive income (OCI) primarily due to the net actuarial losses that resulted from the annual remeasurement of pension assets and liabilities, and an increase of $34 included in additional paid-in capital.

Until the Company generates sustained levels of profitability, additional valuation allowances may have to be recorded with corresponding adverse impacts on earnings and/or OCI.

The Tax Cuts and Jobs Act one-time repatriation tax and Global Intangible Low Tax Income liabilities effectively taxed the undistributed earnings previously deferred from U.S. income taxes. We have not provided for deferred income taxes on the undistributed earnings from certain non-U.S. subsidiaries because such earnings are considered to be indefinitely reinvested. If such earnings were to be distributed, any deferred income taxes would not be significant.

As of December 31, 2023 and 2022, the amounts accrued for the payment of income tax-related interest and penalties included in the Consolidated Statements of Financial Position were not significant. The amounts of interest included in the Consolidated Statements of Operations were not significant for 2023, 2022 and 2021.

A reconciliation of the beginning and ending amount of unrecognized tax benefits is as follows:

	2023	2022	2021
Unrecognized tax benefits – January 1	**$915**	$858	$966
Gross increases – tax positions in prior periods	**38**	17	64
Gross decreases – tax positions in prior periods	**(3)**	(51)	(245)
Gross increases – current period tax positions	**181**	91	73
Gross decreases – current period tax positions			
Unrecognized tax benefits – December 31	**$1,131**	$915	$858

As of December 31, 2023, 2022 and 2021, the total amount of unrecognized tax benefits include $ 1,088, $878 and $790, respectively, that would affect the effective tax rate, if recognized. As of December 31, 2023, these amounts were primarily associated with the amount of research tax credits claimed and various other matters.

Federal income tax audits have been settled for all years prior to 2018. The Internal Revenue Service is currently auditing the 2018-2020 tax years. We are also subject to examination in major state and international jurisdictions for the 2010-2022 tax years. We believe appropriate provisions for all outstanding tax issues have been made for all jurisdictions and all open years.

Audit outcomes and the timing of audit settlements are subject to significant uncertainty. It is reasonably possible that within the next 12 months, unrecognized tax benefits related to federal tax matters under audit may decrease by up to $620 based on current estimates.

The Organization for Economic Co-operation and Development has issued Pillar Two model rules introducing a new global minimum tax of 15% intended to be effective on January 1, 2024. While the US has not yet adopted the Pillar Two rules, various other governments around the world are enacting legislation. As currently designed, Pillar Two will ultimately apply to our worldwide operations. Considering we do not have material operations in jurisdictions with tax rates lower than the Pillar Two minimum, these rules are not expected to materially increase our global tax costs. There remains uncertainty as to the final Pillar Two model rules. We will continue to monitor US and global legislative action related to Pillar Two for potential impacts.

Note 5 – Accounts Receivable, net

Accounts receivable, net at December 31 consisted of the following:

	2023	2022
U.S. government contracts[1]	**$970**	$800
Commercial Airplanes	**57**	293
Global Services[2]	**1,526**	1,390
Defense, Space, & Security[2]	**160**	145
Other	**25**	5
Less valuation allowance	**(89)**	(116)
Total	**$2,649**	$2,517

(1) Includes foreign military sales through the U.S. government

(2) Excludes U.S. government contracts

Note 6 – Allowances for Losses on Financial Assets

The change in allowances for expected credit losses for the years ended December 31, 2023 and 2022 consisted of the following:

	Accounts receivable	**Unbilled receivables**	**Other Current Assets**	**Financing receivables**	**Other Assets**	**Total**
Balance at January 1, 2022	($390)	($91)	($62)	($18)	($186)	($747)
Changes in estimates	2	21	(27)	(37)	(35)	(76)
Write-offs	260	47	4		133	444
Recoveries	12					12
Balance at December 31, 2022	($116)	($23)	($85)	($55)	($88)	($367)

Balance at January 1, 2023	**($116)**	**($23)**	**($85)**	**($55)**	**($88)**	**($367)**
Changes in estimates	**(6)**	**4**	**30**	**4**	**(34)**	**(2)**
Write-offs	**29**		**5**			**34**
Recoveries	**4**					**4**
Balance at December 31, 2023	**($89)**	**($19)**	**($50)**	**($51)**	**($122)**	**($331)**

Note 7 – Inventories

Inventories at December 31 consisted of the following:

	2023	2022
Commercial aircraft programs	**$68,683**	$67,702
Long-term contracts in progress	**686**	582
Capitalized precontract costs[1]	**946**	794
Commercial spare parts, used aircraft, general stock materials and other	**9,426**	9,073
Total	**$79,741**	$78,151

(1) Capitalized precontract costs at December 31, 2023 and 2022 includes amounts related to KC-46A Tanker, Commercial Crew, and T-7A Red Hawk Production Options. See Note 13.

Commercial Aircraft Programs

At December 31, 2023 and 2022, commercial aircraft programs inventory included the following amounts related to the 737 program: deferred production costs of $6,011 and $2,955 and unamortized tooling and other non-recurring costs of $ 792 and $626. At December 31, 2023, $ 6,767 of 737 deferred production costs, unamortized tooling and other non-recurring costs are expected to be recovered from units included in the program accounting quantity that have firm orders, and $36 is expected to be recovered from units included in the program accounting quantity that represent expected future orders.

At December 31, 2023 and 2022, commercial aircraft programs inventory included the following amounts related to the 777X program: $4,638 and $4,059 of work in process, $1,792 and $ 1,330 of deferred production costs, and $ 4,063 and $3,774 of unamortized tooling and other non-recurring costs. In April 2022, we decided to pause production of the 777X-9 during 2022 and 2023, which resulted in abnormal production costs of $513 and $325 during the years ended December 31, 2023 and 2022. In the fourth quarter of 2023, the 777X program resumed production.

At December 31, 2023 and 2022, commercial aircraft programs inventory included the following amounts related to the 787 program: deferred production costs of $12,384 and $12,689, $1,764 and $1,831 of supplier advances, and $ 1,480 and $1,722 of unamortized tooling and other non-recurring costs. At December 31, 2023, $12,384 of 787 deferred production costs, unamortized tooling and other non-recurring costs are expected to be recovered from units included in the program accounting quantity that have firm orders, and $1,480 is expected to be recovered from units included in the program accounting quantity that represent expected future orders. We produced at abnormally low production rates resulting in abnormal production costs that were expensed as incurred from the third quarter of 2021 through the third quarter of 2023. We expensed abnormal production costs of $1,014, $1,240, and $468 during the years ended December 31, 2023, 2022 and 2021. The remaining abnormal costs associated with rework are not expected to be significant.

Commercial aircraft programs inventory included amounts credited in cash or other consideration (early issue sales consideration) to airline customers totaling $4,126 and $3,586 at December 31, 2023 and 2022.

Note 8 – Contracts with Customers

Unbilled receivables decreased from $ 8,634 at December 31, 2022 to $ 8,317 at December 31, 2023, primarily driven by an increase in billings at BDS and BGS.

The following table summarizes our contract assets under long-term contracts that were unbillable or related to outstanding claims as of December 31:

	Unbilled		Claims	
	2023	2022	**2023**	2022
Current	**$6,565**	$6,478	**$6**	
Expected to be collected after one year	**1,771**	2,179	**40**	$16
Less valuation allowance	**(19)**	(23)		
Total	**$8,317**	$8,634	**$46**	$16

Unbilled receivables related to commercial customer incentives expected to be collected after one year were $ 42 and $117 at December 31, 2023 and 2022. Unbilled receivables related to claims are items that we believe are earned, but are subject to uncertainty concerning their determination or ultimate realization.

Advances and progress billings increased from $ 53,081 at December 31, 2022 to $ 56,328 at December 31, 2023, primarily driven by advances on orders received at BCA, partially offset by revenue recognized at BDS.

Revenues recognized for the years ended December 31, 2023 and 2022 from amounts recorded as Advances and progress billings at the beginning of each year were $15,298 and $12,087.

Note 9 – Financing Receivables and Operating Lease Equipment

Financing receivables and operating lease equipment, net consisted of the following at December 31:

	2023	2022
Financing receivables:		
Investment in sales-type leases	**$556**	$804
Notes	**102**	385
Total financing receivables	**658**	1,189
Less allowance for losses on receivables	**51**	55
Financing receivables, net	**607**	1,134
Operating lease equipment, at cost, less accumulated depreciation of $ 70 and $76	**352**	470
Total	**$959**	$1,604

Financing arrangements typically range in terms from 1 to 12 years and may include options to extend or terminate. Certain leases include provisions to allow the lessee to purchase the underlying aircraft at a specified price. At December 31, 2023 and 2022, $44 and $405 were determined to be uncollectible financing receivables and placed on non-accrual status. The allowance for losses on receivables remained largely unchanged during the year ended December 31, 2023.

The components of investment in sales-type leases at December 31 were as follows:

	2023	2022
Gross lease payments receivable	**$697**	$924
Unearned income	**(162)**	(206)
Net lease payments receivable	**535**	718
Unguaranteed residual assets	**21**	86
Total	**$556**	$804

Financing interest income received for the years ended December 31, 2023 and 2022 was $ 108 and $13.

Financing receivables that were past due as of December 31, 2023 totaled $ 9.

Our financing receivable balances at December 31, 2023 by internal credit rating category and year of origination consisted of the following:

Rating categories	Current	2022	2021	2020	2019	Prior	Total
BBB						$13	$13
BB	$73	$32	$198	$103	$36	53	495
B					12	94	106
CCC			35			9	44
Total carrying value of financing receivables	$73	$32	$233	$103	$48	$169	$658

At December 31, 2023, our allowance for losses related to receivables with ratings of CCC, B, BB and BBB. We applied default rates that averaged 100.0%, 0.0%, 2.4% and 0.1%, respectively, to the exposure associated with those receivables.

Financing Receivables Exposure

The majority of our financing receivables and operating lease equipment portfolio is concentrated in the following aircraft models at December 31:

	2023	2022
717 Aircraft ($0 and $45 accounted for as operating leases)	**$478**	$563
747-8 Aircraft (Accounted for as sales-type leases)	**129**	394
737 Aircraft ($148 and $174 accounted for as operating leases)	**156**	186
777 Aircraft ($194 and $209 accounted for as operating leases)	**194**	209
MD-80 Aircraft (Accounted for as sales-type leases)		96
757 Aircraft (Accounted for as sales-type leases)		107
747-400 Aircraft (Accounted for as sales-type leases)	**43**	46

Operating lease equipment primarily includes large commercial jet aircraft.

Impairment charges related to operating lease assets were $ 0, $7, and $31 for the years ended December 31, 2023, 2022 and 2021.

Lease income recorded in Sales of services on the Consolidated Statements of Operations for the years ended December 31, 2023, 2022 and 2021 included $55, $69, and $54 of interest income from

sales-type leases, and $60, $65, and $68 from operating lease payments. Profit at the commencement of sales-type leases was recorded in Sales of services for the years ended December 31, 2023, 2022 and 2021 in the amount of $32, $28, and $78.

As of December 31, 2023, undiscounted cash flows for notes receivable, sales-type and operating leases over the next five years and thereafter are as follows:

	Notes receivable	Sales-type leases	Operating leases
Year 1	$17	$149	$69
Year 2	9	102	61
Year 3	10	109	52
Year 4	11	127	47
Year 5	12	137	45
Thereafter	43	73	48
Total financing receipts	102	697	322
Less imputed interest		(162)	
Estimated unguaranteed residual values		21	
Total	$102	$556	$322

At December 31, 2023 and December 31, 2022, unguaranteed residual values were $ 21 and $86.

Note 10 – Property, Plant and Equipment

Property, plant and equipment at December 31 consisted of the following:

	2023	2022
Land	**$377**	$376
Buildings and land improvements	**14,795**	14,404
Machinery and equipment	**16,055**	15,844
Construction in progress	**1,679**	1,368
Gross property, plant and equipment	**32,906**	31,992
Less accumulated depreciation	**(22,245)**	(21,442)
Total	**$10,661**	$10,550

Depreciation expense was $1,328, $1,396 and $1,488 for 2023, 2022 and 2021, respectively. Interest capitalized in 2023, 2022 and 2021 totaled $ 101, $89 and $76, respectively.

During 2023 and 2022, we acquired $ 124 and $101 of property, plant and equipment through non-cash investing and financing transactions. Accounts payable related to purchases of property, plant and equipment were $498 and $396 for the years ended December 31, 2023 and 2022.

Note 11 – Investments

Our investments, which are recorded in Short-term and other investments or Investments, consisted of the following at December 31:

	2023	2022
Time deposits [1]	**$2,753**	$2,093
Equity method investments [2]	**966**	948
Available-for-sale debt investments [1]	**499**	479
Equity and other investments	**69**	36
Restricted cash & cash equivalents [1][3]	**22**	33
Total	**$4,309**	$3,589

[1] Included in Short-term and other investments on our Consolidated Statements of Financial Position.

[2] Dividends received were $31 and $111 during 2023 and 2022. Retained earnings at December 31, 2023 include undistributed earnings from our equity method investments of $110.

[3] Reflects amounts restricted in support of our property sales, workers' compensation programs and insurance premiums.

Contributions to investments and Proceeds from investments on our Consolidated Statements of Cash Flows primarily relate to time deposits and available-for-sale debt investments. Cash used for the purchase of time deposits during 2023, 2022 and 2021 was $15,794, $4,358 and $34,905, respectively. Cash proceeds from the maturities of time deposits during 2023, 2022 and 2021 were $15,140, $9,943 and $44,372, respectively.

Allowance for losses on available-for-sale debt investments are assessed quarterly. All instruments are considered investment grade, and we have not recognized an allowance for credit losses as of December 31, 2023.

Equity Method Investments

Our equity method investments consisted of the following at December 31:

	Segment	Ownership Percentages	Investment Balance	
			2023	2022
United Launch Alliance	BDS	50%	**$582**	$587
Other	BCA, BDS, BGS and Other		**384**	361
Total equity method investments			**$966**	$948

Note 12 – Leases

Our operating lease assets primarily represent manufacturing and research and development facilities, warehouses and offices. Total operating lease expense was $457 and $421 for the years ended December 31, 2023 and 2022, of which $ 76 and $75 was attributable to variable lease expenses.

For the years ended December 31, 2023 and 2022, cash payments against operating lease liabilities totaled $ 323 and $294 and non-cash transactions totaled $488 and $245 to recognize operating assets and liabilities for new leases.

Supplemental Consolidated Statement of Financial Position information related to leases consisted of the following at December 31:

	2023	2022
Operating leases:		
Operating lease right-of-use assets	**$1,690**	$1,451
Current portion of lease liabilities	**296**	276
Non-current portion of lease liabilities	**1,518**	1,305
Total operating lease liabilities	**$1,814**	$1,581
Weighted average remaining lease term *(years)*	**11**	12
Weighted average discount rate	**3.21%**	4.13%

Operating lease assets are included in Other assets, net, with the related liabilities included in Accrued liabilities and Other long-term liabilities.

Maturities of operating lease liabilities for the next five years are as follows:

	Operating leases
2024	$358
2025	317
2026	279
2027	230
2028	182
Thereafter	1,032
Total lease payments	2,398
Less imputed interest	(584)
Total	$1,814

As of December 31, 2023, we have entered into leases that have not yet commenced of $ 430, primarily for a maintenance, repair and overhaul hangar that will support military aircraft programs. These leases will commence in 2024 with lease terms of 2 years to 27 years.

Note 13 – Liabilities, Commitments and Contingencies

Accrued Liabilities

Accrued liabilities at December 31 consisted of the following:

	2023	2022
Accrued compensation and employee benefit costs	**$6,721**	$6,351
737 MAX customer concessions and other considerations	**1,327**	1,864
Other customer concessions and considerations	**1,300**	1,102
Environmental	**844**	752
Product warranties	**2,448**	2,275
Forward loss recognition	**4,699**	4,060
Accrued interest payable	**652**	599
Current portion of lease liabilities	**296**	276
Current portion of retiree healthcare and pension liabilities	**473**	494
Other	**3,571**	3,808
Total	**$22,331**	$21,581

737 MAX Customer Concessions and Other Considerations

The following table summarizes changes in the 737 MAX customer concessions and other considerations liability during 2023 and 2022.

	2023	2022
Beginning balance – January 1	**$1,864**	$2,940
Reductions for payments made	**(449)**	(1,031)
Reductions for concessions and other in-kind considerations	**(61)**	(29)
Changes in estimates	**(27)**	(16)
Ending balance – December 31	**$1,327**	$1,864

At December 31, 2023, $0.1 billion of the liability balance remains subject to negotiations with customers. We expect to pay $ 0.6 billion in 2024 while the remaining amounts are expected to be liquidated by lower customer delivery payments.

Environmental

The following table summarizes changes in environmental remediation liabilities during the years ended December 31, 2023 and 2022.

	2023	2022
Beginning balance – January 1	**$752**	$605
Reductions for payments made, net of recoveries	**(79)**	(43)
Changes in estimates	**171**	190
Ending balance – December 31	**$844**	$752

The liabilities recorded represent our best estimate or the low end of a range of reasonably possible costs expected to be incurred to remediate sites, including operation and maintenance over periods of up to 30 years. It is reasonably possible that we may incur costs that exceed these recorded amounts because of regulatory agency orders and directives, changes in laws and/or regulations, higher than

expected costs and/or the discovery of new or additional contamination. As part of our estimating process, we develop a range of reasonably possible alternate scenarios that includes the high end of a range of reasonably possible cost estimates for all remediation sites for which we have sufficient information based on our experience and existing laws and regulations. There are some potential remediation obligations where the costs of remediation cannot be reasonably estimated. At December 31, 2023 and 2022, the high end of the estimated range of reasonably possible remediation costs exceeded our recorded liabilities by $1,030 and $1,058.

Product Warranties

The following table summarizes changes in product warranty liabilities recorded during the years ended December 31, 2023 and 2022.

	2023	2022
Beginning balance – January 1	**$2,275**	$1,900
Additions for current year deliveries	**164**	202
Reductions for payments made	**(320)**	(403)
Changes in estimates	**329**	576
Ending balance – December 31	**$2,448**	$2,275

Commercial Aircraft Trade-In Commitments

In conjunction with signing definitive agreements for the sale of new aircraft, we have entered into trade-in commitments with certain customers that give them the right to trade in used aircraft at a specified price. The probability that trade-in commitments will be exercised is determined by using both quantitative information from valuation sources and qualitative information from other sources. The probability of exercise is assessed quarterly, or as events trigger a change, and takes into consideration the current economic and airline industry environments. Trade-in commitments, which can be terminated by mutual consent with the customer, may be exercised only during the period specified in the agreement and require advance notice by the customer.

Trade-in commitment agreements at December 31, 2023 have expiration dates from 2024 through 2030. At December 31, 2023 and 2022, total contractual trade-in commitments were $1,415 and $1,117. As of December 31, 2023 and 2022, we estimated that it was probable we would be obligated to perform on certain of these commitments with net amounts payable to customers totaling $407 and $286, and the fair value of the related trade-in aircraft was $407 and $286.

Financing Commitments

Financing commitments related to aircraft on order, including options and those proposed in sales campaigns, and refinancing of delivered aircraft, totaled $17,003 and $16,105 as of December 31, 2023 and 2022. The estimated earliest potential funding dates for these commitments as of December 31, 2023 are as follows:

	Total
2024	$1,946
2025	3,098
2026	4,829
2027	2,421
2028	1,641
Thereafter	3,068
	$17,003

As of December 31, 2023, all of these financing commitments relate to customers we believe have less than investment-grade credit. We have concluded that no reserve for future potential losses is required for these financing commitments based upon the terms, such as collateralization and interest rates, under which funding would be provided.

Other Financial Commitments

We have financial commitments to make additional capital contributions totaling $ 264 related to certain joint ventures over the next nine years.

Standby Letters of Credit and Surety Bonds

We have entered into standby letters of credit and surety bonds with financial institutions primarily relating to the guarantee of our future performance on certain contracts and security agreements. Contingent liabilities on outstanding letters of credit agreements and surety bonds aggregated approximately $4,548 and $5,070 as of December 31, 2023 and 2022.

Company Owned Life Insurance

McDonnell Douglas Corporation insured its executives with Company Owned Life Insurance (COLI), which are life insurance policies with a cash surrender value. Although we do not use COLI currently, these obligations from the merger with McDonnell Douglas are still a commitment at this time. We have loans in place to cover costs paid or incurred to carry the underlying life insurance policies. As of December 31, 2023 and 2022, the cash surrender value was $360 and $376 and the total loans were $ 334 and $346. As we have the right to offset the loans against the cash surrender value of the policies, we present the net asset in Other assets on the Consolidated Statements of Financial Position as of December 31, 2023 and 2022.

Supply Chain Financing Programs

The Company has supply chain financing programs in place under which participating suppliers may elect to obtain payment from an intermediary. The Company confirms the validity of invoices from participating suppliers and agrees to pay the intermediary an amount based on invoice totals. The majority of amounts payable under these programs are due within 30 to 90 days but may extend up to 12 months. At December 31, 2023 and 2022, Accounts payable included $2.9 billion and $2.5 billion payable to suppliers who have elected to participate in these programs. We do not believe that future changes in the availability of supply chain financing would have a significant impact on our liquidity.

Government Assistance

Certain states and localities in which we operate offer or have offered various business incentives related to investment and/or job creation. Between 2010 and 2016, we received cash grants totaling $346 related to our investment in operations in South Carolina. The grants were recorded in Accrued liabilities and are being amortized, primarily to inventory, over the useful life of the Property, plant and equipment extending through 2052. During 2023 and 2022, we amortized $10 and $11 to Inventories, and recorded a benefit of $ 12 and $5 in cost of sales. At December 31, 2023 and 2022, Inventories included a benefit of $62 and $64 and Accrued liabilities included a balance of $ 97 and $106.

We are eligible to claim tax refunds from the State of Missouri and City of Irving, Texas primarily related to job creation and retention through 2031. During 2023 and 2022, we received $22 and $30 in cash and recorded a benefit of $ 28 and $21 in cost of sales. At December 31, 2023 and 2022, Other current assets includes receivables of $26 and $20. As of December 31, 2023, $ 60 of refunds, plus interest, is subject to clawback if we fail to meet certain conditions, including employment levels.

We are eligible to claim cash grants through 2032 of up to $ 62, related to operations in Queensland, Australia. During 2023 and 2022, $ 5 and $7 cash was received and recorded as a benefit in cost of sales. At December 31, 2023, $4 is subject to clawback if we fail to meet certain conditions, including employment levels.

Industrial Revenue Bonds (IRB) issued by St. Louis County and the city of St. Charles, Missouri were used to finance the purchase and/or construction of real and personal property at our St. Louis and St. Charles sites. Tax benefits associated with IRBs include twelve-year property tax abatements and sales tax exemptions from St. Louis County and a 22 year property tax abatement and sales tax exemption from the city of St. Charles. We record these properties on our Consolidated Statements of Financial Position. We have also purchased the IRBs, and therefore, are the bondholders as well as the borrower/lessee of the properties purchased with the IRB proceeds. The liabilities and IRB assets are equal and are reported net in the Consolidated Statements of Financial Position. As of December 31, 2023 and 2022, the assets and liabilities associated with the IRBs were $333 and $271.

Recoverable Costs on Government Contracts

Our final incurred costs for each year are subject to audit and review for allowability by the U.S. government, which can result in payment demands related to costs they believe should be disallowed. We work with the U.S. government to assess the merits of claims and where appropriate reserve for amounts disputed. If we are unable to satisfactorily resolve disputed costs, we could be required to record an earnings charge and/or provide refunds to the U.S. government.

Fixed-Price Contracts

Long-term contracts that are contracted on a fixed-price basis could result in losses in future periods. Certain of the fixed-price contracts are for the development of new products, services and related technologies. This development work scope is inherently uncertain and subject to significant variability in estimates of the cost and time required to complete the work by us and our suppliers. The operational and technical complexities of fixed-price development contracts create financial risk, which could trigger additional earnings charges, termination provisions, order cancellations, or other financially significant exposure.

VC-25B Presidential Aircraft

The Company's firm fixed-price contract for the Engineering and Manufacturing Development (EMD) effort on the U.S. Air Force's (USAF) VC-25B Presidential Aircraft, commonly known as Air Force One, is a $4 billion program to develop and modify two 747-8 commercial aircraft. During 2022, we increased

the reach-forward loss on the contract by $ 1,452. This year we made progress completing engineering and production requirements. During 2023, we increased the reach-forward loss on the contract by $482 driven by engineering changes to support the build and installation process; the resolution of supplier negotiations; and factory performance related to labor instability. While we have provisioned for all of our anticipated costs to complete the contract, risk remains that we may record additional losses in future periods.

KC-46A Tanker

In 2011, we were awarded a contract from the USAF to design, develop, manufacture, and deliver four next generation aerial refueling tankers as well as priced options for 13 annual production lots totaling 179 aircraft. Since 2016, the USAF has authorized ten low rate initial production (LRIP) lots for a total of 139 aircraft, including lots 9 and 10 that were authorized in 2023. The EMD contract and authorized LRIP lots total approximately $ 27 billion as of December 31, 2023.

During 2022, we increased the reach-forward loss on the KC-46A Tanker program by $ 1,374. During 2023, we increased the reach-forward loss on the KC-46A Tanker program by $ 309 primarily resulting from factory disruption and additional rework due to a supplier quality issue. As of December 31, 2023, we had approximately $125 of capitalized precontract costs and $ 48 of potential termination liabilities to suppliers related to unexercised future lots. Risk remains that we may record additional losses in future periods.

MQ-25

In the third quarter of 2018, we were awarded the MQ-25 EMD contract by the U.S. Navy. The contract is a fixed-price contract that now includes development and delivery of seven aircraft and test articles at a contract price of $ 890. In connection with winning the competition, we recognized a reach-forward loss of $291 in the third quarter of 2018. During 2022, we increased the MQ-25 reach-forward loss by $ 579. During 2023, we increased the reach-forward loss by $231 primarily driven by production and flight testing delays as well as higher than anticipated production costs to complete EMD aircraft attributable to recent factory performance. Risk remains that we may record additional losses in future periods.

T-7A Red Hawk EMD Contract & Production Options

In 2018, we were awarded the T-7A Red Hawk program. The EMD portion of the contract is a $ 860 fixed-price contract and includes five aircraft and seven simulators. During the year ended December 31, 2022, we recorded earnings charges of $ 203 related to the T-7A Red Hawk fixed-price EMD contract, which had a reach-forward loss at December 31, 2022. The production portion of the contract includes 11 production lots for aircraft and related services for 346 T-7A Red Hawk aircraft that we believe are probable of being exercised. We expect the first production and support contract option to be exercised in 2025. During 2022, we increased the reach-forward loss by $552. During 2023, we increased the reach-forward loss by $ 275 primarily reflecting higher estimated production costs. At December 31, 2023, we had approximately $185 of capitalized precontract costs and $ 249 of potential termination liabilities to suppliers related to future production lots. Risk remains that we may record additional losses in future periods.

Commercial Crew

National Aeronautics and Space Administration (NASA) has contracted us to design and build the CST-100 Starliner spacecraft to transport crews to the International Space Station and in the second quarter of 2022 we successfully completed the uncrewed Orbital Flight Test. During 2022, we increased the reach-forward loss by $288. During 2023, we also increased the reach-forward loss by $ 288 primarily as a result of delaying the crewed flight test previously scheduled for July 2023 following

notification by a parachute supplier of an issue identified through testing. A crewed flight test is now planned for April 2024. At December 31, 2023, we had approximately $226 of capitalized precontract costs and $ 160 of potential termination liabilities to suppliers related to unauthorized future missions. Risk remains that we may record additional losses in future periods.

Note 14 – Arrangements with Off-Balance Sheet Risk

We enter into arrangements with off-balance sheet risk in the normal course of business, primarily in the form of guarantees.

The following table provides quantitative data regarding our third party guarantees. The maximum potential payments represent a “worst-case scenario” and do not necessarily reflect amounts that we expect to pay. The carrying amount of liabilities represents the amount included in Accrued liabilities.

	Maximum Potential Payments		Estimated Proceeds from Collateral/ Recourse		Carrying Amount of Liabilities	
December 31,	**2023**	2022	**2023**	2022	**2023**	2022
Contingent repurchase commitments	**$404**	$514	**$404**	$514		
Credit guarantees	**15**	45			**$14**	$27

Contingent Repurchase Commitments In conjunction with signing a definitive agreement for the sale of commercial aircraft, we have entered into contingent repurchase commitments with certain customers wherein we agree to repurchase the sold aircraft at a specified price, generally 10 to 15 years after delivery. Our repurchase of the aircraft is contingent upon entering into a mutually acceptable agreement for the sale of additional new aircraft in the future. The commercial aircraft repurchase price specified in contingent repurchase commitments is generally lower than the expected fair value at the specified repurchase date. Estimated proceeds from collateral/recourse in the table above represent the lower of the contracted repurchase price or the expected fair value of each aircraft at the specified repurchase date.

If a future sale agreement is reached and a customer elects to exercise its right under a contingent repurchase commitment, the contingent repurchase commitment becomes a trade-in commitment. Our historical experience is that contingent repurchase commitments infrequently become trade-in commitments.

Credit Guarantees We have issued credit guarantees where we are obligated to make payments to a guaranteed party in the event that the original lessee or debtor does not make payments or perform certain specified services. Generally, these guarantees have been extended on behalf of guaranteed parties with less than investment-grade credit. Current outstanding credit guarantees expire through 2036.

Other Indemnifications In conjunction with our sales of Electron Dynamic Devices, Inc. and Rocketdyne Propulsion and Power businesses and our BCA facilities in Wichita, Kansas and Tulsa and McAlester, Oklahoma, we agreed to indemnify, for an indefinite period, the buyers for costs relating to pre-closing environmental conditions and certain other items. We are unable to assess the potential number of future claims that may be asserted under these indemnifications, nor the amounts thereof (if any). As a result, we cannot estimate the maximum potential amount of future payments under these indemnities. To the extent that claims have been made under these indemnities and/or are probable and reasonably estimable, liabilities associated with these indemnities are included in the environmental liability disclosure in Note 13.

Note 15 – Debt

In the third quarter of 2023, we entered into a $ 3,000 five-year revolving credit agreement expiring in August 2028 and a $ 800 364-day revolving credit agreement expiring in August 2024. The 364-day credit facility has a one-year term out option which allows us to extend the maturity of any borrowings until August 2025. The legacy three-year revolving credit agreement expiring in August 2025, which consists of $ 3,000 of total commitments, and the legacy five-year revolving credit agreement expiring in October 2024, as amended, which consists of $ 3,200 of total commitments, each remain in effect. As of December 31, 2023, we had $10,000 currently available under credit line agreements. We continue to be in full compliance with all covenants contained in our debt or credit facility agreements.

Interest incurred, including amounts capitalized, was $2,560, $2,650 and $2,790 for the years ended December 31, 2023, 2022 and 2021, respectively. Total Company interest payments, net of amounts capitalized, were $2,408, $2,572 and $2,583 for the years ended December 31, 2023, 2022 and 2021, respectively.

Short-term debt and current portion of long-term debt at December 31 consisted of the following:

	2023	2022
Unsecured debt	**$5,072**	$5,103
Finance lease obligations	**77**	65
Other notes	**55**	22
Total	**$5,204**	$5,190

Debt at December 31 consisted of the following:

	2023	2022
Unsecured debt		
1.17% - 2.50% due through 2026	**$10,135**	11,846
2.60% - 3.20% due through 2030	**6,071**	6,412
3.25% - 3.90% due through 2059	**9,584**	9,576
3.95% - 5.15% due through 2059	**11,024**	14,035
5.71% - 6.63% due through 2060	**13,015**	13,011
6.88% - 8.75% due through 2043	**1,855**	1,854
Other debt and notes		
Finance lease obligations due through 2044	**253**	206
Other notes	**370**	61
Total debt	**$52,307**	$57,001

Scheduled principal payments for debt and minimum finance lease obligations for the next five years are as follows:

	2024	2025	2026	2027	2028
Debt and other notes	$5,128	$4,581	$7,983	$3,300	$1,800
Minimum finance lease obligations	$84	$76	$55	$24	$3

Note 16 – Postretirement Plans

Many of our employees have earned benefits under defined benefit pension plans. The majority of employees that had participated in defined benefit pension plans have transitioned to a company-funded defined contribution retirement savings plan.

We fund our major pension plans through trusts. Pension assets are placed in trust solely for the benefit of the plans' participants and are structured to maintain liquidity that is sufficient to pay benefit obligations as well as to keep pace over the long-term with the growth of obligations for future benefit payments.

We also have other postretirement benefits (OPB) other than pensions which consist principally of health care coverage for eligible retirees and qualifying dependents, and to a lesser extent, life insurance to certain groups of retirees. Retiree health care is provided principally until age 65 for approximately three-fourths of those participants who are eligible for retiree health care coverage. Certain employee groups, including employees covered by most United Auto Workers bargaining agreements, are provided lifetime health care coverage.

The funded status of the plans is measured as the difference between the plan assets at fair value and the projected benefit obligation (PBO). We have recognized the aggregate of all overfunded plans in Other assets and the aggregate of all underfunded plans in either Accrued retiree health care or Accrued pension plan liability, net. The portion of the amount by which the actuarial present value of benefits included in the PBO exceeds the fair value of plan assets, payable in the next 12 months, is reflected in Accrued liabilities.

The components of net periodic benefit (income)/cost were as follows:

	Pension			Other Postretirement Benefits		
Years ended December 31,	**2023**	2022	2021	**2023**	2022	2021
Service cost	**$2**	$3	$3	**$49**	$72	$87
Interest cost	**2,820**	2,080	1,988	**148**	98	97
Expected return on plan assets	**(3,441)**	(3,789)	(3,848)	**(9)**	(10)	(7)
Amortization of prior service credits	**(81)**	(81)	(80)	**(22)**	(35)	(35)
Recognized net actuarial loss/(gain)	**173**	913	1,219	**(175)**	(111)	(56)
Settlement/curtailment (gain)/loss		(4)	193			
Net periodic benefit (income)/cost	**($527)**	($878)	($525)	**($9)**	$14	$86
Net periodic benefit cost included in Loss from operations	**$2**	$3	$3	**$62**	$79	$90
Net periodic benefit income included in Other income, net	**(529)**	(881)	(528)	**(58)**	(58)	(1)
Net periodic benefit (income)/cost included in Loss before income taxes	**($527)**	($878)	($525)	**$4**	$21	$89

The following tables show changes in the benefit obligation, plan assets and funded status of both pensions and OPB for the years ended December 31, 2023 and 2022. Benefit obligation balances presented below reflect the PBO for our pension plans and accumulated postretirement benefit obligations (APBO) for our OPB plans.

	Pension		Other Postretirement Benefits	
	2023	2022	**2023**	2022
Change in benefit obligation				
Beginning balance	**$55,117**	$75,635	**$2,978**	$4,092
Service cost	**2**	3	**49**	72
Interest cost	**2,820**	2,080	**148**	98
Amendments		1		
Actuarial loss/(gain)	**1,217**	(17,605)	**(152)**	(914)
Gross benefits paid	**(4,837)**	(4,971)	**(375)**	(406)
Subsidies			**2**	39
Exchange rate adjustment	**6**	(26)	**1**	(3)
Ending balance	**$54,325**	$55,117	**$2,651**	$2,978
Change in plan assets				
Beginning balance at fair value	**$49,825**	$67,813	**$140**	$172
Actual return on plan assets	**3,756**	(13,141)	**23**	(27)
Company contribution		2		
Plan participants' contributions			**4**	6
Benefits paid	**(4,698)**	(4,824)	**(4)**	(11)
Exchange rate adjustment	**8**	(25)		
Ending balance at fair value	**$48,891**	$49,825	**$163**	$140
Amounts recognized in statement of financial position at December 31 consist of:				
Other assets	**$1,219**	$987	**$81**	$21
Accrued liabilities	**(137)**	(138)	**(336)**	(356)
Accrued retiree health care			**(2,233)**	(2,503)
Accrued pension plan liability, net	**(6,516)**	(6,141)		
Net amount recognized	**($5,434)**	($5,292)	**($2,488)**	($2,838)

Amounts recognized in Accumulated other comprehensive loss (AOCI) at December 31 were as follows:

	Pension		Other Postretirement Benefits	
	2023	2022	**2023**	2022
Net actuarial loss/(gain)	**$18,175**	$17,448	**($1,852)**	($1,862)
Prior service credits	**(1,143)**	(1,224)	**(19)**	(41)
Total recognized in AOCI	**$17,032**	$16,224	**($1,871)**	($1,903)

The accumulated benefit obligation (ABO) for all pension plans was $ 53,671 and $54,481 at December 31, 2023 and 2022. Key information for our plans with ABO and PBO in excess of plan assets as of December 31 was as follows:

	2023	2022
Accumulated benefit obligation	**$47,665**	$48,134
Fair value of plan assets	**41,666**	42,491

	2023	2022
Projected benefit obligation	**$48,320**	$48,770
Fair value of plan assets	**41,666**	42,491

Assumptions

The following assumptions, which are the weighted average for all plans, are used to calculate the benefit obligation at December 31 of each year and the net periodic benefit cost for the subsequent year.

December 31,	**2023**	2022	2021
Discount rate:			
Pension	**5.10 %**	5.40 %	2.80 %
Other postretirement benefits	**5.00 %**	5.30 %	2.50 %
Expected return on plan assets	**6.00 %**	6.00 %	6.30 %
Rate of compensation increase	**4.30 %**	4.30 %	4.30 %
Interest crediting rates for cash balance plans	**5.00 %**	5.00 %	5.00 %

The discount rate for each plan is determined based on the plans' expected future benefit payments using a yield curve developed from high quality bonds that are rated as Aa or better by at least half of the four rating agencies utilized as of the measurement date. The yield curve is fitted to yields developed from bonds at various maturity points. Bonds with the ten percent highest and the ten percent lowest yields are omitted. The present value of each plan's benefits is calculated by applying the discount rates to projected benefit cash flows.

The pension fund's expected return on plan assets assumption is derived from a review of actual historical returns achieved by the pension trust and anticipated future long-term performance of individual asset classes. While consideration is given to historical returns, the assumption represents a long-term, prospective return. The expected return on plan assets component of the net periodic benefit cost for the upcoming plan year is determined based on the expected return on plan assets assumption and the market-related value of plan assets (MRVA). Since our adoption of the accounting standard for pensions in 1987, we have determined the MRVA based on a five-year moving average of plan assets. As of December 31, 2023, the MRVA was approximately $8,466 more than the fair market value of assets.

Assumed health care cost trend rates were as follows:

December 31,	**2023**	2022	2021
Health care cost trend rate assumed next year	**5.50 %**	5.50 %	4.50 %
Ultimate trend rate	**4.50 %**	4.50 %	4.50 %
Year that trend reaches ultimate rate	**2028**	2028	2021

Plan Assets

Investment Strategy The overall objective of our pension assets is to earn a rate of return over time to satisfy the benefit obligations of the pension plans and to maintain sufficient liquidity to pay benefits and address other cash requirements of the pension fund. Specific investment objectives for our long-term investment strategy include reducing the volatility of pension assets relative to pension liabilities, achieving a competitive total investment return, achieving diversification between and within asset classes and managing other risks. Investment objectives for each asset class are determined based on specific risks and investment opportunities identified.

We periodically update our long-term, strategic asset allocations. We use various analytics to determine the optimal asset mix and consider plan liability characteristics, liquidity characteristics, funding requirements, expected rates of return and the distribution of returns. A key element of our strategy is to de-risk the plan as the funded status of the plan increases. During 2023, we completed a strategy review including an asset/liability study and, as a result, target allocations were updated with a modest increase to risk assets. The changes in the asset allocation are reflected in the table below. We identify investment benchmarks to evaluate performance for the asset classes in the strategic asset allocation that are market-based and investable where possible. Actual allocations to each asset class vary from target allocations due to periodic investment strategy changes, market value fluctuations, the length of time it takes to fully implement investment allocation positions, and the timing of benefit payments and contributions. Short-term investments and exchange-traded derivatives are used to rebalance the actual asset allocation to the target asset allocation. The asset allocation is monitored and rebalanced frequently. The actual and target allocations by asset class for the pension assets at December 31 were as follows:

	Actual Allocations		Target Allocations	
Asset Class	**2023**	2022	**2023**	2022
Fixed income	**60 %**	63 %	**59 %**	63 %
Global equity	**19**	14	**20**	20
Private equity	**8**	8	**7**	4
Real estate and real assets	**7**	8	**7**	7
Hedge funds	**6**	7	**7**	6
Total	**100 %**	100 %	**100 %**	100 %

Fixed income securities are invested primarily in a diversified portfolio of long duration instruments as well as Emerging Market, Structured, High Yield and Private Debt. Global equity securities are invested in a diversified portfolio of U.S. and non-U.S. companies, across various industries and market capitalizations.

Private equity investment vehicles are primarily limited partnerships (LPs) that mainly invest in U.S. and non-U.S. leveraged buyout, venture capital, growth and special situation strategies. Real estate and real assets include global private investments that may be held through investments in LPs or other fund structures. Real estate includes, but is not limited to, investments in office, retail, apartment and industrial properties. Real assets include, but are not limited to, investments in natural resources (such as energy, farmland and timber), commodities and infrastructure.

Hedge fund investments seek to capitalize on inefficiencies identified across and within different asset classes or markets. Hedge fund strategy types include, but are not limited to, directional, event driven, relative value and long-short.

Investment managers are retained for explicit investment roles specified by contractual investment guidelines. Certain investment managers are authorized to use derivatives, such as equity or bond

futures, swaps, options and currency futures or forwards. Derivatives are used to achieve the desired market exposure of a security or an index, transfer value-added performance between asset classes, achieve the desired currency exposure, adjust portfolio duration or rebalance the total portfolio to the target asset allocation.

As a percentage of total pension assets, derivative net notional amounts were 38.3% and 37.1% for fixed income, including to-be-announced mortgage-backed securities and treasury forwards, and 2.1% and (5.6%) for global equity and commodities at December 31, 2023 and 2022.

In November 2020, the Company elected to contribute $ 3,000 of our common stock to the pension fund. An independent fiduciary was retained to manage and liquidate the stock over time at its discretion. At December 31, 2022, plan assets included $1,782 of our common stock, which was liquidated during 2023.

Risk Management In managing the pension assets, we review and manage risk associated with funded status risk, interest rate risk, market risk, counterparty risk, liquidity risk and operational risk. Liability matching and asset class diversification are central to our risk management approach and are integral to the overall investment strategy. Further, asset classes are constructed to achieve diversification by investment strategy, by investment manager, by industry or sector and by holding. Investment manager guidelines for publicly traded assets are specified and are monitored regularly through the custodian. Credit parameters for counterparties have been established for managers permitted to trade over-the-counter derivatives. Valuation is governed through several types of procedures, including reviews of manager valuation policies, custodian valuation processes, pricing vendor practices, pricing reconciliation and periodic, security-specific valuation testing.

Fair Value Measurements The following table presents our plan assets using the fair value hierarchy as of December 31, 2023 and 2022. The fair value hierarchy has three levels based on the reliability of the inputs used to determine fair value. Level 1 refers to fair values determined based on quoted prices in active markets for identical assets. Level 2 refers to fair values estimated using significant other observable inputs and Level 3 includes fair values estimated using significant unobservable inputs.

	December 31, 2023				December 31, 2022			
	Total	**Level 1**	**Level 2**	**Level 3**	Total	Level 1	Level 2	Level 3
Fixed income securities:								
Corporate	**$17,809**		**$17,750**	**$59**	$15,095		$15,025	$70
U.S. government and agencies	**6,822**		**6,822**		7,827		7,827	
Mortgage backed and asset backed	**505**		**344**	**161**	664		502	162
Municipal	**816**		**816**		843		811	32
Sovereign	**720**		**720**		706		706	
Other	**9**	**$6**		**3**	8	$8		
Derivatives:								
Assets	**69**		**69**		36		36	
Liabilities					(87)		(87)	
Cash equivalents and other short-term investments	**326**		**326**		571		571	
Equity securities:								
U.S. common and preferred stock	**3,391**	**3,391**			2,931	2,931		
Non-U.S. common and preferred stock	**2,204**	**2,204**			2,023	2,023		
Boeing company stock					1,782	1,782		
Derivatives:								
Assets								
Liabilities					(1)		(1)	
Private equity								
Real estate and real assets:								
Real estate								
Real assets	**385**	**349**	**33**	**3**	362	310	47	5
Derivatives:								
Assets					1		1	
Liabilities					(8)		(7)	(1)
Total	**$33,056**	**$5,950**	**$26,880**	**$226**	$32,753	$7,054	$25,431	$268
Fixed income common/collective/pooled funds	**$1,378**				$1,511			
Fixed income other	**1,364**				832			
Equity common/collective/ pooled funds	**2,702**				2,757			
Private equity	**4,102**				4,239			
Real estate and real assets	**3,138**				3,525			
Hedge funds	**2,751**				3,391			
Total investments measured at NAV as a practical expedient	**$15,435**				$16,255			
Cash	**$86**				$409			
Receivables	**438**				541			
Payables	**(124)**				(133)			
Total	**$48,891**				$49,825			

Fixed income securities are primarily valued upon a market approach, using matrix pricing and considering a security's relationship to other securities for which quoted prices in an active market may be available, or an income approach, converting future cash flows to a single present value amount.

Inputs used in developing fair value estimates include reported trades, broker quotes, benchmark yields and base spreads.

Common/collective/pooled funds are typically common or collective trusts valued at their net asset values (NAVs) that are calculated by the investment manager or sponsor of the fund and have daily or monthly liquidity.

Derivatives included in the table above are over-the-counter and are primarily valued using an income approach with inputs that include benchmark yields, swap curves, cash flow analysis, rating agency data and interdealer broker rates. Exchange-traded derivative positions are reported in accordance with changes in daily variation margin which is settled daily and therefore reflected in the payables and receivables portion of the table.

Cash equivalents and other short-term investments (which are used to pay benefits) are held in a separate account which consists of a commingled fund (with daily liquidity) and separately held short-term securities and cash equivalents. All of the investments in this cash vehicle are valued daily using a market approach with inputs that include quoted market prices for similar instruments. In the event a market price is not available for instruments with an original maturity of one year or less, amortized cost is used as a proxy for fair value. Common and preferred stock equity securities are primarily valued using a market approach based on the quoted market prices of identical instruments.

Private equity and private debt NAV valuations are based on the valuation of the underlying investments, which include inputs such as cost, operating results, discounted future cash flows and market based comparable data. For those investments reported on a one-quarter lagged basis (primarily LPs) we use NAVs, adjusted for subsequent cash flows and significant events.

Real estate and real asset NAVs are based on the valuation of the underlying investments, which include inputs such as cost, discounted future cash flows, independent appraisals and market based comparable data. For those investments reported on a one-quarter lagged basis (primarily LPs), NAVs are adjusted for subsequent cash flows and significant events. Publicly traded infrastructure stocks are valued using a market approach based on quoted market prices of identical instruments. Exchange-traded commodities futures positions are reported in accordance with changes in daily variation margin which is settled daily and therefore reflected in the payables and receivables portion of the table.

Hedge fund NAVs are generally based on the valuation of the underlying investments. This is primarily done by applying a market or income valuation methodology depending on the specific type of security or instrument held.

Investments in private equity, private debt, real estate, real assets and hedge funds are primarily calculated and reported by the General Partner, fund manager or third-party administrator. Additionally, some investments in fixed income and equity are made via commingled vehicles and are valued in a similar fashion. Pension assets invested in commingled and LP structures rely on the NAV of these investments as the practical expedient for the valuations.

The following tables summarizes the changes of Level 3 assets, reconciled by asset class, held during the years ended December 31, 2023 and 2022. Transfers into and out of Level 3 are reported at the beginning-of-year values.

	January 1 2023 Balance	Net Realized and Unrealized Gains/(Losses)	Net Purchases, Issuances and Settlements	Net Transfers Into/(Out of) Level 3	December 31 2023 Balance
Fixed income securities:					
Corporate	**$70**	**$5**	**($16)**		**$59**
U.S. government and agencies			**(1)**	**$1**	
Mortgage backed and asset backed	**162**	**7**	**10**	**(18)**	**161**
Municipal	**32**		**(5)**	**(27)**	
Other		**3**			**3**
Real assets	**4**	**(1)**			**3**
Total	**$268**	**$14**	**($12)**	**($44)**	**$226**

	January 1 2022 Balance	Net Realized and Unrealized Gains/(Losses)	Net Purchases, Issuances and Settlements	Net Transfers Into/(Out of) Level 3	December 31 2022 Balance
Fixed income securities:					
Corporate	$53	($19)	$3	$33	$70
Mortgage backed and asset backed	102	(11)	16	55	162
Municipal	29	(14)	9	8	32
Sovereign	9			(9)	
Equity securities:					
Non-U.S. common and preferred stock	5	(45)	(2)	42	
Real assets		(1)	5		4
Total	$198	($90)	$31	$129	$268

For the year ended December 31, 2023, the changes in unrealized gains/(losses) for Level 3 assets still held at December 31, 2023 were $ 2 for corporate fixed income securities, $6 for mortgage backed and asset backed fixed income securities, and $ 3 for other fixed income securities. For the year ended December 31, 2022, the changes in unrealized gains/(losses) for Level 3 assets still held at December 31, 2022 were ($16) for corporate fixed income securities, ($11) for mortgage backed and asset backed fixed income securities, ($ 14) for municipal fixed income securities, and ($ 1) for real asset securities.

OPB Plan Assets The majority of OPB plan assets are invested in a balanced index fund which is comprised of approximately 60% equities and 40% debt securities. The index fund is valued using a market approach based on the quoted market price of an identical instrument (Level 1). The expected rate of return on these assets does not have a material effect on the net periodic benefit cost.

Cash Flows

Contributions Required pension contributions under the Employee Retirement Income Security Act (ERISA), as well as rules governing funding of our non-US pension plans, are not expected to be significant in 2024. We do not expect to make discretionary contributions to our pension plans in 2024.

Estimated Future Benefit Payments The table below reflects the total pension benefits expected to be paid from the plans or from our assets, including both our share of the benefit cost and the participants' share of the cost, which is funded by participant contributions. OPB payments reflect our portion only.

Year(s)	2024	2025	2026	2027	2028	2029-2033
Pensions	$4,524	$4,425	$4,345	$4,241	$4,143	$19,106
Other postretirement benefits:						
Gross benefits paid	358	341	319	295	269	1,004
Subsidies	(12)	(13)	(13)	(13)	(13)	(61)
Net other postretirement benefits	$346	$328	$306	$282	$256	$943

Termination Provisions

Certain of the pension plans provide that, in the event there is a change in control of the Company which is not approved by the Board of Directors and the plans are terminated within five years thereafter, the assets in the plan first will be used to provide the level of retirement benefits required by ERISA, and then any surplus will be used to fund a trust to continue present and future payments under the postretirement medical and life insurance benefits in our group insurance benefit programs.

Should we terminate certain pension plans under conditions in which the plan's assets exceed that plan's obligations, the U.S. government will be entitled to a fair allocation of any of the plan's assets based on plan contributions that were reimbursed under U.S. government contracts.

Defined Contribution Plans

We provide certain defined contribution plans to all eligible employees. The principal plans are the Company-sponsored 401(k) plans. The expense for these defined contribution plans was $1,564, $1,260 and $1,268 in 2023, 2022 and 2021, respectively.

Note 17 – Share-Based Compensation and Other Compensation Arrangements

Share-Based Compensation

Our 2023 Incentive Stock Plan, permits awards of incentive and non-qualified stock options, stock appreciation rights, restricted stock or units, performance restricted stock or units, and other stock and cash-based awards to our employees, officers, directors, consultants, and independent contractors. The aggregate number of shares of our stock authorized for issuance under the plan is 12,900,000, plus shares that remain available, undelivered, or retained under our 2003 Incentive Stock Plan, as amended and restated. Following approval of our 2023 Incentive Stock Plan in 2023, no further awards have been or may be granted under our 2003 Incentive Stock Plan.

Shares issued as a result of stock option exercises or conversion of stock unit awards will be funded out of treasury shares, except to the extent there are insufficient treasury shares, in which case new shares will be issued. We believe we currently have adequate treasury shares to satisfy these issuances during 2024.

Share-based plans expense is primarily included in Total costs and expenses and General and administrative expense, as well as a portion allocated to production as inventoried costs. The share-based plans expense and related income tax benefit were as follows:

Years ended December 31,	**2023**	2022	2021
Restricted stock units and other awards	**$697**	$726	$840
Income tax benefit (before consideration of valuation allowance)	**$157**	$178	$148

Stock Options

Options have been granted to our executive officers that are scheduled to vest and become exercisable three years after the grant date and expire ten years after the grant date. If an executive terminates employment because of retirement, layoff, disability, or death, the executive (or beneficiary) may receive some or all of their stock options depending on certain age and service conditions. The fair values of the stock options granted were estimated using a Monte-Carlo simulation model using the assumptions presented below. The model includes no expected dividend yield.

Stock options granted during 2023 were not material.

On February 16, 2022, we granted 348,769 premium-priced stock options to our executive officers as part of our long-term incentive program. These stock options have an exercise price equal to 120% of the fair market value of our stock on the date of grant. If certain performance measures are met, the exercise price is reduced to 110% of the grant date fair market value of our stock.

On February 17, 2021, we granted 342,986 premium-priced stock options to our executive officers as part of our long-term incentive program. These stock options have an exercise price equal to 120% of the fair market value of our stock on the date of grant. During 2021, we also granted 148,322 stock options to certain executives, of which 40,322 had an exercise price equal to 120% of the fair market value of our stock on the date of grant, and the remaining 108,000 had an exercise price equal to the fair market value of our stock on the date of grant. The grant date fair market values of these awards were not significant.

Grant Year	Grant Date	Expected Life	Expected Volatility	Risk Free Interest Rate	Grant Date Fair Value Per Option
2022	2/16/2022	6.8 years	36.6 %	2.0 %	$83.04
2021	2/17/2021	6.6 years	37.8 %	1.3 %	$74.63

Options granted through January 2014 had an exercise price equal to the fair market value of our stock on the date of grant and expire 10 years after the date of grant. These stock options vested over a period of three years and were fully vested as of December 31, 2017.

Stock option activity for the year ended December 31, 2023 was as follows:

	Shares	Weighted Average Exercise Price Per Option	Weighted Average Remaining Contractual Life (Years)	Aggregate Intrinsic Value
Number of shares under option:				
Outstanding at beginning of year	1,390,769	$178.18		
Granted	30,000	210.68		
Exercised	(597,030)	77.06		
Forfeited	(31,077)	260.26		
Outstanding at end of year	792,662	$252.35	7.7	$7
Exercisable at end of year	7,953	$197.07	3.1	$1

The total intrinsic value of options exercised during the years ended December 31, 2023, 2022 and 2021 was $ 80, $75 and $84, with a related tax benefit of $18, $17 and $19, respectively. At December 31, 2023, there was $ 11 of total unrecognized compensation cost related to options which is expected to be recognized over a weighted average period of 1.2 years.

Restricted Stock Units

In February 2023, 2022 and 2021, we granted to our executives 327,523, 1,804,541 and 980,077 restricted stock units (RSUs) as part of our long-term incentive program with grant date fair values of $214.35, $217.48 and $215.70 per unit, respectively. On July 29, 2022, we also granted 2,568,112 RSUs with a grant date fair value of $157.69 per unit as part of our long-term incentive program, accelerating awards planned for 2023 to retain executives. The RSUs granted under this program will generally vest and settle in common stock (on a one-for-one basis) on the third anniversary of the grant date. If an executive terminates employment because of retirement, layoff, disability, or death, the executive (or beneficiary) may receive some or all of their stock units depending on certain age and service conditions. In all other cases, the RSUs will not vest and all rights to the stock units will terminate. These RSUs are labeled executive long-term incentive program in the table below.

In addition to RSUs awarded under our long-term incentive programs, we granted RSUs to certain executives and employees. These RSUs are labeled other RSUs in the table below.

The fair values of all RSUs are estimated using the average of the high and low stock prices on the date of grant.

RSU activity for the year ended December 31, 2023 was as follows:

	Executive Long-Term Incentive Program	Employee Long-Term Incentive Program	Other
Number of units:			
Outstanding at beginning of year	6,117,900	4,373,807	958,694
Granted	411,134		142,711
Forfeited	(229,226)	(106,087)	(32,253)
Distributed	(950,318)	(4,242,199)	(303,642)
Outstanding at end of year	5,349,490	25,521	765,510
Undistributed vested units	1,630,233	25,407	34,600
Unrecognized compensation cost	$366		$55
Weighted average remaining amortization period *(years)*	1.5		1.6

Performance Restricted Stock Units

On February 16, 2023, we granted 199,899 performance restricted stock units (PRSU) to our executive officers as part of our long-term incentive program that will result in that number of PRSUs being paid out if the target performance metric is achieved. The PRSUs granted under this program have a grant date fair value of $214.35 per unit. The award payout can range from 0% to 200% of the initial PRSU grant based on cumulative free cash flow achievement over the period January 1, 2023 through December 31, 2025 as compared to the target set at the start of the performance period. The PRSUs granted under this program will vest at the payout amount determined on the third anniversary of the grant date and settle in common stock (on a one-for-one basis). If an executive terminates employment because of retirement, layoff, disability, or death, the executive (or beneficiary) remains eligible under the award and, if the award is earned, may receive some or all of their stock units depending on certain age and service conditions. In all other cases, the PRSUs will not vest and all rights to the stock units will terminate. During the year ended December 31, 2023, there were no forfeitures or distributions. At December 31, 2023, unrecognized compensation cost was $31, and the weighted average remaining amortization period was 2.1 years.

Performance-Based Restricted Stock Units

Performance-Based Restricted Stock Units (PBRSUs) are stock units that pay out based on the Company's total shareholder return (TSR) as compared to a group of peer companies over a three-year period. The award payout can range from 0% to 200% of the initial PBRSU grant. During 2023, these performance awards expired with a payout of 0%.

Employee Stock Purchase Plan

The Company has an employee stock purchase plan which permits eligible employees to purchase Boeing stock at 95% of the fair market value on the last trading day of each three-month period using payroll deduction. The aggregate number of shares of our stock authorized for issuance under the plan is 12,000,000. During the year ended December 31, 2023, approximately 216,719 shares were purchased at an average price of $ 193.52 per share.

Deferred Compensation

The Company has deferred compensation plans which permit certain employees and executives to defer a portion of their salary, bonus, certain other incentive awards and retirement contributions.

Participants can diversify these amounts among 23 investment funds including a Boeing stock unit account.

Total expense/(income) related to deferred compensation was $ 188, ($117) and $126 in 2023, 2022 and 2021, respectively. As of December 31, 2023 and 2022, the deferred compensation liability which is being marked to market was $1,640 and $1,499.

Note 18 – Shareholders' Equity

As of December 31, 2023 and 2022, there were 1,200,000,000 shares of common stock and 20,000,000 shares of preferred stock authorized. No preferred stock has been issued.

Changes in Share Balances

The following table shows changes in each class of shares:

	Common Stock	Treasury Stock
Balance at January 1, 2021	1,012,261,159	429,941,021
Issued		(6,904,556)
Acquired		307,242
Balance at December 31, 2021	1,012,261,159	423,343,707
Issued		(8,877,047)
Acquired		204,723
Balance at December 31, 2022	**1,012,261,159**	**414,671,383**
Issued		**(13,651,201)**
Acquired		**1,725,954**
Balance at December 31, 2023	**1,012,261,159**	**402,746,136**

Additional Paid-in Capital

During the year ended December 31, 2023, Additional paid-in capital included a decrease of $ 267 related to a non-cash transaction to purchase shares in a consolidated subsidiary from the noncontrolling interests.

Accumulated Other Comprehensive Loss

Changes in AOCI by component for the years ended December 31, 2023, 2022 and 2021 were as follows:

	Currency Translation Adjustments	Unrealized Gains and Losses on Certain Investments	Unrealized Gains and Losses on Derivative Instruments	Defined Benefit Pension Plans & Other Postretirement Benefits	Total [1]
Balance at January 1, 2021	($30)	$1	($43)	($17,061)	($17,133)
Other comprehensive (loss)/income before reclassifications	(75)		55	4,268 [2]	4,248
Amounts reclassified from AOCI			(6)	1,232 [3]	1,226
Net current period Other comprehensive (loss)/income	(75)		49	5,500	5,474
Balance at December 31, 2021	($105)	$1	$6	($11,561)	($11,659)
Other comprehensive (loss)/income before reclassifications	(62)	(1)	(40)	1,529 [2]	1,426
Amounts reclassified from AOCI			10 [4]	673 [3]	683
Net current period Other comprehensive (loss)/income	(62)	(1)	(30)	2,202	2,109
Balance at December 31, 2022	**($167)**		**($24)**	**($9,359)**	**($9,550)**
Other comprehensive income/(loss) before reclassifications	**33**	**2**	**41**	**(722)** [2]	**(646)**
Amounts reclassified from AOCI			**(5)**	**(104)** [3]	**(109)**
Net current period Other comprehensive income/(loss)	**33**	**2**	**36**	**(826)**	**(755)**
Balance at December 31, 2023	**($134)**	**$2**	**$12**	**($10,185)**	**($10,305)**

[1] Net of tax.

[2] Primarily related to remeasurement of assets and benefit obligations related to the Company's pension and other postretirement benefit plans resulting in an actuarial (loss)/gain of ($722), $1,533 and $4,262 (net of tax of $13, ($22) and ($32)) for the years ended December 31, 2023, 2022 and 2021. See Note 16.

[3] Amounts reclassified from AOCI for the year ended December 31, 2023, primarily related to amortization of prior service credits totaling ($ 102) (net of tax of $1). Amounts reclassified from AOCI for the years ended December 31, 2022 and 2021, primarily related to amortization of actuarial losses totaling $791 and $1,155 (net of tax of ($11) and ($8)). These are included in net periodic pension cost. See Note 16.

[4] Included losses of $39 (net of tax of ($11)) from cash flow hedges reclassified to Other income, net because the forecasted transactions are not probable of occurring.

Note 19 – Derivative Financial Instruments

Cash Flow Hedges

Our cash flow hedges include foreign currency forward contracts, commodity swaps and commodity purchase contracts. We use foreign currency forward contracts to manage currency risk associated with certain expected sales and purchased through 2031. We use commodity derivatives, such as fixed-price purchase commitments and swaps to hedge against potentially unfavorable price changes for commodities used in production. Our commodity contracts hedge forecasted transactions through 2028.

Derivative Instruments Not Receiving Hedge Accounting Treatment

We have entered into agreements to purchase and sell aluminum to address long-term strategic sourcing objectives and non-U.S. business requirements. These agreements are derivative instruments for accounting purposes. The quantities of aluminum in these agreements offset and are priced at prevailing market prices. We also hold certain foreign currency forward contracts and commodity swaps which do not qualify for hedge accounting treatment.

Notional Amounts and Fair Values

The notional amounts and fair values of derivative instruments in the Consolidated Statements of Financial Position as of December 31 were as follows:

	Notional amounts[1]		Other assets		Accrued liabilities	
	2023	2022	**2023**	2022	**2023**	2022
Derivatives designated as hedging instruments:						
Foreign exchange contracts	**$4,120**	$2,815	**$85**	$23	**($63)**	($122)
Commodity contracts	**514**	602	**83**	115	**(8)**	(9)
Derivatives not receiving hedge accounting treatment:						
Foreign exchange contracts	**254**	462	**1**	5	**(32)**	(42)
Commodity contracts	**115**	412		2	**(2)**	(1)
Total derivatives	**$5,003**	$4,291	**169**	145	**(105)**	(174)
Netting arrangements			**(47)**	(33)	**47**	33
Net recorded balance			**$122**	$112	**($58)**	($141)

[1] Notional amounts represent the gross contract/notional amount of the derivatives outstanding.

Gains/(losses) associated with our hedging transactions and forward points recognized in Other comprehensive income are presented in the following table:

Years ended December 31,	**2023**	2022	2021
Recognized in Other comprehensive income, net of taxes:			
Foreign exchange contracts	**$61**	($118)	($47)
Commodity contracts	**(20)**	78	102

Gains/(losses) associated with our hedging transactions and forward points reclassified from AOCI to earnings are presented in the following table:

Years ended December 31,	**2023**	2022	2021
Foreign exchange contracts			
Revenues		$1	
Costs and expenses	**($15)**	7	$13
General and administrative	**(17)**	(12)	8
Commodity contracts			
Costs and expenses	**$31**	$31	($18)
General and administrative expense	**7**	10	5

During the twelve months ended December 31, 2022, we reclassified losses associated with certain cash flow hedges of $ 50 from AOCI to Other income, net because it became probable the forecasted transactions would not occur. Gains/(losses) related to undesignated derivatives on foreign exchange and commodity cash flow hedging transactions recognized in Other income, net were insignificant for the years ended December 31, 2023, 2022 and 2021.

Based on our portfolio of cash flow hedges, we expect to reclassify losses of $ 39 (pre-tax) out of AOCI into earnings during the next 12 months.

We have derivative instruments with credit-risk-related contingent features. If we default on our five-year credit facility, our derivative counterparties could require settlement for foreign exchange and certain commodity contracts with original maturities of at least five years. The fair value of those contracts in a net liability position at December 31, 2023 was $16. For other particular commodity contracts, our counterparties could require collateral posted in an amount determined by our credit ratings. At December 31, 2023, there was no collateral posted related to our derivatives.

Note 20 – Fair Value Measurements

The fair value hierarchy has three levels based on the reliability of the inputs used to determine fair value. Level 1 refers to fair values determined based on quoted prices in active markets for identical assets. Level 2 refers to fair values estimated using significant other observable inputs, and Level 3 includes fair values estimated using significant unobservable inputs. The following table presents our assets and liabilities that are measured at fair value on a recurring basis and are categorized using the fair value hierarchy.

	December 31, 2023			December 31, 2022		
	Total	**Level 1**	**Level 2**	Total	Level 1	Level 2
Assets						
Money market funds	**$1,514**	**$1,514**		$1,797	$1,797	
Available-for-sale debt investments:						
Commercial paper	**291**		**$291**	256		$256
Corporate notes	**183**		**183**	195		195
U.S. government agencies	**25**		**25**	47		47
Other equity investments	**44**	**44**		10	10	
Derivatives	**122**		**122**	112		112
Total assets	**$2,179**	**$1,558**	**$621**	$2,417	$1,807	$610
Liabilities						
Derivatives	**($58)**		**($58)**	($141)		($141)
Total liabilities	**($58)**		**($58)**	($141)		($141)

Money market funds, available-for-sale debt investments and equity securities are valued using a market approach based on the quoted market prices or broker/dealer quotes of identical or comparable instruments.

Derivatives include foreign currency and commodity contracts. Our foreign currency forward contracts are valued using an income approach based on the present value of the forward rate less the contract rate multiplied by the notional amount. Commodity derivatives are valued using an income approach based on the present value of the commodity index prices less the contract rate multiplied by the notional amount.

Certain assets have been measured at fair value on a nonrecurring basis, using significant unobservable inputs (Level 3). The following table presents the nonrecurring losses recognized for the years ended December 31 due to long-lived asset impairment, and the fair value and asset classification of the related assets as of the impairment date:

	2023		2022	
	Fair Value	**Total Losses**	Fair Value	Total Losses
Investments		**($18)**		($31)
Operating lease equipment			$47	(7)
Property, plant and equipment	**$14**	**(26)**		(19)
Other Assets		**(2)**	15	(55)
Total	**$14**	**($46)**	$62	($112)

Investments, Property, plant and equipment, and Other assets were primarily valued using an income approach based on the discounted cash flows associated with the underlying assets. The fair value of the impaired operating lease equipment is derived by calculating a median collateral value from a consistent group of third party aircraft value publications. The values provided by the third party aircraft publications are derived from their knowledge of market trades and other market factors. Management reviews the publications quarterly to assess the continued appropriateness and consistency with market trends. Under certain circumstances, we adjust values based on the attributes and condition of the specific aircraft or equipment, usually when the features or use of the aircraft vary significantly from the more generic aircraft attributes covered by third party publications, or on the expected net sales price for the aircraft.

Fair Value Disclosures

The fair values and related carrying values of financial instruments that are not required to be remeasured at fair value on the Consolidated Statements of Financial Position at December 31 were as follows:

	December 31, 2023				
	Carrying Amount	**Total Fair Value**	**Level 1**	**Level 2**	**Level 3**
Assets					
Notes receivable, net	**$257**	**$270**		**$270**	
Liabilities					
Debt, excluding finance lease obligations	**(52,055)**	**(51,039)**		**(51,039)**	

	December 31, 2022				
	Carrying Amount	Total Fair Value	Level 1	Level 2	Level 3
Assets					
Notes receivable, net	$385	$403		$403	
Liabilities					
Debt, excluding finance lease obligations	(56,794)	(52,856)		(52,856)	

The fair values of notes receivable are estimated with discounted cash flow analysis using interest rates currently offered on loans with similar terms to borrowers of similar credit quality. The fair value of our debt that is traded in the secondary market is classified as Level 2 and is based on current market yields. For our debt that is not traded in the secondary market, the fair value is classified as Level 2 and is based on our indicative borrowing cost derived from dealer quotes or discounted cash flows. With regard to other financial instruments with off-balance sheet risk, it is not practicable to estimate the fair value of our indemnifications and financing commitments because the amount and timing of those arrangements are uncertain. Items not included in the above disclosures include cash, restricted cash, time deposits and other deposits, commercial paper, money market funds, Accounts receivable, Unbilled receivables, Other current assets, Accounts payable and long-term payables. The carrying values of those items, as reflected in the Consolidated Statements of Financial Position, approximate their fair value at December 31, 2023 and 2022. The fair value of assets and liabilities whose carrying value approximates fair value is determined using Level 2 inputs, with the exception of cash (Level 1).

Note 21 – Legal Proceedings

Various legal proceedings, claims and investigations related to products, contracts, employment and other matters are pending against us. In addition, we are subject to various government inquiries and investigations from which civil, criminal or administrative proceedings could result or have resulted in the past. Such proceedings involve or could involve claims by the government for fines, penalties, compensatory and treble damages, restitution and/or forfeitures. Under U.S. government regulations, a company, or one or more of its operating divisions or subdivisions, can also be suspended or debarred from government contracts, or lose its export privileges, based on the results of investigations. We believe, based upon current information, that the outcome of any currently pending legal proceeding, claim, or government dispute, inquiry or investigation will not have a material effect on our financial position, results of operations or cash flows. With respect to the matters set forth below, we cannot reasonably estimate a range of loss in excess of recorded amounts, if any.

Multiple legal actions and inquiries were initiated as a result of the October 29, 2018 accident of Lion Air Flight 610 and the March 10, 2019 accident of Ethiopian Airlines Flight 302. On January 7, 2021, we entered into a Deferred Prosecution Agreement (DPA) with the U.S. Department of Justice that resolved the Department of Justice's investigation into us regarding the evaluation of the 737 MAX by the Federal Aviation Administration (FAA). Among other obligations, the DPA includes a three-year reporting period, which ended earlier this month. The Department is currently considering whether we fulfilled our obligations under the DPA and whether to move to dismiss the information, which motion will require court approval.

During 2019, we entered into agreements with Embraer S.A. (Embraer) to establish joint ventures that included the commercial aircraft and services operations of Embraer, of which we were expected to acquire an 80 percent ownership stake for $4,200, as well as a joint venture to promote and develop new markets for the C-390 Millennium. In 2020, we exercised our contractual right to terminate these agreements based on Embraer's failure to meet certain required closing conditions. Embraer has disputed our right to terminate the agreements, and the dispute is currently in arbitration, which we currently expect to be resolved in 2024.

Note 22 – Segment and Revenue Information

Segment results reflect the realignment of the Boeing Customer Financing team and portfolio into the BCA segment during the first quarter of 2023. Interest and debt expense now includes interest and debt expense previously attributable to Boeing Capital and classified as a component of Total Costs and Expenses ("Cost of Sales"). Prior period amounts have been reclassified to conform to current period presentation.

Our primary profitability measurement to review segment operating results is Loss from operations. We operate in three reportable segments: BCA, BDS, and BGS. All other activities fall within Unallocated items, eliminations and other. See page 56 for the Summary of Business Segment Data, which is an integral part of this note.

BCA develops, produces and markets commercial jet aircraft principally to the commercial airline industry worldwide. Revenue on commercial aircraft contracts is recognized at the point in time when an aircraft is completed and accepted by the customer.

BDS engages in the research, development, production and modification of the following products and related services: manned and unmanned military aircraft and weapons systems, surveillance and engagement, strategic defense and intelligence systems, satellite systems and space exploration. BDS revenue is generally recognized over the contract term (over time) as costs are incurred.

BGS provides parts, maintenance, modifications, logistics support, training, data analytics and information-based services to commercial and government customers worldwide. BGS segment revenue and costs include certain products and services provided to other segments. Revenue on commercial spare parts contracts is recognized at the point in time when a spare part is delivered to the customer. Revenue on other contracts is generally recognized over the contract term (over time) as costs are incurred.

While our principal operations are in the United States, Canada and Australia, some key suppliers and subcontractors are located in Europe and Japan. Revenues, including foreign military sales, are reported by customer location and consisted of the following:

Years ended December 31,	**2023**	2022	2021
Europe	**$10,520**	$7,916	$8,967
Asia	**10,013**	8,393	5,845
Middle East	**6,594**	5,047	4,653
Oceania	**1,655**	1,576	1,147
Canada	**1,256**	1,612	969
Africa	**825**	418	239
Latin America, Caribbean and other	**1,524**	2,412	1,376
Total non-U.S. revenues	**32,387**	27,374	23,196
United States	**45,380**	39,218	39,076
Estimated potential concessions and other considerations to 737 MAX customers	**27**	16	14
Total revenues	**$77,794**	$66,608	$62,286

Revenues from the U.S. government (including foreign military sales through the U.S. government), primarily recorded at BDS and BGS, represented 37%, 40% and 49% of consolidated revenues for 2023, 2022 and 2021, respectively. Approximately 4% of operating assets were located outside the United States as of December 31, 2023 and 2022.

The following tables present BCA, BDS and BGS revenues from contracts with customers disaggregated in a number of ways, such as geographic location, contract type and the method of revenue recognition. We believe these best depict how the nature, amount, timing and uncertainty of our revenues and cash flows are affected by economic factors.

BCA revenues by customer location consisted of the following:

Years ended December 31,	**2023**	2022	2021
Revenue from contracts with customers:			
Asia	**$6,328**	$4,488	$2,816
Europe	**6,172**	4,085	4,387
Middle East	**4,311**	2,003	1,098
Other non-U.S.	**2,431**	3,042	1,683
Total non-U.S. revenues	**19,242**	13,618	9,984
United States	**14,501**	12,275	9,614
Estimated potential concessions and other considerations to 737 MAX customers	**27**	16	14
Total revenues from contracts with customers	**33,770**	25,909	19,612
Intersegment revenues, eliminated on consolidation	**131**	117	102
Total segment revenues	**$33,901**	$26,026	$19,714
Revenue recognized on fixed-price contracts	**100 %**	100 %	100 %
Revenue recognized at a point in time	**99 %**	99 %	99 %

BDS revenues on contracts with customers, based on the customer's location, consisted of the following:

Years ended December 31,	**2023**	2022	2021
Revenue from contracts with customers:			
U.S. customers	**$20,051**	$17,144	$19,869
Non-U.S. customers[(1)]	**4,882**	6,018	6,671
Total segment revenue from contracts with customers	**$24,933**	$23,162	$26,540
Revenue recognized over time	**99 %**	99 %	99 %
Revenue recognized on fixed-price contracts	**58 %**	60 %	68 %
Revenue from the U.S. government[(1)]	**91 %**	89 %	89 %

(1) Includes revenues earned from foreign military sales through the U.S. government.

BGS revenues consisted of the following:

Years ended December 31,	**2023**	2022	2021
Revenue from contracts with customers:			
Commercial	**$11,020**	$9,560	$7,527
Government	**7,751**	7,681	8,553
Total revenues from contracts with customers	**18,771**	17,241	16,080
Intersegment revenues eliminated on consolidation	**356**	370	248
Total segment revenues	**$19,127**	$17,611	$16,328
Revenue recognized at a point in time	**51 %**	50 %	45 %
Revenue recognized on fixed-price contracts	**87 %**	88 %	86 %
Revenue from the U.S. government [(1)]	**30 %**	33 %	40 %

(1) Includes revenues earned from foreign military sales through the U.S. government.

Earnings in Equity Method Investments

During the years ended December 31, 2023, 2022, and 2021, our share of income from equity method investments was $ 70, $56, and $40, respectively. In 2023 and 2021, earnings in equity method investments were primarily driven by investments held at our BDS segment. In 2022, earnings in equity method investments were primarily driven by investments held in Unallocated items, eliminations and other.

Backlog

Our total backlog includes contracts that we and our customers are committed to perform. The value in backlog represents the estimated transaction prices on performance obligations to our customers for which work remains to be performed. Backlog is converted into revenue, primarily based on the cost incurred or at delivery and acceptance of products, depending on the applicable revenue recognition model.

Our backlog at December 31, 2023 was $ 520,195. We expect approximately 16% to be converted to revenue through 2024 and approximately 62% through 2027, with the remainder thereafter. There is significant uncertainty regarding the timing of when backlog will convert into revenue due to timing of 737 and 787 deliveries from inventory and timing of entry into service of the 777X, 737-7 and/or 737-10.

Unallocated Items, Eliminations and other

Unallocated items, eliminations and other include common internal services that support Boeing's global business operations and eliminations of certain sales between segments. We generally allocate costs to business segments based on the U.S. Government Cost Accounting Standards (CAS). Components of Unallocated items, eliminations and other (expense)/income are shown in the following table.

Years ended December 31,	**2023**	2022	2021
Share-based plans	**$62**	($114)	($174)
Deferred compensation	**(188)**	117	(126)
Amortization of previously capitalized interest	**(95)**	(95)	(107)
Research and development expense, net	**(315)**	(278)	(184)
Eliminations and other unallocated items	**(1,223)**	(1,134)	(636)
Unallocated items, eliminations and other	**($1,759)**	($1,504)	($1,227)

Pension and Other Postretirement Benefit Expense

Pension costs are allocated to BDS and BGS businesses supporting government customers using CAS, which employ different actuarial assumptions and accounting conventions than GAAP. These costs are allocable to government contracts. Other postretirement benefit costs are allocated to business segments based on CAS, which is generally based on benefits paid. FAS/CAS service cost adjustment represents the difference between the Financial Accounting Standards (FAS) pension and postretirement service costs calculated under GAAP and costs allocated to the business segments. Non-operating pension and postretirement expenses represent the components of net periodic benefit costs other than service cost. These expenses are included in Other income, net. Components of FAS/CAS service cost adjustment are shown in the following table:

Years ended December 31,	**2023**	2022	2021
Pension FAS/CAS service cost adjustment	**$799**	$849	$882
Postretirement FAS/CAS service cost adjustment	**257**	294	291
FAS/CAS service cost adjustment	**$1,056**	$1,143	$1,173

Assets

Segment assets are summarized in the table below.

December 31,	**2023**	2022
Commercial Airplanes	**$77,047**	$76,825
Defense, Space & Security	**14,921**	14,426
Global Services	**16,193**	16,149
Unallocated items, eliminations and other	**28,851**	29,700
Total	**$137,012**	$137,100

Assets included in Unallocated items, eliminations and other primarily consist of Cash and cash equivalents, Short-term and other investments, tax assets, capitalized interest and assets managed centrally on behalf of the three principal business segments and intercompany eliminations.

Capital Expenditures

Years ended December 31,	**2023**	2022	2021
Commercial Airplanes	**$420**	$218	$177
Defense, Space & Security	**192**	202	199
Global Services	**127**	130	94
Unallocated items, eliminations and other	**788**	672	510
Total	**$1,527**	$1,222	$980

Capital expenditures for Unallocated items, eliminations and other relate primarily to assets managed centrally on behalf of the three principal business segments.

Depreciation and Amortization

Years ended December 31,	**2023**	2022	2021
Commercial Airplanes	**$464**	$554	$594
Defense, Space & Security	**219**	238	233
Global Services	**320**	346	414
Centrally Managed Assets [(1)]	**858**	841	903
Total	**$1,861**	$1,979	$2,144

[(1)] Amounts shown in the table represent depreciation and amortization expense recorded by the individual business segments. Depreciation and amortization for centrally managed assets are included in segment operating earnings based on usage and occupancy. In 2023, $650 was included in the primary business segments, of which $311, $264 and $75 was included in BCA, BDS and BGS, respectively. In 2022, $ 644 was included in the primary business segments, of which $361, $230 and $53 was included in BCA, BDS and BGS, respectively. In 2021, $ 669 was included in the primary business segments, of which $387, $222 and $60 was included in BCA, BDS and BGS, respectively.

Note 23 – Subsequent Events

On January 5, 2024, an Alaska Airlines 737-9 flight made an emergency landing after a mid-exit door plug detached in flight. Following the accident, the Federal Aviation Administration (FAA) grounded and required inspections of all 737-9 aircraft with a mid-exit door plug, which constitute the large majority of the approximately 220 737-9 aircraft in the in-service fleet. On January 24, 2024, the FAA approved an

enhanced maintenance and inspection process that must be performed on each of the grounded 737-9 aircraft. Our 737-9 operators have begun returning their fleets to service, and many 737-9s have completed inspections and resumed revenue flights. All 737-9 aircraft in production will undergo this same enhanced inspection process prior to delivery.

On January 10, 2024, the FAA notified Boeing that the FAA has initiated an investigation into Boeing's quality control system.

On January 24, 2024, the FAA stated that it will not approve production rate increases or additional production lines for the 737 MAX until it is satisfied that Boeing is in full compliance with required quality control procedures.

We are currently unable to reasonably estimate what impact the accident and the related FAA actions will have on our financial position, results of operations and cash flows.

REPORT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

To the shareholders and the Board of Directors of The Boeing Company

Opinion on the Financial Statements

We have audited the accompanying consolidated statements of financial position of The Boeing Company and subsidiaries (the "Company") as of December 31, 2023 and 2022, the related consolidated statements of operations, comprehensive income, equity, and cash flows, for each of the three years in the period ended December 31, 2023, and the related notes (collectively referred to as the "financial statements"). In our opinion, the financial statements present fairly, in all material respects, the financial position of the Company as of December 31, 2023 and 2022, and the results of its operations and its cash flows for each of the three years in the period ended December 31, 2023, in conformity with accounting principles generally accepted in the United States of America.

We have also audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the Company's internal control over financial reporting as of December 31, 2023, based on criteria established in *Internal Control — Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission and our report dated January 31, 2024, expressed an unqualified opinion on the Company's internal control over financial reporting.

Basis for Opinion

These financial statements are the responsibility of the Company's management. Our responsibility is to express an opinion on the Company's financial statements based on our audits. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the financial statements are free of material misstatement, whether due to error or fraud. Our audits included performing procedures to assess the risks of material misstatement of the financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the financial statements. We believe that our audits provide a reasonable basis for our opinion.

Critical Audit Matters

The critical audit matters communicated below are matters arising from the current-period audit of the financial statements that were communicated or required to be communicated to the audit committee and that (1) relate to accounts or disclosures that are material to the financial statements and (2) involved our especially challenging, subjective, or complex judgments. The communication of critical audit matters does not alter in any way our opinion on the financial statements, taken as a whole, and we are not, by communicating the critical audit matters below, providing separate opinions on the critical audit matters or on the accounts or disclosures to which they relate.

Cost Estimates for Fixed-Price Development Contracts - Refer to Notes 1 and 13 to the financial statements

Critical Audit Matter Description

As more fully described in Notes 1 and 13 to the consolidated financial statements, the Company recognizes revenue over time for long-term contracts as goods are produced or services are rendered. The Company uses costs incurred as the method for determining progress, and revenue is recognized based on costs incurred to date plus an estimate of margin at completion. The process of estimating margin at completion involves estimating the costs to complete production of goods or rendering of services and comparing those costs to the estimated final revenue amount. Margins on fixed-price development contracts are inherently uncertain in that revenue is fixed while the estimates of costs required to complete these contracts are subject to significant variability. The operational and technical complexities of fixed-price development contracts create financial risk, which could increase the estimates of costs and result in lower margins or material reach-forward losses. The ongoing effects of supply chain and labor disruption compound these complexities and related financial risks.

Given the operational and technical complexities of certain of the Company's fixed-price development contracts, including the KC-46A Tanker, Commercial Crew, VC-25B Presidential Aircraft, T-7A Redhawk, and MQ-25 contracts and the limited amount of historical data available in certain instances and significant judgments necessary to estimate future costs at completion, auditing these estimates involved extensive audit effort, a high degree of auditor judgment, and required audit professionals with specialized industry experience.

How the Critical Audit Matter Was Addressed in the Audit

Our audit procedures related to the cost estimates for the KC-46A Tanker, Commercial Crew, VC-25B Presidential Aircraft, T-7A Redhawk, and MQ-25 fixed-price development contracts included the following, among others:

- We evaluated the appropriateness and consistency of management's methods used in developing its estimates.
- We evaluated the reasonableness of judgments made and significant assumptions used by management relating to key cost and schedule estimates, including the effects of supply chain and labor disruptions. We also evaluated the range and probabilities of reasonably possible outcomes, and where management set its point estimate within the range.
- We evaluated the appropriateness of the timing of the incorporation of changes to key estimates, including evaluating the timeline of key events and knowledge points that led to management's determination that a change in estimate was necessary.
- We inquired of project management, engineers, supply chain leadership, and others directly involved with the execution of contracts to evaluate management's ability to achieve the key

cost and schedule estimates, as well as evaluate project status and challenges which may affect total estimated costs to complete.

- We observed the project work site when key estimates related to tangible or physical progress of the project.
- We tested the accuracy and completeness of the key data used in developing estimates. We developed independent expectations of reasonable outcomes using, in part, the program's data and compared our expectations to management's estimates.
- We performed retrospective reviews when evaluating the thoroughness and precision of management's estimation process and effectiveness of the related internal controls by comparing actual outcomes to previous estimates and the related financial statement impact, and evaluating key judgements made by management when determining the timing of changes to key estimates.
- We tested the effectiveness of internal controls including, those over the review of significant judgments made and assumptions used to develop key estimates, key data used in developing the estimates and the mathematical extrapolation of such data.

Program Accounting Estimates for the 777X Program — Refer to Notes 1 and 7 to the financial statements

Critical Audit Matter Description

The introduction of new aircraft programs involves increased risk associated with meeting development, certification, and production schedules. The Company uses program accounting to compute cost of sales and margin for each commercial airplane sold. The use of program accounting requires estimating the costs over the expected life of each program. In particular, the level of effort to meet regulatory requirements and achieve certification may be challenging to predict, including potential delays in the timing of achieving certification that would delay entry into service and corresponding increases in estimated costs. Changes to the cost estimates related to regulatory requirements to achieve certification could occur, resulting in lower margins or material reach-forward losses. Auditing these estimated costs for the 777X program involved extensive audit effort, a high degree of auditor judgment, and required audit professionals with specialized industry experience.

How the Critical Audit Matter Was Addressed in the Audit

Our audit procedures related to the estimated costs for the 777X program included the following, among others:

- We evaluated the appropriateness and consistency of management's methods used in developing its cost estimates related to regulatory requirements to achieve certification.
- We performed procedures to evaluate new changes in estimated costs driven by changes in regulatory requirements to achieve certification.
- We evaluated the timeline of key events and knowledge points that informs management's determination to change the cost estimate related to regulatory requirements to achieve certification.
- We inquired of those directly involved with the certification of the aircraft to evaluate project status and challenges which may affect total estimated costs to certify the aircraft.

- We obtained and evaluated communications with regulatory bodies for consistency between management's certification timeline assumptions and cost estimates related to regulatory requirements.
- We tested the effectiveness of internal controls, including those over the data used in developing the cost estimates, the mathematical extrapolation of such data, and management's judgment regarding the range of possible outcomes relating to the specific cost estimates in the current regulatory environment.

/s/ Deloitte & Touche LLP

Chicago, Illinois

January 31, 2024

We have served as the Company's auditor since at least 1934; however, an earlier year could not be reliably determined.

REPORT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

To the shareholders and the Board of Directors of The Boeing Company

Opinion on Internal Control over Financial Reporting

We have audited the internal control over financial reporting of The Boeing Company and subsidiaries (the "Company") as of December 31, 2023, based on criteria established in *Internal Control — Integrated Framework (2013*) issued by the Committee of Sponsoring Organizations of the Treadway Commission (COSO). In our opinion, the Company maintained, in all material respects, effective internal control over financial reporting as of December 31, 2023, based on criteria established in *Internal Control — Integrated Framework (2013)* issued by COSO.

We have also audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the consolidated financial statements as of and for the year ended December 31, 2023 of the Company, and our report dated January 31, 2024 expressed an unqualified opinion on those financial statements.

Basis for Opinion

The Company's management is responsible for maintaining effective internal control over financial reporting and for its assessment of the effectiveness of internal control over financial reporting, included in the accompanying Management's Report on Internal Control Over Financial Reporting. Our responsibility is to express an opinion on the Company's internal control over financial reporting based on our audit. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audit in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether effective internal control over financial reporting was maintained in all material respects. Our audit included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, testing and evaluating the design and operating effectiveness of internal control based on the assessed risk, and performing such other procedures as we considered necessary in the circumstances. We believe that our audit provides a reasonable basis for our opinion.

Definition and Limitations of Internal Control over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (1) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

/s/ Deloitte & Touche LLP

Chicago, Illinois

January 31, 2024

Item 9. Changes in and Disagreements with Accountants on Accounting and Financial Disclosure

None.

Item 9A. Controls and Procedures

(a) Evaluation of Disclosure Controls and Procedures.

Our Chief Executive Officer and Chief Financial Officer have evaluated our disclosure controls and procedures as of December 31, 2023 and have concluded that these disclosure controls and procedures are effective to ensure that information required to be disclosed by us in the reports that we file or submit under the Securities Exchange Act of 1934 is recorded, processed, summarized and reported within the time periods specified in the Securities and Exchange Commission's rules and forms and is accumulated and communicated to our management, including the Chief Executive Officer and Chief Financial Officer, as appropriate to allow timely decisions regarding required disclosure.

(b) Management's Report on Internal Control Over Financial Reporting.

Our management is responsible for establishing and maintaining adequate internal control over financial reporting, as such term is defined in Exchange Act Rule 13a-15(f). Our management conducted an evaluation of the effectiveness of our internal control over financial reporting based on the framework in Internal Control – Integrated Framework (2013) issued by the Committee of Sponsoring Organizations of the Treadway Commission. Based on this evaluation under the framework in Internal Control – Integrated Framework, our management concluded that our internal control over financial reporting was effective as of December 31, 2023.

Our internal control over financial reporting as of December 31, 2023, has been audited by Deloitte & Touche LLP, an independent registered public accounting firm, as stated in their report which is included in Item 8 of this report and is incorporated by reference herein.

(c) Changes in Internal Controls Over Financial Reporting.

There were no changes in our internal control over financial reporting that occurred during the fourth quarter of 2023 that have materially affected or are reasonably likely to materially affect our internal control over financial reporting.

Item 9B. Other Information

During the three months ended December 31, 2023, none of our directors or officers adopted, modified or terminated a "Rule 10b5-1 trading arrangement" or a "non-Rule 10b5-1 trading arrangement" as such terms are defined under Item 408 of Regulation S-K.

During the three months ended December 31, 2023, the Company did not adopt, modify or terminate a "Rule 10b5-1 trading arrangement" as such term is defined under Item 408 of Regulation S-K.

Item 9C. Disclosure Regarding Foreign Jurisdictions that Prevent Inspections

None.

Part III

Item 10. Directors, Executive Officers and Corporate Governance

Our executive officers and their ages as of January 31, 2024, are as follows:

Name	Age	Principal Occupation or Employment/Other Business Affiliations
Brian R. Besanceney	51	Chief Communications Officer and Senior Vice President, Communications since August 2022. Prior to joining Boeing, Mr. Besanceney served as Senior Vice President and Chief Communications Officer for Walmart Inc. from April 2016 to August 2022. Prior to that he held executive-level positions for The Walt Disney Company including Senior Vice President of Public Affairs from 2010 to 2016 and Vice President of Public Affairs and Business Development for Disney's Parks and Resorts division from 2009 to 2010.
Stephen E. Biegun	60	Senior Vice President, Global Public Policy since April 2023. Prior to joining Boeing, Mr. Biegun served as senior Advisor to Macro Advisory Partners from August 2021 to April 2023, Deputy Secretary of State for the U.S. Department of State from December 2019 to January 2021, Special Representative for North Korea for the U.S. Department of State from September 2018 to January 2021 and Vice President, International Governmental Relations at Ford Motor Company from April 2004 to November 2018.
David L. Calhoun	66	President and Chief Executive Officer since January 2020 and a member of the Board of Directors since June 2009. Previously, Mr. Calhoun served as Senior Managing Director & Head of Private Equity Portfolio Operations at The Blackstone Group from January 2014 to January 2020. Prior to that, Mr. Calhoun served as Chairman of the Board of Nielsen Holdings plc from January 2014 to January 2016, as Chief Executive Officer of Nielsen Holdings plc from May 2010 to January 2014, and as Chairman of the Executive Board and Chief Executive Officer of The Nielsen Company B.V. from August 2006 to January 2014. Prior to joining Nielsen, he served as Vice Chairman of General Electric Company and President and Chief Executive Officer of GE Infrastructure. During his 26-year tenure at GE, he ran multiple business units including GE Transportation, GE Aircraft Engines, GE Employers Reinsurance Corporation, GE Lighting and GE Transportation Systems. Mr. Calhoun also serves on the board of directors of Caterpillar Inc.

Name	Age	Principal Occupation or Employment/Other Business Affiliations
Theodore Colbert III	50	Executive Vice President, President and Chief Executive Officer, Boeing Defense, Space & Security since April 2022. Mr. Colbert previously served as Executive Vice President, President and Chief Executive Officer, Boeing Global Services from October 2019 to March 2022; Chief Information Officer and Senior Vice President, Information Technology & Data Analytics from April 2016 to October 2019; Chief Information Officer and Vice President of Information Technology from November 2013 to April 2016; Vice President of Information Technology Infrastructure from December 2011 to November 2013; and Vice President of IT Business Systems from September 2010 to December 2011. Mr. Colbert serves on the board of directors of Archer-Daniels-Midland Company.
Michael D'Ambrose	66	Chief Human Resources Officer and Executive Vice President, Human Resources since June 2021. Prior to joining Boeing in July 2020 as Executive Vice Present, Human Resources, Mr. D'Ambrose served as Senior Vice President and Chief Human Resources Officer for Archer-Daniels-Midland Company from October 2006 to June 2020. Previously, he served in a series of executive-level business and human resources positions, including chief human resources officer at Citigroup, First Data Corporation and Toys 'R' Us, Inc.
Stanley A. Deal	59	Executive Vice President, President and Chief Executive Officer, Boeing Commercial Airplanes since October 2019. Mr. Deal joined Boeing in 1986, and his previous positions include Executive Vice President, President and Chief Executive Officer, Boeing Global Services from November 2016 to October 2019; Senior Vice President of Commercial Aviation Services from March 2014 to November 2016; Vice President and General Manager of Supply Chain Management and Operations for Commercial Airplanes from September 2011 to February 2014; Vice President of Supplier Management from February 2010 to August 2011; and Vice President of Asia Pacific Sales from December 2006 to January 2010.
Susan Doniz	54	Chief Information Officer and Senior Vice President, Information Technology & Data Analytics since May 2020. Prior to joining Boeing, Ms. Doniz served as Global Chief Information Officer of Qantas Airways Limited from January 2017 to April 2020; as strategic advisor to the Global CEO of SAP SE on transformation and technology issues in support of customers from September 2015 to January 2017; and Global Product, Digital Strategy and Chief Information Officer of AIMIA Inc. from June 2011 to January 2015.
Brett C. Gerry	52	Chief Legal Officer and Executive Vice President, Global Compliance since May 2020. Mr. Gerry previously served as Senior Vice President and General Counsel from May 2019 to May 2020; President of Boeing Japan from February 2016 to May 2019; Vice President and General Counsel, Boeing Commercial Airplanes from March 2009 to March 2016; and Chief Counsel, Network and Space Systems from September 2008 to March 2009.

Name	Age	Principal Occupation or Employment/Other Business Affiliations
Howard E. McKenzie	57	Chief Engineer and Executive Vice President, Engineering, Test & Technology since March 2023. Mr. McKenzie joined Boeing in 1987 and his previous positions include Vice President and Chief Engineer of Boeing Commercial Airplanes from August 2021 to March 2023; Vice President and Chief Engineer of Boeing Global Services from June 2020 to August 2021; Vice President of Boeing Test and Evaluation from June 2019 to June 2020; and Vice President and Chief Project Engineer for the 777 program from October 2017 to June 2019.
Brendan J. Nelson	65	Senior Vice President and President, Boeing International since January 2023. Dr. Nelson previously served as President of Boeing Australia, New Zealand and South Pacific from February 2020 to January 2023. Prior to joining Boeing, he served as the Director of the Australian War Memorial from December 2012 to December 2019 and as the Australian Ambassador to Belgium, Luxembourg, the European Union and NATO from February 2010 to November 2012.
Ziad S. Ojakli	56	Executive Vice President, Government Operations since October 2021. Prior to joining Boeing, Mr. Ojakli served as a managing partner and Senior Vice President of Global Government Affairs at SoftBank Group Corp. from August 2018 to September 2020. Prior to that, he served as Group Vice President, Government & Community Relations at Ford Motor Company from January 2004 to July 2018.
Stephanie F. Pope	51	Executive Vice President and Chief Operating Officer since January 2024. Ms. Pope joined Boeing in 1994, and her previous positions include Executive Vice President, President and Chief Executive Officer, Boeing Global Services from April 2022 to December 2023; Vice President and Chief Financial Officer of Boeing Commercial Airplanes from December 2020 to March 2022; Vice President and Chief Financial Officer of Boeing Global Services from January 2017 to December 2020; Vice President of Finance and Controller for Boeing Defense, Space & Security from August 2016 to December 2016; and Vice President, Financial Planning & Analysis from February 2013 to July 2016.
D. Christopher Raymond	59	Executive Vice President, President and Chief Executive Officer, Boeing Global Services since January 2024. Mr. Raymond joined Boeing in 1986 and his previous positions include Senior Vice President and Chief Sustainability Officer from October 2020 to December 2023; Vice President of Sustainability, Strategy and Corporate Development from April 2019 to October 2020; Vice President and General Manager of Autonomous Systems, a division within Boeing Defense, Space & Security, from April 2015 to July 2018; and a series of other Vice President and General Manager of several businesses for Boeing Defense, Space & Security.

Name	Age	Principal Occupation or Employment/Other Business Affiliations
Brian J. West	54	Executive Vice President and Chief Financial Officer since August 2021. Prior to joining Boeing, Mr. West served as Chief Financial Officer of Refinitiv Holdings (a London Stock Exchange Group business and provider of financial markets data and infrastructure) from November 2018 to June 2021. Prior to that, he served as Chief Financial Officer and Executive Vice President of Operations of Oscar Insurance Corporation from January 2016 to October 2018. Mr. West served as Chief Operating Officer of Nielsen Holdings plc from March 2014 to December 2015 and as Chief Financial Officer of Nielsen Holdings plc (or its predecessor) from February 2007 to March 2014. Prior to joining Nielsen, Mr. West was employed by the General Electric Company as the Chief Financial Officer of its GE Aviation division from June 2005 to February 2007 and Chief Financial Officer of its GE Aviation Services division from March 2004 to June 2005. Prior to that, Mr. West held several senior financial positions across General Electric Company businesses, including Plastics, NBC, Energy and Transportation.

Codes of Ethics. We have adopted (1) The Boeing Company Code of Ethical Business Conduct for the Board of Directors; and (2) The Boeing Code of Conduct that applies to all employees, including our CEO (collectively, the Codes of Conduct). The Codes of Conduct are posted on our website, www.boeing.com/company/general-info/corporate-governance.page. We intend to disclose promptly on our website any amendments to, or waivers of, the Codes of Conduct covering our CEO, CFO and/or Controller.

No family relationships exist among any of the executive officers, directors or director nominees.

Additional information required by this item will be included under the captions “Election of Directors,” “Stock Ownership Information” and “Board Committees” in our proxy statement, which will be filed with the SEC no later than 120 days after December 31, 2023 (the “2024 Proxy Statement”), and that information is incorporated by reference herein.

Item 11. Executive Compensation

The information required by this item will be included under the captions “Compensation Discussion and Analysis,” “Compensation of Executive Officers,” “Compensation of Directors,” in the 2024 Proxy Statement, and that information is incorporated by reference herein.

Item 12. Security Ownership of Certain Beneficial Owners and Management and Related Stockholder Matters

Equity Compensation Plan Information

Our equity compensation plans approved by our shareholders provide for the issuance of common stock to officers and other employees, directors and consultants. The following table sets forth information regarding outstanding options and units, and shares available for future issuance under these plans as of December 31, 2023:

Plan Category	Number of shares to be issued upon exercise of outstanding options and units	Weighted-average exercise price of outstanding options	Number of securities remaining available for future issuance under equity compensation plans (excluding shares reflected in column (a))
	(a)	(b)	(c)
Equity compensation plans approved by shareholders[1]			
Stock options	792,662	$252.35	
Deferred compensation	604,179		
Other stock units[2]	6,540,319		
Equity compensation plans not approved by shareholders	None	None	None
Total	7,937,160	$252.35	29,851,662 [3][4]

[1] Includes the employee stock purchase plan and the 2023 Incentive Stock Plan and its predecessor plan.

[2] Includes 399,798 shares issuable in respect of Performance Restricted Stock Units. The shares included represent the maximum number of shares that may be issued upon vesting if the maximum performance goal is achieved for the three-year performance period.

[3] Includes 11,783,281 shares issuable under our employee stock purchase plan. There were 60,365 shares subject to purchase under the employee stock purchase plan as of December 31, 2023.

[4] Excludes shares of common stock that may be offered and sold under our 401(k) Plan. On February 5, 2021, 30,000,000 shares of common stock were registered for this purpose, of which 12,998,806 remained available as of December 31, 2023.

For further information, see Note 17 to our Consolidated Financial Statements.

The additional information required by this item will be included under the caption “Stock Ownership Information” in the 2024 Proxy Statement, and that information is incorporated by reference herein.

Item 13. Certain Relationships and Related Transactions, and Director Independence

The information required by this item will be included under the captions “Related Person Transactions” and “Director Independence” in the 2024 Proxy Statement, and that information is incorporated by reference herein.

Item 14. Principal Accountant Fees and Services

Our independent registered public accounting firm is Deloitte & Touche LLP (PCAOB ID No. 34).

The information required by this item will be included under the caption “Independent Auditor Fees” in the 2024 Proxy Statement, and that information is incorporated by reference herein.

PART IV

Item 15. Exhibits and Financial Statement Schedules

(a) List of documents filed as part of this report:

1. **Financial Statements**

   Our consolidated financial statements are as set forth under Item 8 of this report on Form 10-K.

2. **Financial Statement Schedules**

   All schedules are omitted because they are not applicable, not required or the information is included in the consolidated financial statements.

3. **Exhibits**

3.1	Amended and Restated Certificate of Incorporation of The Boeing Company dated May 5, 2006 (Exhibit 3.1 to the Company’s Current Report on Form 8-K dated May 1, 2006)
3.2	By-Laws of The Boeing Company, as amended and restated, effective August 29, 2023 (Exhibit 3.1 to the Company's Form 10-Q for the quarter ended September 30, 2023)
4.1	Description of The Boeing Company Securities Registered under Section 12 of the Exchange Act (Exhibit 4.1 to the Company’s Form 10-K for the year ended December 31, 2019)
10.1	364-Day Credit Agreement, dated as of August 24, 2023, among The Boeing Company for itself and on behalf of its Subsidiaries, as a Borrower, the Lenders party hereto, Citibank, N.A., as administrative agent, JPMorgan Chase Bank, N.A. as syndication agent and Citibank, N.A. and JPMorgan Chase Bank N.A., as joint lead arrangers and joint book managers (Exhibit 10.1 to the Company’s Form 10-Q for the quarter ended September 30, 2023)
10.2	Five-Year Credit Agreement, dated as of August 24, 2023, among The Boeing Company for itself and on behalf of its Subsidiaries, as a Borrower, the Lenders party hereto, Citibank, N.A., as administrative agent, JPMorgan Chase Bank, N.A., as syndication agent and Citibank N.A. and JPMorgan Chase Bank, N.A., as joint lead arrangers and joint book managers (Exhibit 10.2 to the Company’s Form 10-Q for the quarter ended September 30, 2023)
10.3	Three-Year Credit Agreement, dated as of August 25, 2022, among The Boeing Company for itself and on behalf of its Subsidiaries, as a Borrower, the Lenders party hereto, Citibank, N.A., as administrative agent, JPMorgan Chase Bank, N.A., as syndication agent and Citibank, N.A. and JPMorgan Chase Bank, N.A., as joint lead arrangers and joint book managers (Exhibit 10.2 to the Company’s Current Report on Form 8-K, dated August 25, 2022)

10.4	Five-Year Credit Agreement, dated as of October 30, 2019, among The Boeing Company, for itself and on behalf of its Subsidiaries, as a Borrower, the Lenders party hereto, Citibank, N.A., as administrative agent, JPMorgan Chase Bank, N.A., as syndication agent and Citibank N.A. and JPMorgan Chase Bank, N.A., as joint lead arrangers and joint book managers (Exhibit 10.2 to the Company's Current Report on Form 8-K dated October 30, 2019
10.5	Amendment No. 1, dated as of August 25, 2022, to Five-Year Credit Agreement, dated as of October 30, 2019, among The Boeing Company for itself and on behalf of its Subsidiaries, as a Borrower, the Lenders party hereto, Citibank, N.A., as administrative agent, JPMorgan Chase Bank, N.A., as syndication agent and Citibank N.A. and JPMorgan Chase Bank, N.A., as joint lead arrangers and joint book managers (Exhibit 10.3 to the Company's Current Report on Form 8-K, dated August 25, 2022)
10.6	Joint Venture Master Agreement, dated as of May 2, 2005, by and among Lockheed Martin Corporation, The Boeing Company and United Launch Alliance, L.L.C. (Exhibit (10)(i) to the Company's Form 10-Q for the quarter ended June 30, 2005)
10.7	Delta Inventory Supply Agreement, dated as of December 1, 2006, by and between United Launch Alliance, L.L.C. and The Boeing Company (Exhibit (10)(vi) to the Company's Form 10-K for the year ended December 31, 2006)
10.8	Deferred Prosecution Agreement dated January 6, 2021 (Exhibit 10.1 to the Company's Current Report on Form 8-K dated January 6, 2021)
10.9	Summary of Non employee Director Compensation (Exhibit 10.6 to the Company's Form 10-K for the year ended December 31, 2019)*
10.10	Deferred Compensation Plan for Directors of The Boeing Company, as amended and restated effective January 1, 2008 (Exhibit 10.2 to the Company's Current Report on Form 8-K dated October 28, 2007)*
10.11	The Boeing Company Annual Incentive Plan, as amended and restated February 24, 2020 (formerly known as the Incentive Compensation Plan for Employees of The Boeing Company and Subsidiaries) (Exhibit 10.2 to the Company's Form 10-Q for the quarter ended March 31, 2020)*
10.12	The Boeing Company 1997 Incentive Stock Plan, as amended effective May 1, 2000 and further amended effective January 1, 2008 (Exhibit 10.5 to the Company's Current Report on Form 8-K dated October 28, 2007)*
10.13	Supplemental Executive Retirement Plan for Employees of The Boeing Company, as amended and restated as of June 1, 2021*
10.14	The Boeing Company Executive Supplemental Savings Plan, as amended and restated effective January 1, 2022* (Exhibit 10.13 to the Company's Form 10-K for the year ended December 31, 2022)
10.15	The Boeing Company Executive Layoff Benefits Plan, as amended and restated effective January 1, 2017 (Exhibit (10)(xviii) to the Company's Form 10-K for the year ended December 31, 2016)*
10.16	The Boeing Company 2003 Incentive Stock Plan, as amended and restated effective December 9, 2021 (Exhibit 10.16 to the Company's Form 10-K for the year ended December 31, 2021)*
10.17	The Boeing Company 2023 Incentive Stock Plan, effective April 18, 2023 (Exhibit 10.9 to the Company's Form 10-Q for the quarter ended March 31, 2023)*

10.18	Form of U.S. Notice of Terms of Non-Qualified Stock Option (Exhibit 10.1 to the Company's 10-Q for the quarter ended March 31, 2021)*
10.19	Form of International Notice of Terms of Non-Qualified Stock Option (Exhibit 10.2 to the Company's 10-Q for the quarter ended March 31, 2021)*
10.20	Form of U.S. Notice of Terms of Non-Qualified Stock Option for CEO (Exhibit 10.3 to the Company's 10-Q for the quarter ended March 31, 2021)*
10.21	Form of U.S. Notice of Terms of Restricted Stock Units (Exhibit 10.4 to the Company's 10-Q for the quarter ended March 31, 2021)*
10.22	Form of International Notice of Terms of Restricted Stock Units (Exhibit 10.5 to the Company's 10-Q for the quarter ended March 31, 2021)*
10.23	Form of U.S. Notice of Terms of Restricted Stock Units for CEO (Exhibit 10.6 to the Company's 10-Q for the quarter ended March 31, 2021)*
10.24	Form of Notice of Terms of Supplemental Restricted Stock Units (Exhibit 10.2 to the Company's Current Report on Form 8-K dated June 29, 2021)*
10.25	Form of Notice of Terms of Supplemental Non-Qualified Stock Option (Exhibit 10.3 to the Company's Current Report on Form 8-K dated June 29, 2021)*
10.26	U.S. Notice of Terms of Non-Qualified Premium-Priced Stock Option for CEO, dated February 16, 2022 (Exhibit 10.1 to the Company's Form 10-Q for the quarter ended March 31, 2022)*
10.27	U.S. Notice of Terms of Long-Term Incentive Restricted Stock Units for CEO, dated February 16, 2022 (Exhibit 10.2 to the Company's Form 10-Q for the quarter ended March 31, 2022)*
10.28	Form of U.S. Notice of Terms of Non-Qualified Premium-Priced Stock Option (Exhibit 10.3 to the Company's Form 10-Q for the quarter ended March 31, 2022)*
10.29	Form of U.S. Notice of Terms of Long-Term Incentive Restricted Stock Units (Exhibit 10.4 to the Company's Form 10-Q for the quarter ended March 31, 2022)*
10.30	Form of International Notice of Terms of Non-Qualified Premium-Priced Stock Option (Exhibit 10.5 to the Company's Form 10-Q for the quarter ended March 31, 2022)*
10.31	Form of International Notice of Terms of Long-Term Incentive Restricted Stock Units (Exhibit 10.6 to the Company's Form 10-Q for the quarter ended March 31, 2022)*
10.32	Form of U.S. Notice of Terms of Long-Term Incentive Restricted Stock Units – CEO (Exhibit 10.1 to the Company's Form 10-Q for the quarter ended March 31, 2023)*
10.33	Form of U.S. Notice of Terms of Long-Term Incentive Performance Restricted Stock Units – CEO (Exhibit 10.2 to the Company's Form 10-Q for the quarter ended March 31, 2023)*
10.34	Form of U.S. Notice of Terms of Long-Term Incentive Restricted Stock Units (Exhibit 10.3 to the Company's Form 10-Q for the quarter ended March 31, 2023)*
10.35	Form of International Notice of Terms of Long-Term Incentive Restricted Stock Units (Stock-Settled) (Exhibit 10.4 to the Company's Form 10-Q for the quarter ended March 31, 2023)*
10.36	Form of U.S. Notice of Terms of Long-Term Incentive Performance Restricted Stock Units (Exhibit 10.5 to the Company's Form 10-Q for the quarter ended March 31, 2023)*
10.37	Form of International Notice of Terms of Long-Term Incentive Performance Restricted Stock Units (Stock-Settled) (Exhibit 10.6 to the Company's Form 10-Q for the quarter ended March 31, 2023)*

10.38	U.S. Notice of Terms of Supplemental Restricted Stock Units (Exhibit 10.7 to the Company's Form 10-Q for the quarter ended March 31, 2023)*
10.39	U.S. Notice of Terms of Special Restricted Stock Units - CEO, dated February 16, 2023 (Exhibit 10. 1 to the Company's Current Report on Form 8-K dated February 16, 2023)*
10.40	Employment Agreement between Boeing Canada Operations LTD and Susan Doniz (Exhibit 10.1 to the Company's Form 10-Q for the quarter ended June 30, 2020)*
21	List of Company Subsidiaries
23	Consent of Independent Registered Public Accounting Firm
31.1	Certification of Chief Executive Officer pursuant to Section 302 of Sarbanes-Oxley Act of 2002
31.2	Certification of Chief Financial Officer pursuant to Section 302 of Sarbanes-Oxley Act of 2002
32.1	Certification of Chief Executive Officer pursuant to Section 906 of Sarbanes-Oxley Act of 2002
32.2	Certification of Chief Financial Officer pursuant to Section 906 of Sarbanes-Oxley Act of 2002
97	The Boeing Company Clawback Policy
99.1	Commercial Program Method of Accounting (Exhibit (99)(i) to the Company's Form 10-K for the year ended December 31, 1997)
101.SCH	XBRL Taxonomy Extension Schema Document
101.CAL	XBRL Taxonomy Extension Calculation Linkbase Document
101.DEF	XBRL Taxonomy Extension Definition Linkbase Document
101.LAB	XBRL Taxonomy Extension Label Linkbase Document
101.PRE	XBRL Taxonomy Extension Presentation Linkbase Document
104	Cover Page Interactive Data File - the cover page XBRL tags are embedded within the Inline XBRL document contained in Exhibit 101

* Management contract or compensatory plan

In accordance with Item 601(b)(4)(iii)(A) of Regulation S-K, copies of certain instruments defining the rights of holders of long-term debt of the Company are not filed herewith. Pursuant to this regulation, we hereby agree to furnish a copy of any such instrument to the SEC upon request.

Item 16. Form 10-K Summary

None.

Signatures

Pursuant to the requirements of Section 13 of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized, on January 31, 2024.

THE BOEING COMPANY
(Registrant)

By: /s/ Michael J. Cleary
Michael J. Cleary – Senior Vice President and Controller

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following persons on behalf of the registrant and in the capacities indicated on January 31, 2024.

/s/ David L. Calhoun	/s/ Akhil Johri
David L. Calhoun – President and Chief Executive Officer and Director (Principal Executive Officer)	Akhil Johri – Director
/s/ Brian J. West	/s/ David L. Joyce
Brian J. West – Executive Vice President and Chief Financial Officer (Principal Financial Officer)	David L. Joyce – Director
/s/ Michael J. Cleary	/s/ Lawrence W. Kellner
Michael J. Cleary – Senior Vice President and Controller (Principal Accounting Officer)	Lawrence W. Kellner – Chair of the Board
/s/ Robert A. Bradway	/s/ Steven M. Mollenkopf
Robert A. Bradway – Director	Steven M. Mollenkopf – Director
/s/ Lynne M. Doughtie	/s/ John M. Richardson
Lynne M. Doughtie – Director	John M. Richardson – Director
/s/ David L. Gitlin	/s/ Sabrina Soussan
David L. Gitlin – Director	Sabrina Soussan – Director
/s/ Lynn J. Good	/s/ Ronald A. Williams
Lynn J. Good – Director	Ronald A. Williams – Director
/s/ Stayce D. Harris	
Stayce D. Harris – Director	